

Technical Information

Proline Promass E 300

Coriolis flowmeter



The flowmeter with minimized total cost of ownership and a compact, easily accessible transmitter

Application

- Measuring principle operates independently of physical fluid properties such as viscosity or density
- Accurate measurement of liquids and gases for a wide range of standard applications

Device properties

- Compact dual-tube sensor
- Medium temperature up to +150 °C (+302 °F)
- Process pressure: up to 100 bar (1 450 psi)
- Compact dual-compartment housing with up to 3 I/Os
- Backlit display with touch control and WLAN access
- Remote display available

Your benefits

- Cost-effective – multi-purpose device; an alternative to conventional volumetric flowmeters
- Fewer process measuring points – multivariable measurement (flow, density, temperature)
- Space-saving installation – no inlet/outlet run needs
- Full access to process and diagnostic information – numerous, freely combinable I/Os and fieldbuses
- Reduced complexity and variety – freely configurable I/O functionality
- Integrated verification – Heartbeat Technology

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About this document

Symbols used

Electrical symbols

Symbol	Meaning
	Direct current
	Alternating current
	Direct current and alternating current
	Ground connection A grounded terminal which, as far as the operator is concerned, is grounded via a grounding system.
	Protective Earth (PE) A terminal which must be connected to ground prior to establishing any other connections. The ground terminals are situated inside and outside the device: ■ Inner ground terminal: Connects the protective earth to the mains supply. ■ Outer ground terminal: Connects the device to the plant grounding system.

Communication symbols

Symbol	Meaning
	Wireless Local Area Network (WLAN) Communication via a wireless, local network.
	LED Light emitting diode is off.
	LED Light emitting diode is on.
	LED Light emitting diode is flashing.

Symbols for certain types of information

Symbol	Meaning
	Permitted Procedures, processes or actions that are permitted.
	Preferred Procedures, processes or actions that are preferred.
	Forbidden Procedures, processes or actions that are forbidden.
	Tip Indicates additional information.
	Reference to documentation.
	Reference to page.
	Reference to graphic.
	Visual inspection.

Symbols in graphics

Symbol	Meaning
1, 2, 3, ...	Item numbers
1, 2, 3, ...	Series of steps
A, B, C, ...	Views
A-A, B-B, C-C, ...	Sections
	Hazardous area
	Safe area (non-hazardous area)
	Flow direction

Function and system design

Measuring principle

The measuring principle is based on the controlled generation of Coriolis forces. These forces are always present in a system when both translational and rotational movements are superimposed.

$$F_c = 2 \cdot \Delta m (v \cdot \omega)$$

F_c = Coriolis force

Δm = moving mass

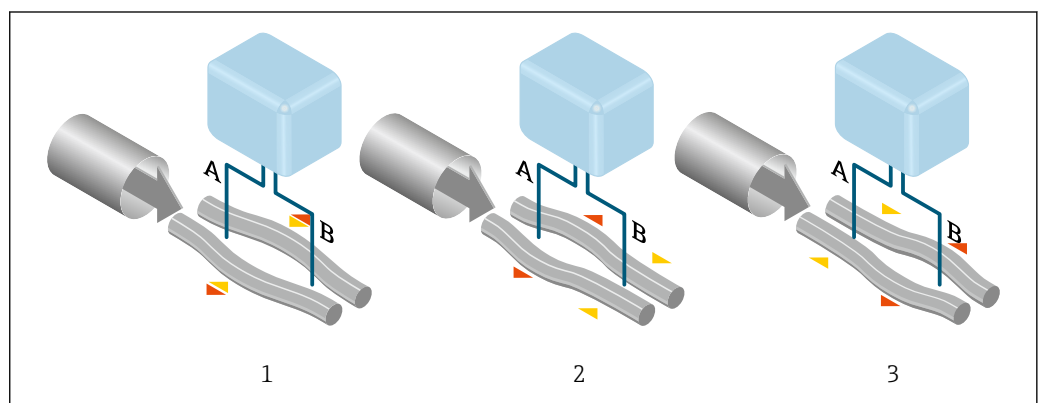
ω = rotational velocity

v = radial velocity in rotating or oscillating system

The amplitude of the Coriolis force depends on the moving mass Δm , its velocity v in the system and thus on the mass flow. Instead of a constant rotational velocity ω , the sensor uses oscillation.

In the sensor, two parallel measuring tubes containing flowing fluid oscillate in antiphase, acting like a tuning fork. The Coriolis forces produced at the measuring tubes cause a phase shift in the tube oscillations (see illustration):

- At zero flow (when the fluid is at a standstill) the two tubes oscillate in phase (1).
- Mass flow causes deceleration of the oscillation at the inlet of the tubes (2) and acceleration at the outlet (3).



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The phase difference (A-B) increases with increasing mass flow. Electrodynamic sensors register the tube oscillations at the inlet and outlet. System balance is ensured by the antiphase oscillation of the two measuring tubes. The measuring principle operates independently of temperature, pressure, viscosity, conductivity and flow profile.

Density measurement

The measuring tube is continuously excited at its resonance frequency. A change in the mass and thus the density of the oscillating system (comprising measuring tube and fluid) results in a corresponding, automatic adjustment in the oscillation frequency. Resonance frequency is thus a function of medium density. The microprocessor utilizes this relationship to obtain a density signal.

Volume measurement

Together with the measured mass flow, this is used to calculate the volume flow.

Temperature measurement

The temperature of the measuring tube is determined in order to calculate the compensation factor due to temperature effects. This signal corresponds to the process temperature and is also available as an output signal.

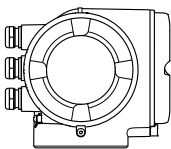
Measuring system

The device consists of a transmitter and a sensor.

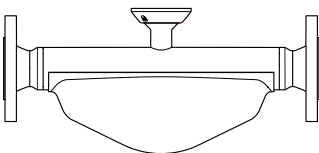
The device is available as a compact version:

The transmitter and sensor form a mechanical unit.

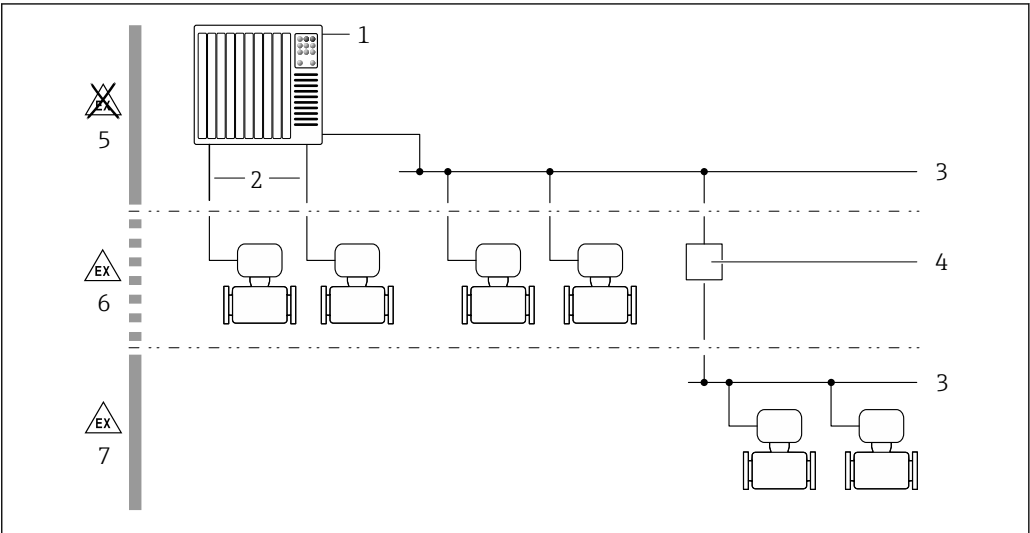
Transmitter

Promass 300  A0026708	Device versions and materials: ■ Transmitter housing - Aluminum, coated: aluminum, AlSi10Mg, coated - Stainless, hygienic: stainless steel, 1.4404 ■ Material of window in transmitter housing: - Aluminum, coated: glass - Stainless, hygienic: polycarbonate Configuration: ■ External operation via 4-line, backlit, graphic local display with touch control and guided menus ("Make-it-run" wizards) for application-specific commissioning. ■ Via service interface or WLAN interface: - Operating tools (e.g. FieldCare, DeviceCare, SmartBlue app) - Web server (access via Web browser, e.g. Microsoft Internet Explorer, Microsoft Edge)
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Sensor

Promass E  A0026712	■ For standard applications where stable and reliable measurements are required ■ Simultaneous measurement of flow, volume flow, density and temperature (multivariable) ■ Immune to process influences ■ Nominal diameter range: DN 8 to 80 ($\frac{3}{8}$ to 3") ■ Materials: - Sensor: stainless steel, 1.4301 (304) - Measuring tubes: stainless steel, 1.4539 (904L) - Process connections: stainless steel, 1.4404 (316/316L)
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Equipment architecture



A0027512

1 Possibilities for integrating measuring devices into a system

- 1 Control system (e.g. PLC)
- 2 Connecting cable (0/4 to 20 mA HART etc.)
- 3 Fieldbus
- 4 Segment coupler
- 5 Non-hazardous area
- 6 Hazardous area: Zone 2; Class I, Division 2
- 7 Hazardous area: Zone 1; Class I, Division 1

Safety

IT security

Our warranty is valid only if the device is installed and used as described in the Operating Instructions. The device is equipped with security mechanisms to protect it against any inadvertent changes to the settings.

IT security measures, which provide additional protection for the device and associated data transfer, must be implemented by the operators themselves in line with their security standards.

Device-specific IT security

The device offers a range of specific functions to support protective measures on the operator's side. These functions can be configured by the user and guarantee greater in-operation safety if used correctly. An overview of the most important functions is provided in the following section.

Function/interface	Factory setting	Recommendation
Write protection via hardware write protection switch → 8	Not enabled.	On an individual basis following risk assessment.
Access code (also applies for Web server login or FieldCare connection) → 8	Not enabled (0000).	Assign a customized access code during commissioning.
WLAN (order option in display module)	Enabled.	On an individual basis following risk assessment.
WLAN security mode	Enabled (WPA2-PSK)	Do not change.
WLAN passphrase (password) → 8	Serial number	Assign a customized access code during commissioning.
WLAN mode	Access Point	On an individual basis following risk assessment.
Web server → 8	Enabled.	On an individual basis following risk assessment.
CDI-RJ45 service interface → 9	–	On an individual basis following risk assessment.

Protecting access via hardware write protection

Write access to the device parameters via the local display, Web browser or operating tool (e.g. FieldCare, DeviceCare) can be disabled via a write protection switch (DIP switch on the motherboard). When hardware write protection is enabled, only read access to the parameters is possible.

Hardware write protection is disabled when the device is delivered.

Protecting access via a password

Different passwords are available to protect write access to the device parameters or access to the device via the WLAN interface.

- **User-specific access code**
Protect write access to the device parameters via the local display, Web browser or operating tool (e.g. FieldCare, DeviceCare). Access authorization is clearly regulated through the use of a user-specific access code.
- **WLAN passphrase**
The network key protects a connection between an operating unit (e.g. notebook or tablet) and the device via the WLAN interface which can be ordered as an option.
- **Infrastructure mode**
When the device is operated in infrastructure mode, the WLAN passphrase corresponds to the WLAN passphrase configured on the operator side.

User-specific access code

Write access to the device parameters via the local display, Web browser or operating tool (e.g. FieldCare, DeviceCare) can be protected by the modifiable, user-specific access code.

WLAN passphrase: Operation as WLAN access point

A connection between an operating unit (e.g. notebook or tablet) and the device via the WLAN interface, which can be ordered as an optional extra, is protected by the network key. The WLAN authentication of the network key complies with the IEEE 802.11 standard.

When the device is delivered, the network key is pre-defined depending on the device. It can be changed via the **WLAN settings** submenu in the **WLAN passphrase** parameter.

Infrastructure mode

A connection between the device and WLAN access point is protected by means of an SSID and passphrase on the system side. Please contact the relevant system administrator for access.

General notes on the use of passwords

- The access code and network key supplied with the device should be changed during commissioning.
- Follow the general rules for generating a secure password when defining and managing the access code or network key.
- The user is responsible for the management and careful handling of the access code and network key.

Access via Web server

The device can be operated and configured via a Web browser with the integrated Web server. The connection is via the service interface (CDI-RJ45) or the WLAN interface. For device versions with the EtherNet/IP and PROFINET communication protocols, the connection can also be established via the terminal connection for signal transmission with EtherNet/IP or PROFINET (RJ45 connector).

The Web server is enabled when the device is delivered. The Web server can be disabled if necessary (e.g. after commissioning) via the **Web server functionality** parameter.

The device and status information can be hidden on the login page. This prevents unauthorized access to the information.



For detailed information on device parameters, see:
The "Description of Device Parameters" document → 102

Access via OPC-UA

The "OPC UA Server" application package is available in the device version with the HART communication protocol → 98.

The device can communicate with OPC UA clients using the "OPC UA Server" application package.

The OPC UA server integrated in the device can be accessed via the WLAN access point using the WLAN interface - which can be ordered as an optional extra - or the service interface (CDI- RJ45) via Ethernet network. Access rights and authorization as per separate configuration.

The following Security Modes are supported as per the OPC UA Specification (IEC 62541):

- None
- Basic128Rsa15 – signed
- Basic128Rsa15 – signed and encrypted

Access via service interface (CDI-RJ45)

The device can be connected to a network via the service interface (CDI-RJ45). Device-specific functions guarantee the secure operation of the device in a network.

The use of relevant industrial standards and guidelines that have been defined by national and international safety committees, such as IEC/ISA62443 or the IEEE, is recommended. This includes organizational security measures such as the assignment of access authorization as well as technical measures such as network segmentation.



The device can be integrated in a ring topology. The device is integrated via the terminal connection for signal transmission (output 1) and the connection to the service interface (CDI-RJ45) →  87.

Input

Measured variable

Direct measured variables

- Mass flow
- Density
- Temperature

Calculated measured variables

- Volume flow
- Corrected volume flow
- Reference density

Measuring range

Measuring range for liquids

DN		Measuring range full scale values $\dot{m}_{\min(F)}$ to $\dot{m}_{\max(F)}$	
[mm]	[in]	[kg/h]	[lb/min]
8	$\frac{3}{8}$	0 to 2 000	0 to 73.50
15	$\frac{1}{2}$	0 to 6 500	0 to 238.9
25	1	0 to 18 000	0 to 661.5
40	$1\frac{1}{2}$	0 to 45 000	0 to 1 654
50	2	0 to 70 000	0 to 2 573
80	3	0 to 180 000	0 to 6 615

Measuring range for gases

The full scale value depends on the density and the sound velocity of the gas used and can be calculated with the formula below:

$$\dot{m}_{\max(G)} = \text{minimum} (\dot{m}_{\max(F)} \cdot \rho_G : x ; \rho_G \cdot c_G \cdot \pi/2 \cdot (d_i)^2 \cdot 3600)$$

$\dot{m}_{\max(G)}$	Maximum full scale value for gas [kg/h]
$\dot{m}_{\max(F)}$	Maximum full scale value for liquid [kg/h]
$\dot{m}_{\max(G)} < \dot{m}_{\max(F)}$	$\dot{m}_{\max(G)}$ can never be greater than $\dot{m}_{\max(F)}$
ρ_G	Gas density in [kg/m ³] at operating conditions
x	Constant dependent on nominal diameter
c_G	Sound velocity (gas) [m/s]
d_i	Measuring tube internal diameter [m]

DN		x
[mm]	[in]	[kg/m ³]
8	$\frac{3}{8}$	85
15	$\frac{1}{2}$	110
25	1	125
40	$1\frac{1}{2}$	125
50	2	125
80	3	155



To calculate the measuring range, use the *Applicator* sizing tool → 101

Calculation example for gas

- Sensor: Promass E, DN 50
- Gas: Air with a density of 60.3 kg/m^3 (at 20°C and 50 bar)
- Measuring range (liquid): 70 000 kg/h
- $x = 125 \text{ kg/m}^3$ (for Promass E, DN 50)

Maximum possible full scale value:

$$\dot{m}_{\max(\text{G})} = \dot{m}_{\max(\text{F})} \cdot \rho_{\text{G}} : x = 70\,000 \text{ kg/h} \cdot 60.3 \text{ kg/m}^3 : 125 \text{ kg/m}^3 = 33\,800 \text{ kg/h}$$

Recommended measuring range

"Flow limit" section →  55

Operable flow range

Over 1000 : 1.

Flow rates above the preset full scale value do not override the electronics unit, with the result that the totalizer values are registered correctly.

Input signal**Input and output versions**

→  13

External measured values

To increase the accuracy of certain measured variables or to calculate the corrected volume flow for gases, the automation system can continuously write different measured values to the measuring device:

- Operating pressure to increase accuracy (Endress+Hauser recommends the use of a pressure measuring device for absolute pressure, e.g. Cerabar M or Cerabar S)
- Medium temperature to increase accuracy (e.g. iTEMP)
- Reference density for calculating the corrected volume flow for gases



Various pressure transmitters and temperature measuring devices can be ordered from Endress +Hauser: see "Accessories" section →  101

It is recommended to read in external measured values to calculate the corrected volume flow.

HART protocol

The measured values are written from the automation system to the measuring device via the HART protocol. The pressure transmitter must support the following protocol-specific functions:

- HART protocol
- Burst mode

Current input

The measured values are written from the automation system to the measuring device via the current input →  11.

Digital communication

The measured values can be written from the automation system to the measuring via:

- FOUNDATION Fieldbus
- PROFIBUS DP
- PROFIBUS PA
- Modbus RS485
- EtherNet/IP
- PROFINET

Current input 0/4 to 20 mA

Current input	0/4 to 20 mA (active/passive)
Current span	■ 4 to 20 mA (active) ■ 0/4 to 20 mA (passive)
Resolution	1 μA
Voltage drop	Typically: 0.6 to 2 V for 3.6 to 22 mA (passive)
Maximum input voltage	$\leq 30 \text{ V}$ (passive)

Open-circuit voltage	≤ 28.8 V (active)
Possible input variables	■ Pressure ■ Temperature ■ Density

Status input

Maximum input values	■ DC -3 to 30 V ■ If status input is active (ON): $R_i > 3 \text{ k}\Omega$
Response time	Adjustable: 5 to 200 ms
Input signal level	■ Low signal: DC -3 to +5 V ■ High signal: DC 12 to 30 V
Assignable functions	■ Off ■ Reset the individual totalizers separately ■ Reset all totalizers ■ Flow override

Output

Output and input variants

Depending on the option selected for output/input 1, different options are available for the other outputs and inputs. Only one option can be selected for each output/input 1 to 3. The table must be read vertically (↓).

Example: If the option BA "4–20 mA HART" was selected for output/input 1, one of the options A, B, D, E, F, H, I or J is available for output 2 and one of the options A, B, D, E, F, H, I or J is available for output 3.

Order code for "Output; input 1" (020) →	Possible options									
Current output 4 to 20 mA HART	BA									
Current output 4 to 20 mA HART Ex i	↓	CA								
FOUNDATION Fieldbus		↓	SA							
FOUNDATION Fieldbus Ex i			↓	TA						
PROFIBUS DP				↓	LA					
PROFIBUS PA					↓	GA				
PROFIBUS PA Ex i						↓	HA			
Modbus RS485							↓	MA		
EtherNet/IP 2-port switch integrated								↓	NA	
PROFINET 2-port switch integrated									↓	RA
Order code for "Output; input 2" (021) →	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
Not assigned	A	A	A	A	A	A	A	A	A	A
Current output 0/4 to 20 mA	B		B		B	B		B	B	B
Current output 0/4 to 20 mA (Ex i)		C		C			C			
User configurable input/output ¹⁾	D		D		D	D		D	D	D
Pulse/frequency/switch output	E		E		E	E		E	E	E
Double pulse output ²⁾	F							F		
Pulse/frequency/switch output (Ex i)		G		G			G			
Relay output	H		H		H	H		H	H	H
Current input 0/4 to 20 mA	I		I		I	I		I	I	I
Status input	J		J		J	J		J	J	J
Order code for "Output; input 3" (022) →	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
Not assigned	A	A	A	A	A	A	A	A	A	A
Current output 0/4 to 20 mA	B				B			B	B	B
Current output 0/4 to 20 mA (Ex i)		C								
User configurable input/output	D				D			D	D	D
Pulse/frequency/switch output	E				E			E	E	E
Double pulse output (slave) ²⁾	F							F		
Pulse/frequency/switch output (Ex i)		G								
Relay output	H				H			H	H	H
Current input 0/4 to 20 mA	I				I			I	I	I
Status input	J				J			J	J	J

1) A specific input or output can be assigned to a user configurable input/output → 17.

2) If double pulse output (F) is selected for output/input 2 (021), only the double pulse output (F) option is available for selection for output/input 3 (022).

Output signal

HART current output

Current output	4 to 20 mA HART
Current span	Can be set to: 4 to 20 mA (active/passive)  Ex-i, passive
Open-circuit voltage	DC 28.8 V (active)
Maximum input voltage	DC 30 V (passive)
Load	250 to 700 Ω
Resolution	0.38 μ A
Damping	Configurable: 0.07 to 999 s
Assignable measured variables	■ Mass flow ■ Volume flow ■ Corrected volume flow ■ Density ■ Reference density ■ Temperature ■ Electronic temperature ■ Oscillation frequency 0 ■ Oscillation damping 0 ■ Signal asymmetry ■ Exciter current 0  The range of options increases if the measuring device has one or more application packages.

PROFIBUS PA

PROFIBUS PA	In accordance with EN 50170 Volume 2, IEC 61158-2 (MBP), galvanically isolated
Data transmission	31.25 kbit/s
Current consumption	10 mA
Permitted supply voltage	9 to 32 V
Bus connection	With integrated reverse polarity protection

PROFIBUS DP

Signal encoding	NRZ code
Data transfer	9.6 kBaud...12 MBaud

EtherNet/IP

Standards	In accordance with IEEE 802.3
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PROFINET

Standards	In accordance with IEEE 802.3
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FOUNDATION Fieldbus

FOUNDATION Fieldbus	H1, IEC 61158-2, galvanically isolated
Data transfer	31.25 kbit/s

Current consumption	10 mA
Permitted supply voltage	9 to 32 V
Bus connection	With integrated reverse polarity protection

Modbus RS485

Physical interface	RS485 in accordance with EIA/TIA-485 standard
Terminating resistor	Integrated, can be activated via DIP switches

Current output 0/4 to 20 mA

Current output	0/4 to 20 mA
Maximum output values	22.5 mA
Current span	Can be set to: ■ 4 to 20 mA (active) ■ 0/4 to 20 mA (passive)  Ex-i, passive
Open-circuit voltage	DC 28.8 V (active)
Maximum input voltage	DC 30 V (passive)
Load	0 to 700 Ω
Resolution	0.38 μ A
Damping	Adjustable: 0.07 to 999 s
Assignable measured variables	■ Mass flow ■ Volume flow ■ Corrected volume flow ■ Density ■ Reference density ■ Temperature ■ Electronic temperature ■ Oscillation frequency 0 ■ Oscillation damping 0 ■ Signal asymmetry ■ Exciter current 0  The range of options increases if the measuring device has one or more application packages.

Pulse/frequency/switch output

Function	Can be set to pulse, frequency or switch output
Version	Open collector Can be set to: ■ Active ■ Passive  Ex-i, passive
Maximum input values	DC 30 V, 250 mA (passive)
Open-circuit voltage	DC 28.8 V (active)
Voltage drop	For 22.5 mA: $\leq$ DC 2 V
Pulse output	
Maximum input values	DC 30 V, 250 mA (passive)
Maximum output current	22.5 mA (active)

Open-circuit voltage	DC 28.8 V (active)
Pulse width	Adjustable: 0.05 to 2 000 ms
Maximum pulse rate	10 000 Impulse/s
Pulse value	Adjustable
Assignable measured variables	■ Mass flow ■ Volume flow ■ Corrected volume flow
Frequency output	
Maximum input values	DC 30 V, 250 mA (passive)
Maximum output current	22.5 mA (active)
Open-circuit voltage	DC 28.8 V (active)
Output frequency	Adjustable: end value frequency 2 to 10 000 Hz ($f_{\max} = 12\,500$ Hz)
Damping	Adjustable: 0 to 999 s
Pulse/pause ratio	1:1
Assignable measured variables	■ Mass flow ■ Volume flow ■ Corrected volume flow ■ Density ■ Reference density ■ Temperature ■ Electronic temperature ■ Oscillation frequency 0 ■ Oscillation damping 0 ■ Signal asymmetry ■ Exciter current 0  The range of options increases if the measuring device has one or more application packages.
Switch output	
Maximum input values	DC 30 V, 250 mA (passive)
Open-circuit voltage	DC 28.8 V (active)
Switching behavior	Binary, conductive or non-conductive
Switching delay	Adjustable: 0 to 100 s
Number of switching cycles	Unlimited
Assignable functions	■ Off ■ On ■ Diagnostic behavior ■ Limit value – Mass flow – Volume flow – Corrected volume flow – Density – Reference density – Temperature – Totalizer 1-3 ■ Flow direction monitoring ■ Status – Partially filled pipe detection – Low flow  The range of options increases if the measuring device has one or more application packages.

Double pulse output

Function	Double pulse
Version	Open collector Can be set to: ■ Active ■ Passive
Maximum input values	DC 30 V, 250 mA (passive)
Open-circuit voltage	DC 28.8 V (active)
Voltage drop	For 22.5 mA: ≤ DC 2 V
Output frequency	Adjustable: 0 to 1 000 Hz
Damping	Adjustable: 0 to 999 s
Pulse/pause ratio	1:1
Assignable measured variables	■ Mass flow ■ Volume flow ■ Corrected volume flow ■ Density ■ Reference density ■ Temperature  The range of options increases if the measuring device has one or more application packages.

Relay output

Function	Switch output
Version	Relay output, galvanically isolated
Switching behavior	Can be set to: ■ NO (normally open), factory setting ■ NC (normally closed)
Maximum switching capacity (passive)	■ DC 30 V, 0.1 A ■ AC 30 V, 0.5 A
Assignable functions	■ Off ■ On ■ Diagnostic behavior ■ Limit value – Mass flow – Volume flow – Corrected volume flow – Density – Reference density – Temperature – Totalizer 1-3 ■ Flow direction monitoring ■ Status – Partially filled pipe detection – Low flow  The range of options increases if the measuring device has one or more application packages.

User configurable input/output

One specific input or output is assigned to a user-configurable input/output (configurable I/O) during device commissioning.

The following inputs and outputs are available for assignment:

- Choice of current output: 4 to 20 mA (active), 0/4 to 20 mA (passive)
- Pulse/frequency/switch output
- Choice of current input: 4 to 20 mA (active), 0/4 to 20 mA (passive)
- Status input

The technical values correspond to those of the inputs and outputs described in this section.

Signal on alarm

Depending on the interface, failure information is displayed as follows:

HART current output

Device diagnostics	Device condition can be read out via HART Command 48
--------------------	--

PROFIBUS PA

Status and alarm messages	Diagnostics in accordance with PROFIBUS PA Profile 3.02
Failure current FDE (Fault Disconnection Electronic)	0 mA

PROFIBUS DP

Status and alarm messages	Diagnostics in accordance with PROFIBUS PA Profile 3.02
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EtherNet/IP

Device diagnostics	Device condition can be read out in Input Assembly
--------------------	--

PROFINET

Device diagnostics	According to "Application Layer protocol for decentralized periphery", Version 2.3
--------------------	--

FOUNDATION Fieldbus

Status and alarm messages	Diagnostics in accordance with FF-891
Failure current FDE (Fault Disconnection Electronic)	0 mA

Modbus RS485

Failure mode	Choose from: ■ NaN value instead of current value ■ Last valid value
--------------	---

Current output 0/4 to 20 mA

4 to 20 mA

Failure mode	Choose from: ■ 4 to 20 mA in accordance with NAMUR recommendation NE 43 ■ 4 to 20 mA in accordance with US ■ Min. value: 3.59 mA ■ Max. value: 22.5 mA ■ Freely definable value between: 3.59 to 22.5 mA ■ Actual value ■ Last valid value
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0 to 20 mA

Failure mode	Choose from: ■ Maximum alarm: 22 mA ■ Freely definable value between: 0 to 20.5 mA
---------------------	---

Pulse/frequency/switch output

Pulse output	
Failure mode	Choose from: ■ Actual value ■ No pulses
Frequency output	
Failure mode	Choose from: ■ Actual value ■ 0 Hz ■ Defined value ($f_{\max}$ 2 to 12 500 Hz)
Switch output	
Failure mode	Choose from: ■ Current status ■ Open ■ Closed

Relay output

Failure mode	Choose from: ■ Current status ■ Open ■ Closed
---------------------	---

Local display

Plain text display	With information on cause and remedial measures
Backlight	Red backlighting indicates a device error.



Status signal as per NAMUR recommendation NE 107

Interface/protocol

- Via digital communication:
 - HART protocol
 - FOUNDATION Fieldbus
 - PROFIBUS PA
 - PROFIBUS DP
 - Modbus RS485
 - EtherNet/IP
 - PROFINET
- Via service interface
 - CDI-RJ45 service interface
 - WLAN interface

Plain text display	With information on cause and remedial measures
---------------------------	---



Additional information on remote operation → 84

Web server

Plain text display	With information on cause and remedial measures
---------------------------	---

Light emitting diodes (LED)

Status information	Status indicated by various light emitting diodes The following information is displayed depending on the device version: ■ Supply voltage active ■ Data transmission active ■ Device alarm/error has occurred ■ EtherNet/IP network available ■ EtherNet/IP connection established ■ PROFINET network available ■ PROFINET connection established ■ PROFINET blinking feature
---------------------------	---

Ex connection data**Safety-related values**

Order code for "Output; input 1"	Output type	Safety-related values "Output; input 1"	
		26 (+)	27 (-)
Option BA	Current output 4 to 20 mA HART	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$	
Option GA	PROFIBUS PA	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$	
Option LA	PROFIBUS DP	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$	
Option MA	Modbus RS485	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$	
Option SA	FOUNDATION Fieldbus	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$	
Option NA	EtherNet/IP	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$	
Option RA	PROFINET	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$	

Order code for "Output; input 2"; "Output; input 3"	Output type	Safety-related values			
		Output; input 2		Output; input 3	
		24 (+)	25 (-)	22 (+)	23 (-)
Option B	Current output 4 to 20 mA	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$			
Option D	User configurable input/ output	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$			
Option E	Pulse/frequency/switch output	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$			
Option F	Double pulse output	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$			
Option H	Relay output	$U_N = 30 V_{DC}$ $I_N = 100 mA_{DC}/500 mA_{AC}$ $U_M = 250 V_{AC}$			
Option I	Current input 4 to 20 mA	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$			
Option J	Status input	$U_N = 30 V_{DC}$ $U_M = 250 V_{AC}$			

Intrinsically safe values

Order code for "Output; input 1"	Output type	Intrinsically safe values "Output; input 1"	
		26 (+)	27 (-)
Option CA	Current output 4 to 20 mA HART Ex i	$U_i = 30\text{ V}$ $I_i = 100\text{ mA}$ $P_i = 1.25\text{ W}$ $L_i = 0$ $C_i = 0$	
Option HA	PROFIBUS PA Ex i	Ex ia ¹⁾ $U_i = 30\text{ V}$ $I_i = 570\text{ mA}$ $P_i = 8.5\text{ W}$ $L_i = 10\text{ }\mu\text{H}$ $C_i = 5\text{ nF}$	Ex ic ²⁾ $U_i = 32\text{ V}$ $I_i = 570\text{ mA}$ $P_i = 8.5\text{ W}$ $L_i = 10\text{ }\mu\text{H}$ $C_i = 5\text{ nF}$
Option TA	FOUNDATION Fieldbus Ex i	Ex ia ¹⁾ $U_i = 30\text{ V}$ $I_i = 570\text{ mA}$ $P_i = 8.5\text{ W}$ $L_i = 10\text{ }\mu\text{H}$ $C_i = 5\text{ nF}$	Ex ic ²⁾ $U_i = 32\text{ V}$ $I_i = 570\text{ mA}$ $P_i = 8.5\text{ W}$ $L_i = 10\text{ }\mu\text{H}$ $C_i = 5\text{ nF}$

1) Only available for the Zone 1; Class I, Division 1 version

2) Only available for the Zone 2; Class I, Division 2 version transmitter

Order code for "Output; input 2"; "Output; input 3"	Output type	Intrinsically safe values or NIFW values			
		Output; input 2		Output; input 3	
		24 (+)	25 (-)	22 (+)	23 (-)
Option C	Current output 4 to 20 mA Ex i	$U_i = 30\text{ V}$ $I_i = 100\text{ mA}$ $P_i = 1.25\text{ W}$ $L_i = 0$ $C_i = 0$			
Option G	Pulse/frequency/switch output Ex i	$U_i = 30\text{ V}$ $I_i = 100\text{ mA}$ $P_i = 1.25\text{ W}$ $L_i = 0$ $C_i = 0$			

Low flow cut off The switch points for low flow cut off are user-selectable.

Galvanic isolation The outputs are galvanically isolated from one another and from earth (PE).

Protocol-specific data **HART**

Manufacturer ID	0x11
Device type ID	0x3B
HART protocol revision	7
Device description files (DTM, DD)	Information and files under: www.endress.com
HART load	Min. 250 Ω
System integration	Information on system integration: Operating Instructions → 102. - Measured variables via HART protocol - Burst Mode functionality

PROFIBUS PA

Manufacturer ID	0x11
Ident number	0x156D
Profile version	3.02
Device description files (GSD, DTM, DD)	Information and files under: ▪ www.endress.com ▪ www.profibus.org
Supported functions	▪ Identification & Maintenance Simplest device identification on the part of the control system and nameplate ▪ PROFIBUS upload/download Reading and writing parameters is up to ten times faster with PROFIBUS upload/download ▪ Condensed status Simplest and self-explanatory diagnostic information by categorizing diagnostic messages that occur
Configuration of the device address	▪ DIP switches on the I/O electronics module ▪ Local display ▪ Via operating tools (e.g. FieldCare)
Compatibility with earlier model	If the device is replaced, the measuring device Promass 300 supports the compatibility of the cyclic data with previous models. It is not necessary to adjust the engineering parameters of the PROFIBUS network with the Promass 300 GSD file. Earlier models: ▪ Promass 80 PROFIBUS PA – ID No.: 1528 (hex) – Extended GSD file: EH3x1528.gsd – Standard GSD file: EH3_1528.gsd ▪ Promass 83 PROFIBUS PA – ID No.: 152A (hex) – Extended GSD file: EH3x152A.gsd – Standard GSD file: EH3_152A.gsd  Description of the function scope of compatibility: Operating Instructions →  102.
System integration	Information regarding system integration: Operating Instructions →  102. ▪ Cyclic data transmission ▪ Block model ▪ Description of the modules

PROFIBUS DP

Manufacturer ID	0x11
Ident number	0x156F
Profile version	3.02
Device description files (GSD, DTM, DD)	Information and files under: ▪ www.endress.com On the product page for the device: Documents/Software → Device drivers ▪ www.profibus.org
Supported functions	▪ Identification & Maintenance Simplest device identification on the part of the control system and nameplate ▪ PROFIBUS upload/download Reading and writing parameters is up to ten times faster with PROFIBUS upload/download ▪ Condensed status Simplest and self-explanatory diagnostic information by categorizing diagnostic messages that occur
Configuration of the device address	▪ DIP switches on the I/O electronics module ▪ Via operating tools (e.g. FieldCare)

Compatibility with earlier model	If the device is replaced, the measuring device Promass 300 supports the compatibility of the cyclic data with previous models. It is not necessary to adjust the engineering parameters of the PROFIBUS network with the Promass 300 GSD file. Previous model: Promass 83 PROFIBUS DP – ID No.: 1529 (hex) – Extended GSD file: EH3x1529.gsd – Standard GSD file: EH3_1529.gsd  Description of the function scope of compatibility: Operating Instructions →  102.
System integration	Information regarding system integration: Operating Instructions →  102. ▪ Cyclic data transmission ▪ Block model ▪ Description of the modules

EtherNet/IP

Protocol	▪ The CIP Networks Library Volume 1: Common Industrial Protocol ▪ The CIP Networks Library Volume 2: EtherNet/IP Adaptation of CIP
Communication type	▪ 10Base-T ▪ 100Base-TX
Device profile	Generic device (product type: 0x2B)
Manufacturer ID	0x11
Device type ID	0x103B
Baud rates	Automatic 10_{100} Mbit with half-duplex and full-duplex detection
Polarity	Auto-polarity for automatic correction of crossed Tx/D and Rx/D pairs
Supported CIP connections	Max. 3 connections
Explicit connections	Max. 6 connections
I/O connections	Max. 6 connections (scanner)
Configuration options for measuring device	▪ DIP switches on the electronics module for IP addressing ▪ Manufacturer-specific software (FieldCare) ▪ Add-on Profile Level 3 for Rockwell Automation control systems ▪ Web browser ▪ Electronic Data Sheet (EDS) integrated in the measuring device
Configuration of the EtherNet interface	▪ Speed: 10 MBit, 100 MBit, auto (factory setting) ▪ Duplex: half-duplex, full-duplex, auto (factory setting)
Configuration of the device address	▪ DIP switches on the electronics module for IP addressing (last octet) ▪ DHCP ▪ Manufacturer-specific software (FieldCare) ▪ Add-on Profile Level 3 for Rockwell Automation control systems ▪ Web browser ▪ EtherNet/IP tools, e.g. RSLinx (Rockwell Automation)
Device Level Ring (DLR)	Yes
System integration	Information regarding system integration: Operating Instructions →  102. ▪ Cyclic data transmission ▪ Block model ▪ Input and output groups

PROFINET

Protocol	"Application layer protocol for decentral device periphery and distributed automation", version 2.3
Communication type	100 MBit/s

Conformity class	Conformance Class B
Netload Class	Netload Class II
Baud rates	Automatic 100 Mbit/s with full-duplex detection
Cycle times	From 8 ms
Polarity	Auto-polarity for automatic correction of crossed TxD and RxD pairs
Media Redundancy Protocol (MRP)	Yes
Device profile	Application interface identifier 0xF600 Generic device
Manufacturer ID	0x11
Device type ID	0x843B
Device description files (GSD, DTM, DD)	Information and files under: ▪ www.endress.com On the product page for the device: Documents/Software → Device drivers ▪ www.profibus.org
Supported connections	▪ 1 x AR (IO Controller AR) ▪ 1 x AR (IO-Supervisor Device AR connection allowed) ▪ 1 x Input CR (Communication Relation) ▪ 1 x Output CR (Communication Relation) ▪ 1 x Alarm CR (Communication Relation)
Configuration options for measuring device	▪ DIP switches on the electronics module, for device name assignment (last part) ▪ Manufacturer-specific software (FieldCare, DeviceCare) ▪ Web browser ▪ Device master file (GSD), can be read out via the integrated Web server of the measuring device
Configuration of the device name	▪ DIP switches on the electronics module, for device name assignment (last part) ▪ DCP protocol ▪ Process Device Manager (PDM) ▪ Integrated Web server
Supported functions	▪ Identification & Maintenance Simple device identification via: – Control system – Nameplate ▪ Measured value status The process variables are communicated with a measured value status ▪ Blinking feature via the onsite display for simple device identification and assignment ▪ Device operation via operating tools (e.g. FieldCare, DeviceCare, SIMATIC PDM)
System integration	Information regarding system integration: Operating Instructions → 102. ▪ Cyclic data transmission ▪ Overview and description of the modules ▪ Status coding ▪ Startup configuration ▪ Factory setting:

FOUNDATION Fieldbus

Manufacturer ID	0x452B48 (hex)
Ident number	0x103B (hex)
Device revision	1
DD revision	Information and files under:
CFF revision	▪ www.endress.com ▪ www.fieldbus.org
Interoperability Test Kit (ITK)	Version 6.2.0

ITK Test Campaign Number	Information: ■ www.endress.com ■ www.fieldbus.org
Link Master capability (LAS)	Yes
Choice of "Link Master" and "Basic Device"	Yes Factory setting: Basic Device
Node address	Factory setting: 247 (0xF7)
Supported functions	The following methods are supported: ■ Restart ■ ENP Restart ■ Diagnostic ■ Set to OOS ■ Set to AUTO ■ Read trend data ■ Read event logbook
Virtual Communication Relationships (VCRs)	
Number of VCRs	44
Number of link objects in VFD	50
Permanent entries	1
Client VCRs	0
Server VCRs	10
Source VCRs	43
Sink VCRs	0
Subscriber VCRs	43
Publisher VCRs	43
Device Link Capabilities	
Slot time	4
Min. delay between PDU	8
Max. response delay	16
System integration	Information regarding system integration: Operating Instructions →  102. ■ Cyclic data transmission ■ Description of the modules ■ Execution times ■ Methods

Modbus RS485

Protocol	Modbus Applications Protocol Specification V1.1
Response times	■ Direct data access: typically 25 to 50 ms ■ Auto-scan buffer (data range): typically 3 to 5 ms
Device type	Slave
Slave address range	1 to 247
Broadcast address range	0
Function codes	■ 03: Read holding register ■ 04: Read input register ■ 06: Write single registers ■ 08: Diagnostics ■ 16: Write multiple registers ■ 23: Read/write multiple registers
Broadcast messages	Supported by the following function codes: ■ 06: Write single registers ■ 16: Write multiple registers ■ 23: Read/write multiple registers

Supported baud rate	1 200 BAUD 2 400 BAUD 4 800 BAUD 9 600 BAUD 19 200 BAUD 38 400 BAUD 57 600 BAUD 115 200 BAUD
Data transfer mode	- ASCII - RTU
Data access	Each device parameter can be accessed via Modbus RS485.  For Modbus register information
Compatibility with earlier model	If the device is replaced, the measuring device Promass 300 supports the compatibility of the Modbus registers for the process variables and the diagnostic information with the previous model Promass 83. It is not necessary to change the engineering parameters in the automation system.  Description of the function scope of compatibility: Operating Instructions →  102.
System integration	Information on system integration: Operating Instructions →  102. - Modbus RS485 information - Function codes - Register information - Response time - Modbus data map

Power supply

Terminal assignment

Transmitter: supply voltage, input/outputs

HART

Supply voltage		Input/output 1		Input/output 2		Input/output 3	
1 (+)	2 (-)	26 (+)	27 (-)	24 (+)	25 (-)	22 (+)	23 (-)
The terminal assignment depends on the specific device version ordered →  13.							

FOUNDATION Fieldbus

Supply voltage		Input/output 1		Input/output 2		Input/output 3	
1 (+)	2 (-)	26 (A)	27 (B)	24 (+)	25 (-)	22 (+)	23 (-)
The terminal assignment depends on the specific device version ordered →  13.							

PROFIBUS PA

Supply voltage		Input/output 1		Input/output 2		Input/output 3	
1 (+)	2 (-)	26 (B)	27 (A)	24 (+)	25 (-)	22 (+)	23 (-)
The terminal assignment depends on the specific device version ordered →  13.							

PROFIBUS DP

Supply voltage		Input/output 1		Input/output 2		Input/output 3	
1 (+)	2 (-)	26 (B)	27 (A)	24 (+)	25 (-)	22 (+)	23 (-)
The terminal assignment depends on the specific device version ordered →  13.							

Modbus RS485

Supply voltage		Input/output 1		Input/output 2		Input/output 3	
1 (+)	2 (-)	26 (B)	27 (A)	24 (+)	25 (-)	22 (+)	23 (-)
The terminal assignment depends on the specific device version ordered → 13.							

PROFINET

Supply voltage		Input/output 1	Input/output 2		Input/output 3	
1 (+)	2 (-)	PROFINET (RJ45 connector)	24 (+)	25 (-)	22 (+)	23 (-)
The terminal assignment depends on the specific device version ordered → 13.						

EtherNet/IP

Supply voltage		Input/output 1	Input/output 2		Input/output 3	
1 (+)	2 (-)	EtherNet/IP (RJ45 connector)	24 (+)	25 (-)	22 (+)	23 (-)
The terminal assignment depends on the specific device version ordered → 13.						



Terminal assignment of the remote display and operating module → 32.

Device plugs available

Device plugs may not be used in hazardous areas!

Device plugs for fieldbus systems:

Order code for "Input; output 1"

- Option **SA** "FOUNDATION Fieldbus" → 27
- Option **GA** "PROFIBUS PA" → 27
- Option **RA** "PROFINET" → 28
- Option **NA** "EtherNet/IP" → 28

Device plug for connecting to the service interface:

Order code for "Accessory mounted"

option **NB**, adapter RJ45 M12 (service interface) → 29

Order code for "Input; output 1", option SA "FOUNDATION Fieldbus"

Order code for "Electrical connection"	Cable entry/connection → 31	
	2	3
M, 3, 4, 5	7/8" connector	–

Order code for "Input; output 1", option GA "PROFIBUS PA"

Order code for "Electrical connection"	Cable entry/connection → 31	
	2	3
L, N, P, U	Connector M12 × 1	–

Order code for "Input; output 1", option RA "PROFINET"

Order code for "Electrical connection"	Cable entry/connection → 31	
	2	3
L, N, P, U	Connector M12 × 1	–
R ^{1) 2)} , S ^{1) 2)} , T ^{1) 2)} , V ^{1) 2)}	Connector M12 × 1	Connector M12 × 1

- 1) Cannot be combined with an external WLAN antenna (order code for "Enclosed accessories", option P8) of an RJ45 M12 adapter for the service interface (order code for "Accessories mounted", option NB) or of the remote display and operating module DKX001.
- 2) Suitable for integrating the device in a ring topology.

Order code for "Input; output 1", option NA "EtherNet/IP"

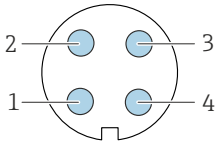
Order code for "Electrical connection"	Cable entry/connection → 31	
	2	3
L, N, P, U	Connector M12 × 1	–
R ^{1) 2)} , S ^{1) 2)} , T ^{1) 2)} , V ^{1) 2)}	Connector M12 × 1	Connector M12 × 1

- 1) Cannot be combined with an external WLAN antenna (order code for "Enclosed accessories", option P8) of an RJ45 M12 adapter for the service interface (order code for "Accessories mounted", option NB) or of the remote display and operating module DKX001
- 2) Suitable for integrating the device in a ring topology.

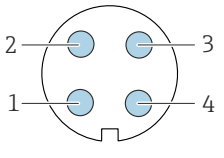
Order code for "Accessory mounted", option NB "Adapter RJ45 M12 (service interface)"

Order code "Accessory mounted"	Cable entry/coupling → 31	
	Cable entry 2	Cable entry 3
NB	Plug M12 × 1	–

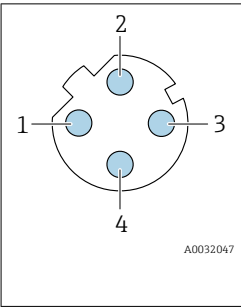
Pin assignment, device plug**FOUNDATION Fieldbus**

	Pin		Assignment	Coding	Plug/socket
	1	+	Signal +	A	Plug
	2	-	Signal -		
	3		Grounding		
	4		Not assigned		

PROFIBUS PA

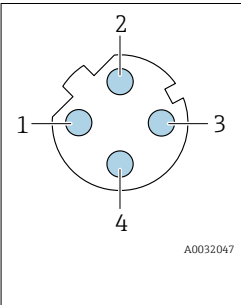
	Pin		Assignment	Coding	Plug/socket
	1	+	PROFIBUS PA +	A	Plug
	2		Grounding		
	3	-	PROFIBUS PA -		
	4		Not assigned		

PROFINET

	Pin	Assignment	
	1	+	TD +
	2	+	RD +
	3	-	TD -
	4	-	RD -
	Coding		Plug/socket
		D	Socket

-  Recommended plug:
- Binder, series 763, part no. 99 3729 810 04
 - Phoenix, part no. 1543223 SACC-M12MSD-4Q
 - When using the device in a hazardous location, use a suitably certified plug.

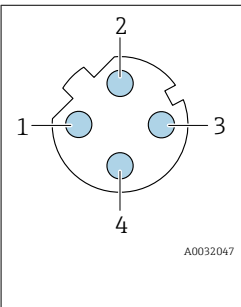
EtherNet/IP

	Pin	Assignment	
	1	+	Tx
	2	+	Rx
	3	-	Tx
	4	-	Rx
	Coding		Plug/socket
		D	Socket

-  Recommended plug:
- Binder, series 763, part no. 99 3729 810 04
 - Phoenix, part no. 1543223 SACC-M12MSD-4Q
 - When using the device in a hazardous location, use a suitably certified plug.

Service interface

Order code for "Accessories mounted", option **NB**: Adapter RJ45 M12 (service interface)

	Pin	Assignment	
	1	+	Tx
	2	+	Rx
	3	-	Tx
	4	-	Rx
	Coding		Plug/socket
		D	Socket

-  Recommended plug:
- Binder, series 763, part no. 99 3729 810 04
 - Phoenix, part no. 1543223 SACC-M12MSD-4Q
 - When using the device in a hazardous location, use a suitably certified plug.

Supply voltage	Order code for "Power supply"		terminal voltage		Frequency range
	Option D		DC24 V	±20%	–
	Option E		AC100 to 240 V	–15...+10%	50/60 Hz
	Option I		DC24 V	±20%	–
			AC100 to 240 V	–15...+10%	50/60 Hz

Power consumption**Transmitter**

Max. 10 W (active power)

switch-on current	Max. 36 A (as per NAMUR Recommendation NE21)
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Current consumption**Transmitter**

- Max. 400 mA (24 V)
- Max. 200 mA (110 V, 50/60 Hz; 230 V, 50/60 Hz)

Power supply failure

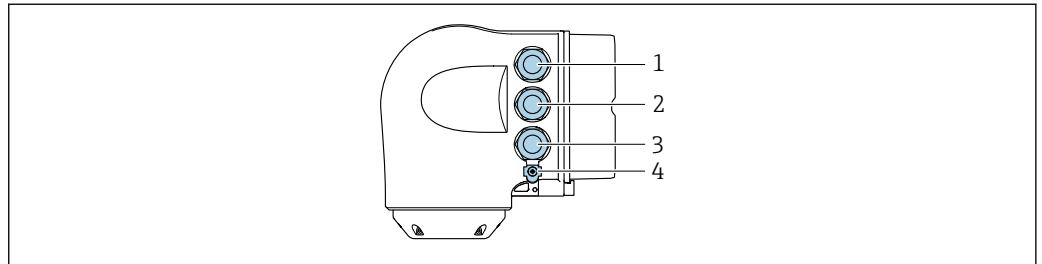
- Totalizers stop at the last value measured.
- Depending on the device version, the configuration is retained in the device memory or in the pluggable data memory (HistoROM DAT).
- Error messages (incl. total operated hours) are stored.

Electrical connection

Connecting the transmitter



- Terminal assignment → 26
- Device plugs available → 27



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- 1 Terminal connection for supply voltage
- 2 Terminal connection for signal transmission, input/output
- 3 Terminal connection for signal transmission, input/output or terminal for network connection via service interface (CDI-RJ45); Optional: terminal connection for external WLAN antenna or connection for remote display and operating module DKX001
- 4 Protective ground (PE)



An adapter for RJ45 and the M12 connector is optionally available:
Order code for "Accessories", option **NB**: "Adapter RJ45 M12 (service interface)"

The adapter connects the service interface (CDI-RJ45) to an M12 connector mounted in the cable entry. Therefore the connection to the service interface can be established via an M12 connector without opening the device.



Network connection via service interface (CDI-RJ45) → 89

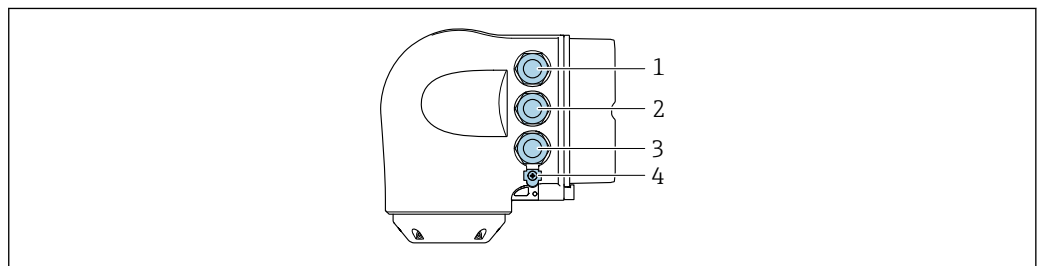
Connecting in a ring topology

Device versions with EtherNet/IP and PROFINET communication protocols can be integrated into a ring topology. The device is integrated via the terminal connection for signal transmission (output 1) and the connection to the service interface (CDI-RJ45).



Integrate the transmitter into a ring topology:

- EtherNet/IP → 87
- PROFINET → 88



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- 1 Terminal connection for supply voltage
- 2 Terminal connection for signal transmission: PROFINET or EtherNet/IP (RJ45 connector)
- 3 Connection to service interface (CDI-RJ45)
- 4 Protective earth (PE)

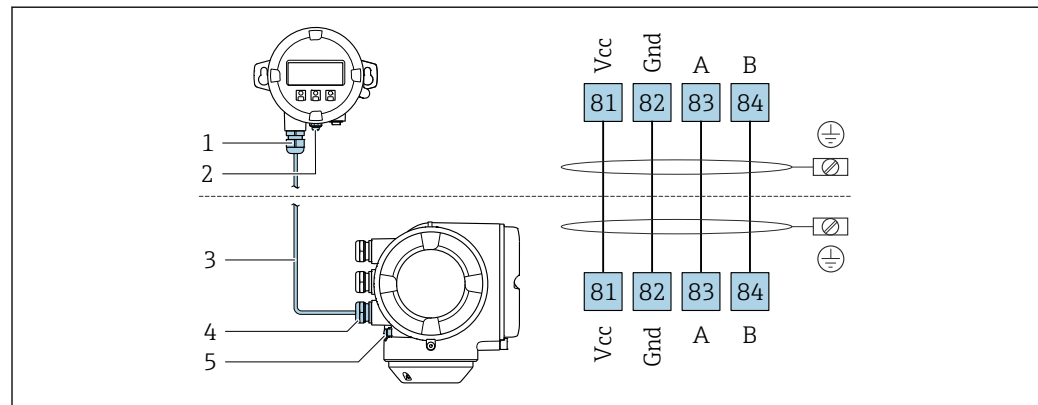


If the device has additional input/outputs, these are routed via the cable entry for the connection to the service interface (CDI-RJ45).

Connecting the remote display and operating module DKX001

i The remote display and operating module DKX001 is available as an optional extra → 99.

- The remote display and operating module DKX001 is only available for the following housing version: order code for "Housing": option A "Aluminum, coated"
- The measuring device is always supplied with a dummy cover when the remote display and operating module DKX001 is ordered directly with the measuring device. Display or operation at the transmitter is not possible in this case.
- If ordered subsequently, the remote display and operating module DKX001 may not be connected at the same time as the existing measuring device display module. Only one display or operation unit may be connected to the transmitter at any one time.

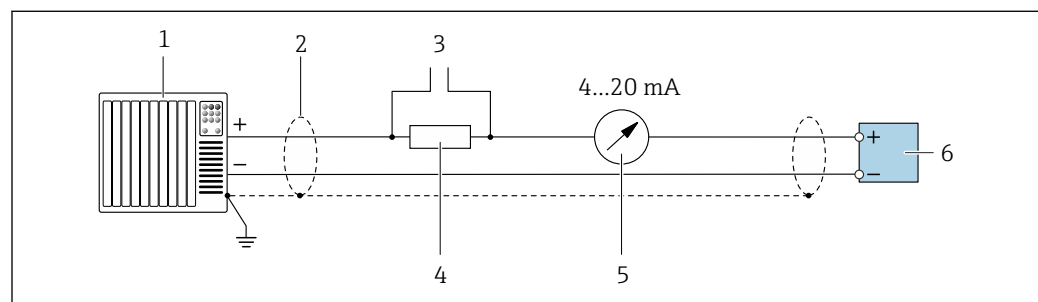


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- 1 Remote display and operating module DKX001
- 2 Protective earth (PE)
- 3 Connecting cable
- 4 Measuring device
- 5 Protective earth (PE)

Connection examples

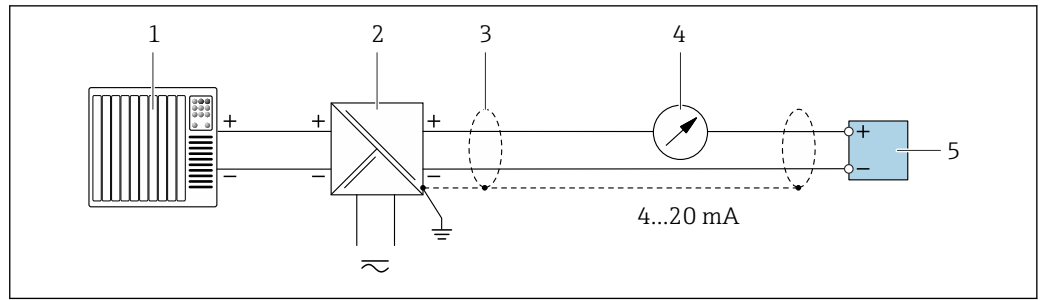
Current output 4 to 20 mA HART



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2 Connection example for 4 to 20 mA HART current output (active)

- 1 Automation system with current input (e.g. PLC)
- 2 Cable shield: the cable shield must be grounded at both ends to comply with EMC requirements; observe cable specifications → 40
- 3 Connection for HART operating devices → 84
- 4 Resistor for HART communication ($\geq 250 \Omega$); observe maximum load → 14
- 5 Analog display unit: observe maximum load → 14
- 6 Transmitter

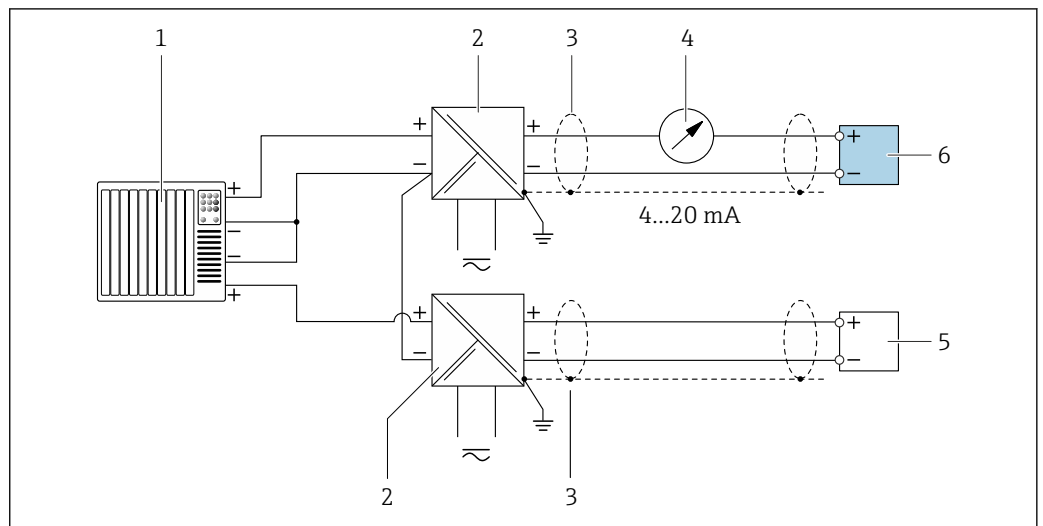


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3 Connection example for 4 to 20 mA HART current output (passive)

- 1 Automation system with current input (e.g. PLC)
- 2 Power supply
- 3 Cable shield: the cable shield must be grounded at both ends to comply with EMC requirements; observe cable specifications → 40
- 4 Analog display unit: observe maximum load → 14
- 5 Transmitter

HART input

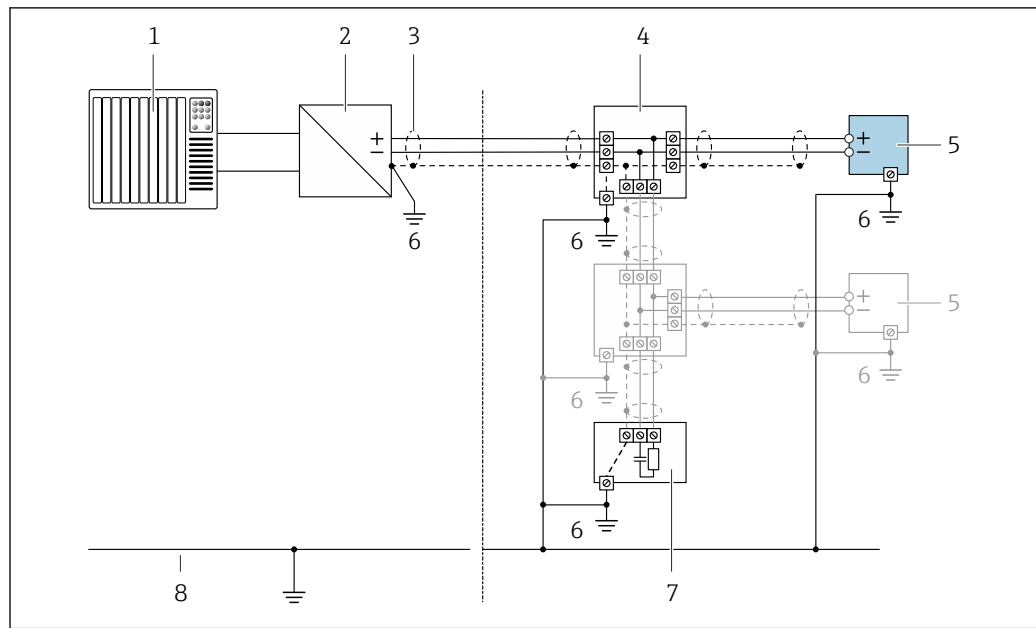


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4 Connection example for HART input with a common negative (passive)

- 1 Automation system with HART output (e.g. PLC)
- 2 Active barrier for power supply (e.g. RN221N)
- 3 Cable shield: the cable shield must be grounded at both ends to comply with EMC requirements; observe cable specifications
- 4 Analog display unit: observe maximum load
- 5 Pressure transmitter (e.g. Cerabar M, Cerabar S): see requirements
- 6 Transmitter

PROFIBUS PA

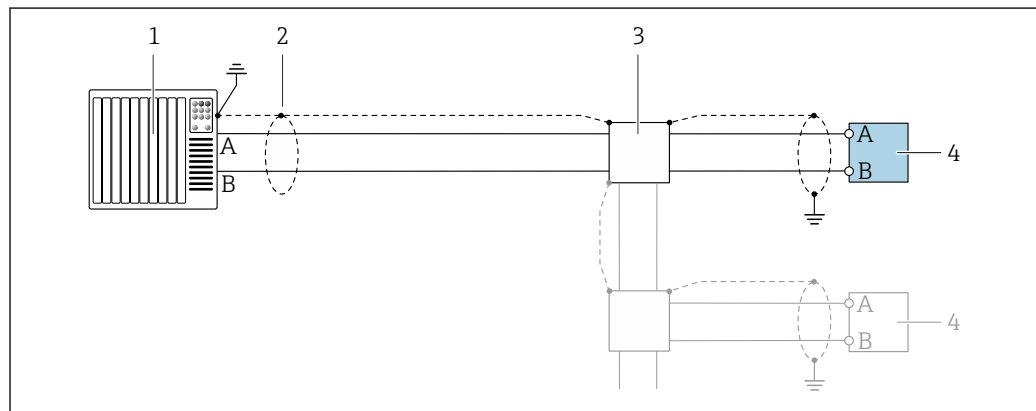


A0028768

5 Connection example for PROFIBUS PA

- 1 Control system (e.g. PLC)
- 2 PROFIBUS PA segment coupler
- 3 Cable shield: the cable shield must be grounded at both ends to comply with EMC requirements; observe cable specifications
- 4 T-box
- 5 Measuring device
- 6 Local grounding
- 7 Bus terminator
- 8 Potential matching line

PROFIBUS DP



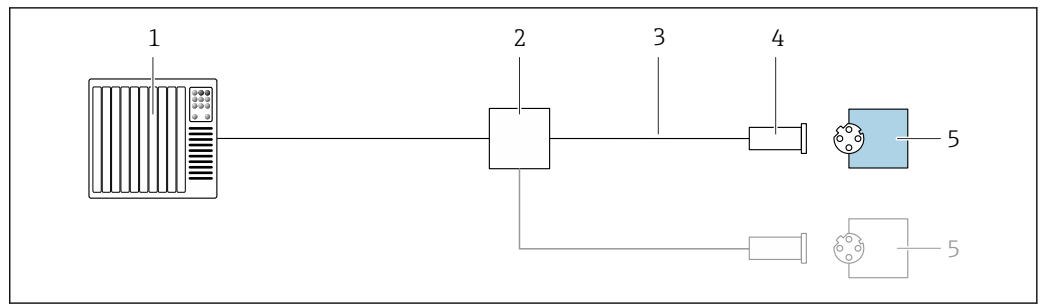
A0028765

6 Connection example for PROFIBUS DP, non-hazardous area and Zone 2/Div. 2

- 1 Control system (e.g. PLC)
- 2 Cable shield: the cable shield must be grounded at both ends to comply with EMC requirements; observe cable specifications
- 3 Distribution box
- 4 Transmitter

i If baud rates > 1.5 MBaud an EMC cable entry must be used and the cable shield must continue as far as the terminal wherever possible.

EtherNet/IP

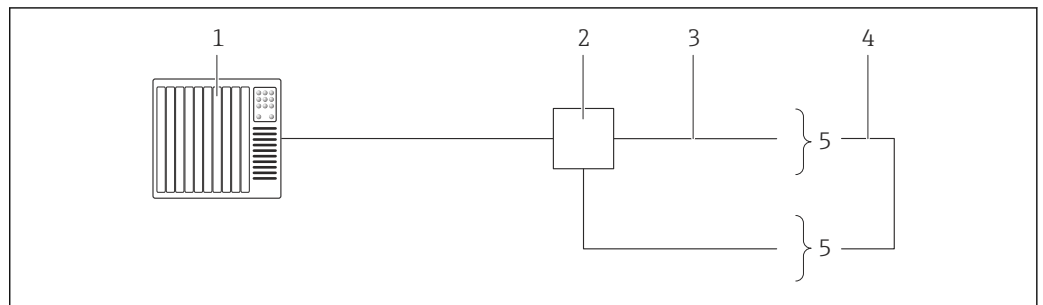


A0028767

7 Connection example for EtherNet/IP

- 1 Control system (e.g. PLC)
- 2 Ethernet switch
- 3 Observe cable specifications
- 4 Device plug
- 5 Transmitter

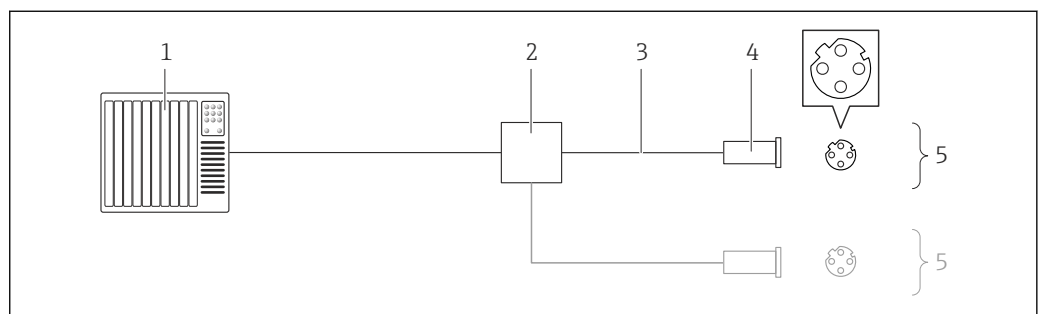
EtherNet/IP: DLR (Device Level Ring)



A0027544

- 1 Control system (e.g. PLC)
- 2 Ethernet switch
- 3 Observe cable specifications → 40
- 4 Connecting cable between the two transmitters
- 5 Transmitter

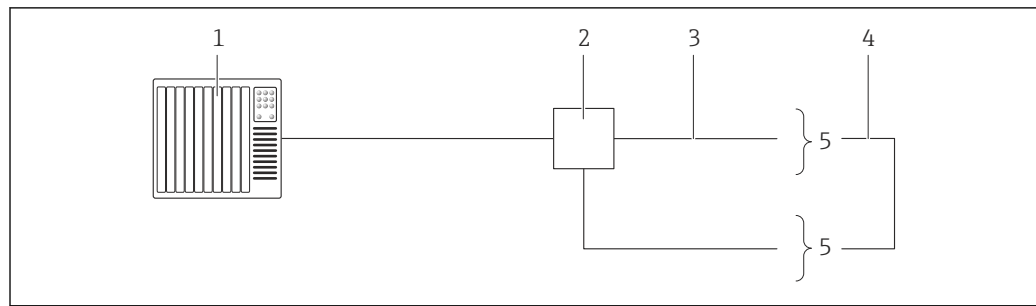
PROFINET



A0016805

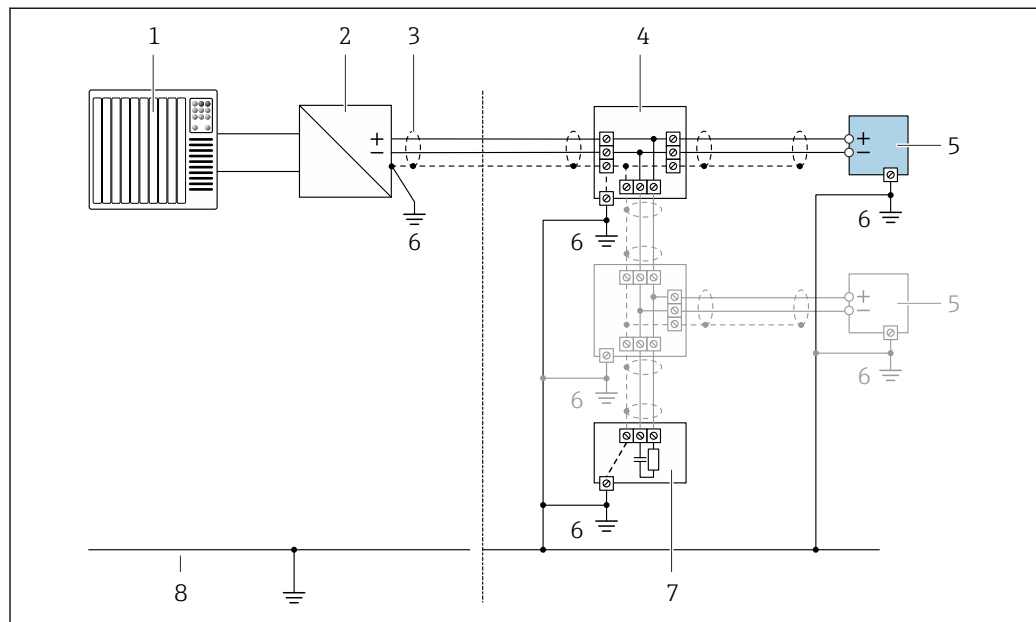
8 Connection example for PROFINET

- 1 Control system (e.g. PLC)
- 2 Ethernet switch
- 3 Observe cable specifications
- 4 Device plug
- 5 Transmitter

PROFINET: MRP (Media Redundancy Protocol)

A0027544

- 1 Control system (e.g. PLC)
- 2 Ethernet switch
- 3 Observe cable specifications → 40
- 4 Connecting cable between the two transmitters
- 5 Transmitter

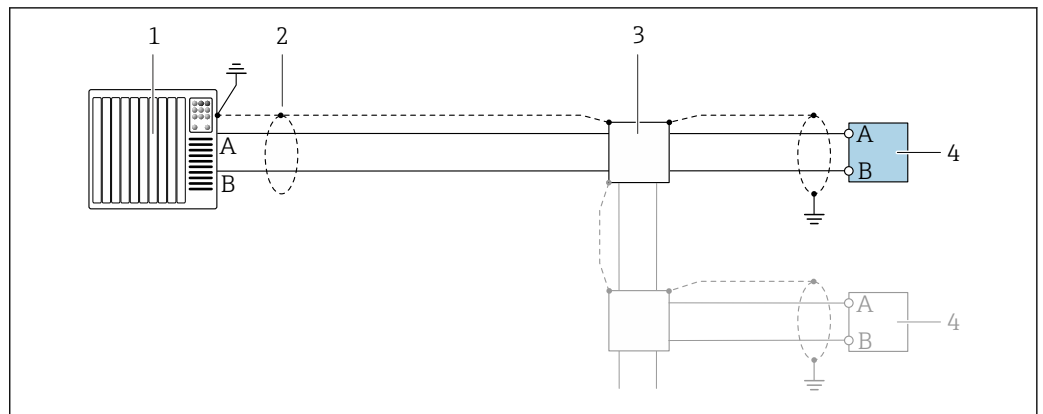
FOUNDATION Fieldbus

A0028768

9 Connection example for FOUNDATION Fieldbus

- 1 Control system (e.g. PLC)
- 2 Power Conditioner (FOUNDATION Fieldbus)
- 3 Cable shield: the cable shield must be grounded at both ends to comply with EMC requirements; observe cable specifications
- 4 T-box
- 5 Measuring device
- 6 Local grounding
- 7 Bus terminator
- 8 Potential matching line

Modbus RS485

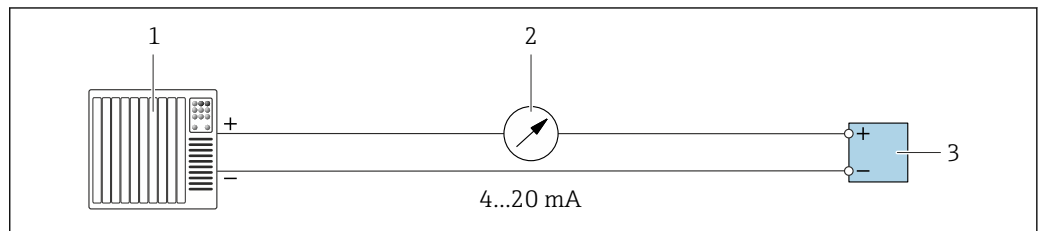


A0028765

10 Connection example for Modbus RS485, non-hazardous area and Zone 2; Class I, Division 2

- 1 Control system (e.g. PLC)
- 2 Cable shield: the cable shield must be grounded at both ends to comply with EMC requirements; observe cable specifications
- 3 Distribution box
- 4 Transmitter

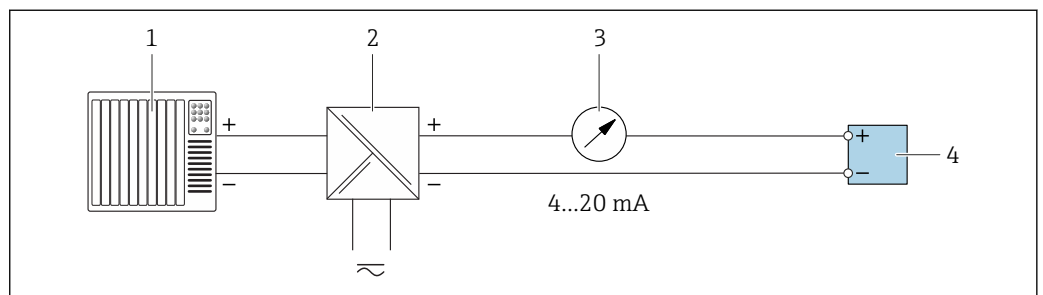
Current output 4-20 mA



A0028758

11 Connection example for 4-20 mA current output (active)

- 1 Automation system with current input (e.g. PLC)
- 2 Analog display unit: observe maximum load
- 3 Transmitter

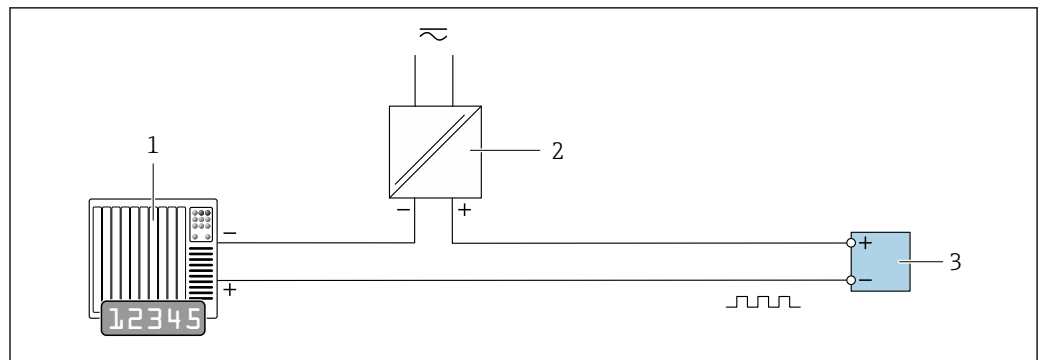


A0028759

12 Connection example for 4-20 mA current output (passive)

- 1 Automation system with current input (e.g. PLC)
- 2 Active barrier for power supply (e.g. RN221N)
- 3 Analog display unit: observe maximum load
- 4 Transmitter

Pulse/frequency output

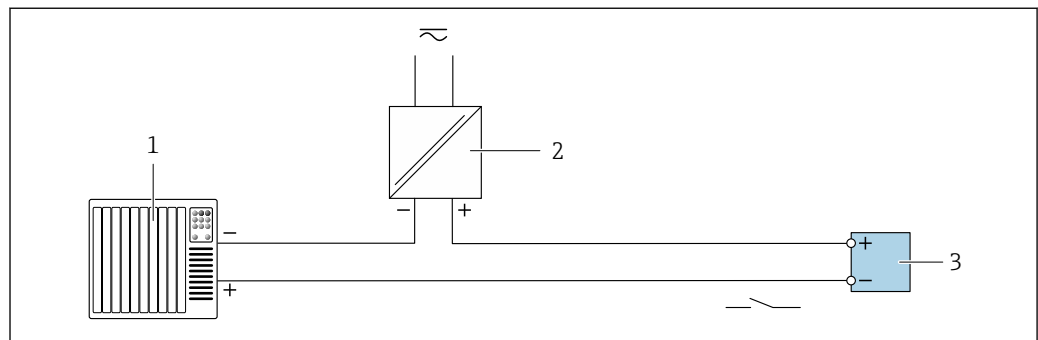


A0028761

13 Connection example for pulse/frequency output (passive)

- 1 Automation system with pulse/frequency input (e.g. PLC)
- 2 Power supply
- 3 Transmitter: Observe input values → 15

Switch output

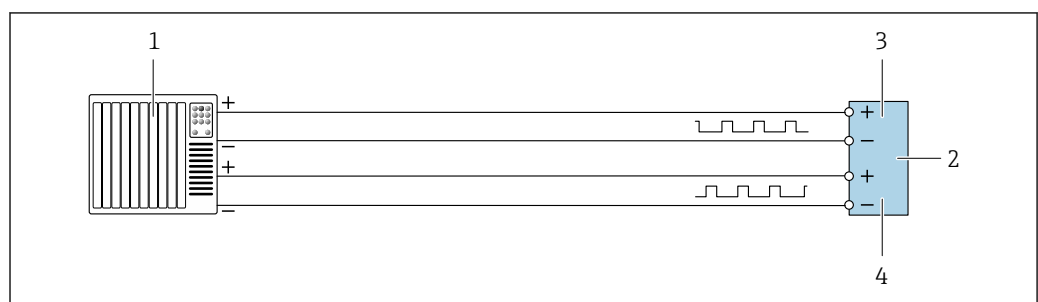


A0028760

14 Connection example for switch output (passive)

- 1 Automation system with switch input (e.g. PLC)
- 2 Power supply
- 3 Transmitter: Observe input values → 15

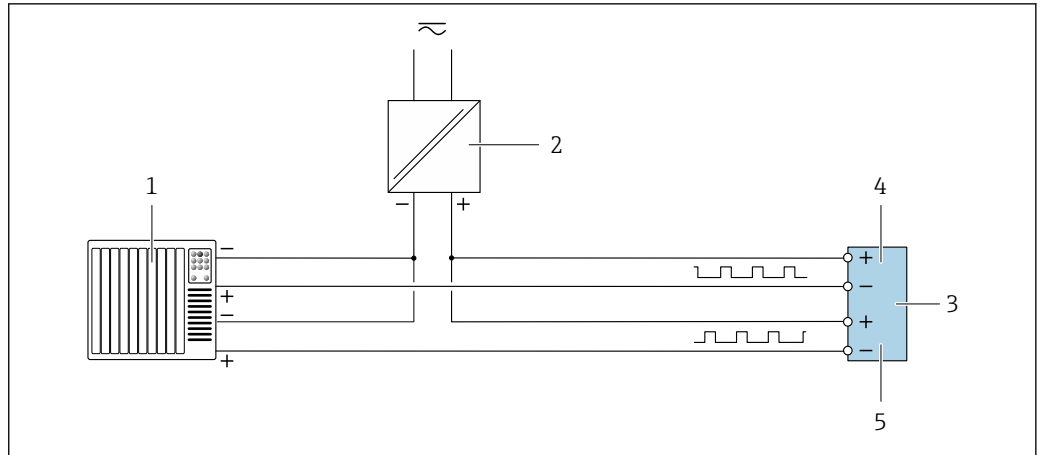
Double pulse output



A0029280

15 Connection example for double pulse output (active)

- 1 Automation system with double pulse input (e.g. PLC)
- 2 Transmitter: Observe input values → 17
- 3 Double pulse output
- 4 Double pulse output (slave), phase-shifted

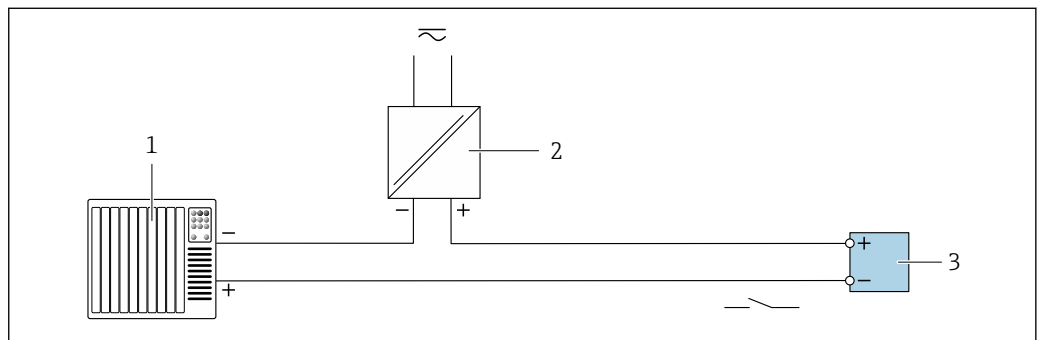


A0029279

16 Connection example for double pulse output (passive)

- 1 Automation system with double pulse input (e.g. PLC)
- 2 Power supply
- 3 Transmitter: Observe input values → 17
- 4 Double pulse output
- 5 Double pulse output (slave), phase-shifted

Relay output

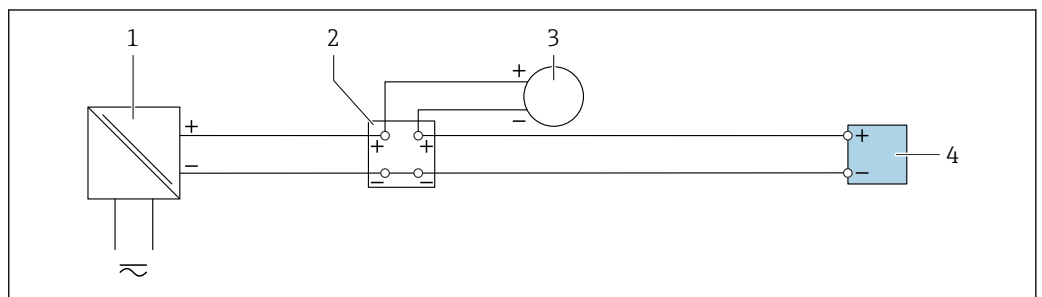


A0028760

17 Connection example for relay output (passive)

- 1 Automation system with relay input (e.g. PLC)
- 2 Power supply
- 3 Transmitter: Observe input values → 17

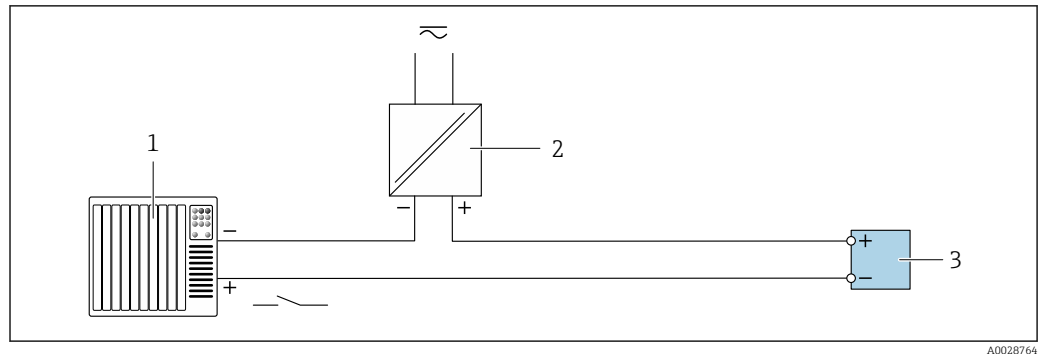
Current input



A0028915

18 Connection example for 4 to 20 mA current input

- 1 Power supply
- 2 Terminal box
- 3 External measuring device (for reading in pressure or temperature, for instance)
- 4 Transmitter

Status input

A0028764

 19 Connection example for status input

- 1 Automation system with status output (e.g. PLC)
- 2 Power supply
- 3 Transmitter

Potential equalization**Requirements**

No special measures for potential equalization are required.

Please consider the following to ensure correct measurement:

- Same electrical potential for the fluid and sensor
- Company-internal grounding concepts

terminals

Spring-loaded terminals: Suitable for strands and strands with ferrules.
Conductor cross-section 0.2 to 2.5 mm² (24 to 12 AWG).

Cable entries

- Cable gland: M20 × 1.5 with cable Ø 6 to 12 mm (0.24 to 0.47 in)
- Thread for cable entry:
 - NPT ½"
 - G ½"
 - M20
- Device plug for digital communication: M12
Only available for certain device versions →  27.

Cable specification**Permitted temperature range**

- The installation guidelines that apply in the country of installation must be observed.
- The cables must be suitable for the minimum and maximum temperatures to be expected.

Power supply cable

Standard installation cable is sufficient.

Signal cable

Current output 4 to 20 mA HART

A shielded cable is recommended. Observe grounding concept of the plant.

PROFIBUS PA

Twisted, shielded two-wire cable. Cable type A is recommended .



For further information on planning and installing PROFIBUS networks see:

- Operating Instructions "PROFIBUS DP/PA: Guidelines for planning and commissioning" (BA00034S)
- PNO Directive 2.092 "PROFIBUS PA User and Installation Guideline"
- IEC 61158-2 (MBP)

PROFIBUS DP

The IEC 61158 standard specifies two types of cable (A and B) for the bus line which can be used for every transmission rate. Cable type A is recommended.

Cable type	A
Characteristic impedance	135 to 165 Ω at a measuring frequency of 3 to 20 MHz
Cable capacitance	< 30 pF/m
Wire cross-section	> 0.34 mm ² (22 AWG)
Cable type	Twisted pairs
Loop resistance	$\leq 110 \Omega/\text{km}$
Signal damping	Max. 9 dB over the entire length of the cable cross-section
Shield	Copper braided shielding or braided shielding with foil shield. When grounding the cable shield, observe the grounding concept of the plant.



For further information on planning and installing PROFIBUS networks see:

- Operating Instructions "PROFIBUS DP/PA: Guidelines for planning and commissioning" (BA00034S)
- PNO Directive 2.092 "PROFIBUS PA User and Installation Guideline"
- IEC 61158-2 (MBP)

EtherNet/IP

The standard ANSI/TIA/EIA-568-B.2 Annex specifies CAT 5 as the minimum category for a cable used for EtherNet/IP. CAT 5e and CAT 6 are recommended.



For more information on planning and installing EtherNet/IP networks, please refer to the "Media Planning and Installation Manual. EtherNet/IP" of ODVA Organization

PROFINET

Standard IEC 61156-6 specifies CAT 5 as the minimum category for a cable used for PROFINET. CAT 5e and CAT 6 are recommended.



For more information on planning and installing PROFINET networks, see: "PROFINET Cabling and Interconnection Technology", Guideline for PROFINET

FOUNDATION Fieldbus

Twisted, shielded two-wire cable.



For further information on planning and installing FOUNDATION Fieldbus networks see:

- Operating Instructions for "FOUNDATION Fieldbus Overview" (BA00013S)
- FOUNDATION Fieldbus Guideline
- IEC 61158-2 (MBP)

Modbus RS485

The EIA/TIA-485 standard specifies two types of cable (A and B) for the bus line which can be used for every transmission rate. Cable type A is recommended.

Cable type	A
Characteristic impedance	135 to 165 Ω at a measuring frequency of 3 to 20 MHz
Cable capacitance	< 30 pF/m
Wire cross-section	> 0.34 mm ² (22 AWG)
Cable type	Twisted pairs
Loop resistance	$\leq 110 \Omega/\text{km}$
Signal damping	Max. 9 dB over the entire length of the cable cross-section
Shield	Copper braided shielding or braided shielding with foil shield. When grounding the cable shield, observe the grounding concept of the plant.

Current output 0/4 to 20 mA

Standard installation cable is sufficient.

Pulse/frequency/switch output

Standard installation cable is sufficient.

Double pulse output

Standard installation cable is sufficient.

Relay output

Standard installation cable is sufficient.

Current input 0/4 to 20 mA

Standard installation cable is sufficient.

Status input

Standard installation cable is sufficient.

Connecting cable for transmitter - remote display and operating module DKX001*Standard cable*

A standard cable can be used as the connecting cable.

Standard cable	4 cores (2 pairs); pair-stranded with common shield
Shielding	Tin-plated copper-braid, optical cover $\geq 85\%$
Capacitance: core/shield	Maximum 1 000 nF for Zone 1; Class I, Division 1
L/R	Maximum 24 $\mu\text{H}/\Omega$ for Zone 1; Class I, Division 1
Cable length	Maximum 300 m (1 000 ft), see the following table

Cross-section	Cable length for use in: ■ Non-hazardous area ■ Hazardous area: Zone 2; Class I, Division 2 ■ Hazardous area: Zone 1; Class I, Division 1
0.34 mm ² (22 AWG)	80 m (270 ft)
0.50 mm ² (20 AWG)	120 m (400 ft)
0.75 mm ² (18 AWG)	180 m (600 ft)
1.00 mm ² (17 AWG)	240 m (800 ft)
1.50 mm ² (15 AWG)	300 m (1 000 ft)

Optionally available connecting cable

Standard cable	2 × 2 × 0.34 mm ² (22 AWG) PVC cable ¹⁾ with common shield (2 pairs, pair-stranded)
Flame resistance	According to DIN EN 60332-1-2
Oil-resistance	According to DIN EN 60811-2-1
Shielding	Tin-plated copper-braid, optical cover $\geq 85\%$
Capacitance: core/shield	≤ 200 pF/m
L/R	≤ 24 $\mu\text{H}/\Omega$
Available cable length	10 m (35 ft)
Operating temperature	When mounted in a fixed position: -50 to +105 °C (-58 to +221 °F); when cable can move freely: -25 to +105 °C (-13 to +221 °F)

1) UV radiation can impair the cable outer sheath. Protect the cable from direct sunshine where possible.

Performance characteristics

Reference operating conditions

- Error limits based on ISO 11631
- Water with +15 to +45 °C (+59 to +113 °F) at 2 to 6 bar (29 to 87 psi)
- Specifications as per calibration protocol
- Accuracy based on accredited calibration rigs that are traced to ISO 17025.



To obtain measured errors, use the *Applicator* sizing tool → 101

Maximum measured error

o.r. = of reading; 1 g/cm³ = 1 kg/l; T = medium temperature

Base accuracy



Design fundamentals → 46

Mass flow and volume flow (liquids)

±0.15 % o.r.

±0.10 % o.r. (order code for "Calibration flow", option A, B, C, for mass flow)

Mass flow (gases)

±0.75 % o.r.

Density (liquids)

Under reference operating conditions [g/cm ³]	Standard density calibration ¹⁾ [g/cm ³]
±0.0005	±0.02

1) Valid over the entire temperature and density range

Temperature

±0.5 °C ± 0.005 · T °C (±0.9 °F ± 0.003 · (T – 32) °F)

Zero point stability

DN		Zero point stability	
[mm]	[in]	[kg/h]	[lb/min]
8	$\frac{3}{8}$	0.20	0.007
15	$\frac{1}{2}$	0.65	0.024
25	1	1.80	0.066
40	1½	4.50	0.165
50	2	7.0	0.257
80	3	18.0	0.6615

Flow values

Flow values as turndown parameter depending on nominal diameter.

SI units

DN	1:1	1:10	1:20	1:50	1:100	1:500
[mm]	[kg/h]	[kg/h]	[kg/h]	[kg/h]	[kg/h]	[kg/h]
8	2 000	200	100	40	20	4
15	6 500	650	325	130	65	13
25	18 000	1 800	900	360	180	36
40	45 000	4 500	2 250	900	450	90
50	70 000	7 000	3 500	1 400	700	140
80	180 000	18 000	9 000	3 600	1 800	360

US units

DN	1:1	1:10	1:20	1:50	1:100	1:500
[inch]	[lb/min]	[lb/min]	[lb/min]	[lb/min]	[lb/min]	[lb/min]
$\frac{3}{8}$	73.50	7.350	3.675	1.470	0.735	0.147
$\frac{1}{2}$	238.9	23.89	11.95	4.778	2.389	0.478
1	661.5	66.15	33.08	13.23	6.615	1.323
$1\frac{1}{2}$	1 654	165.4	82.70	33.08	16.54	3.308
2	2 573	257.3	128.7	51.46	25.73	5.146
3	6 615	661.5	330.8	132.3	66.15	13.23

Accuracy of outputs

The outputs have the following base accuracy specifications.

Current output

Accuracy	$\pm 5 \mu\text{A}$
-----------------	---------------------

Pulse/frequency output

o.r. = of reading

Accuracy	Max. $\pm 50 \text{ ppm}$ o.r. (over the entire ambient temperature range)
-----------------	--

Repeatability

o.r. = of reading; $1 \text{ g/cm}^3 = 1 \text{ kg/l}$; T = medium temperature

Base repeatability

 Design fundamentals →  46

Mass flow and volume flow (liquids)

$\pm 0.075 \%$ o.r.

$\pm 0.05 \%$ o.r. (calibration option, for mass flow)

Mass flow (gases)

$\pm 0.35 \%$ o.r.

Density (liquids)

$\pm 0.00025 \text{ g/cm}^3$

Temperature
 $\pm 0.25\text{ }^{\circ}\text{C} \pm 0.0025 \cdot T\text{ }^{\circ}\text{C}$ ($\pm 0.45\text{ }^{\circ}\text{F} \pm 0.0015 \cdot (T-32)\text{ }^{\circ}\text{F}$)

Response time The response time depends on the configuration (damping).

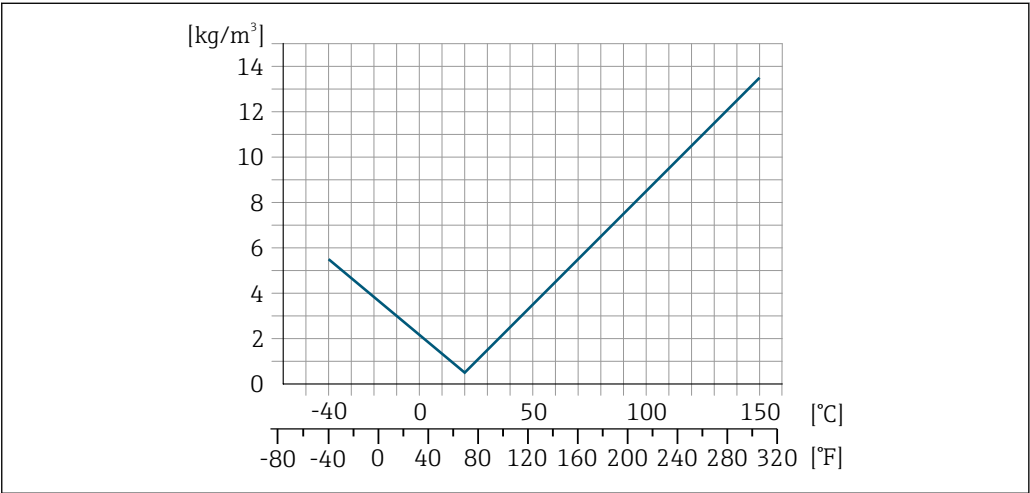
Influence of ambient temperature	Current output	
	<table><tr><td>Temperature coefficient</td><td>Max. 1 $\mu\text{A}/^{\circ}\text{C}$</td></tr></table>	Temperature coefficient
Temperature coefficient	Max. 1 $\mu\text{A}/^{\circ}\text{C}$	

Pulse/frequency output	
Temperature coefficient	No additional effect. Included in accuracy.

Influence of medium temperature

Mass flow and volume flow
o.f.s. = of full scale value
When there is a difference between the temperature for zero point adjustment and the process temperature, the additional measured error of the sensor is typically $\pm 0.0002\text{ \% o.f.s.}/^{\circ}\text{C}$ ($\pm 0.0001\text{ \% o. f.s.}/^{\circ}\text{F}$).
The effect is reduced if zero point adjustment is performed at process temperature.

Density
When there is a difference between the density calibration temperature and the process temperature, the typical measured error of the sensor is $\pm 0.0001\text{ g}/\text{cm}^3\text{ }/^{\circ}\text{C}$ ($\pm 0.00005\text{ g}/\text{cm}^3\text{ }/^{\circ}\text{F}$).
Field density calibration is possible.



20 Field density calibration, for example at +20 $^{\circ}\text{C}$ (+68 $^{\circ}\text{F}$)

Temperature
 $\pm 0.005 \cdot T\text{ }^{\circ}\text{C}$ ($\pm 0.005 \cdot (T - 32)\text{ }^{\circ}\text{F}$)

Influence of medium pressure

The table below shows the effect on accuracy of mass flow due to a difference between calibration pressure and process pressure.

o.r. = of reading

i It is possible to compensate for the effect by:

- Reading in the current pressure measured value via the current input.
- Specifying a fixed value for the pressure in the device parameters.

b Operating Instructions → 102.

DN		[% o.r./bar]	[% o.r./psi]
[mm]	[in]		
8	⅜	no influence	
15	½	no influence	
25	1	no influence	
40	1½	no influence	
50	2	-0.009	-0.0006
80	3	-0.020	-0.0014

Design fundamentals

o.r. = of reading, o.f.s. = of full scale value

BaseAccu = base accuracy in % o.r., BaseRepeat = base repeatability in % o.r.

MeasValue = measured value; ZeroPoint = zero point stability

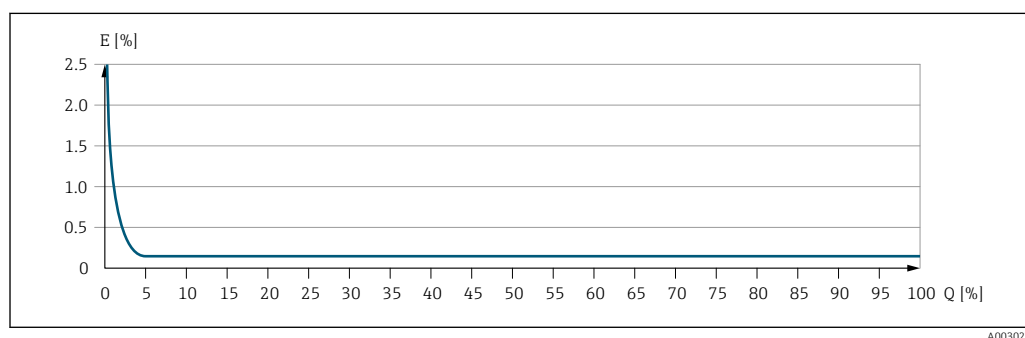
Calculation of the maximum measured error as a function of the flow rate

Flow rate	Maximum measured error in % o.r.
$\geq \frac{\text{ZeroPoint}}{\text{BaseAccu}} \cdot 100$ A0021332	$\pm \text{BaseAccu}$ A0021339
$< \frac{\text{ZeroPoint}}{\text{BaseAccu}} \cdot 100$ A0021333	$\pm \frac{\text{ZeroPoint}}{\text{MeasValue}} \cdot 100$ A0021334

Calculation of the maximum repeatability as a function of the flow rate

Flow rate	Maximum repeatability in % o.r.
$\geq \frac{\frac{1}{2} \cdot \text{ZeroPoint}}{\text{BaseRepeat}} \cdot 100$ A0021335	$\pm \text{BaseRepeat}$ A0021340
$< \frac{\frac{1}{2} \cdot \text{ZeroPoint}}{\text{BaseRepeat}} \cdot 100$ A0021336	$\pm \frac{1}{2} \cdot \frac{\text{ZeroPoint}}{\text{MeasValue}} \cdot 100$ A0021337

Example for maximum measured error

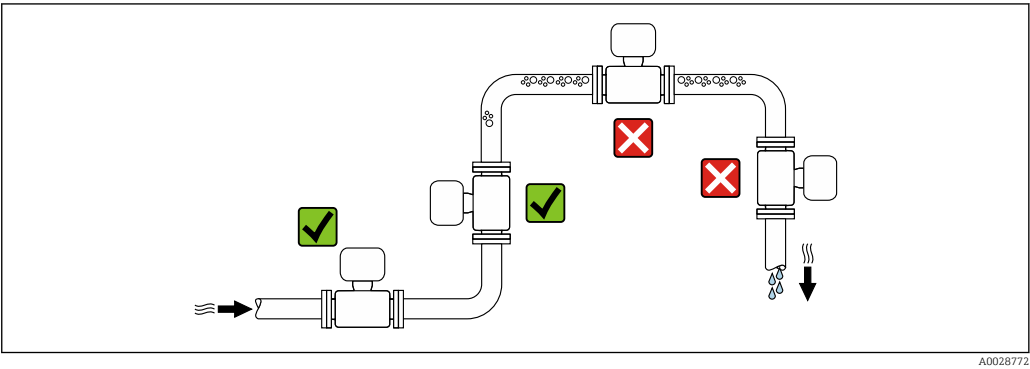


E Maximum measured error in % o.r. (example)
Q Flow rate in % of maximum full scale value

Installation

No special measures such as supports etc. are necessary. External forces are absorbed by the construction of the device.

Mounting location

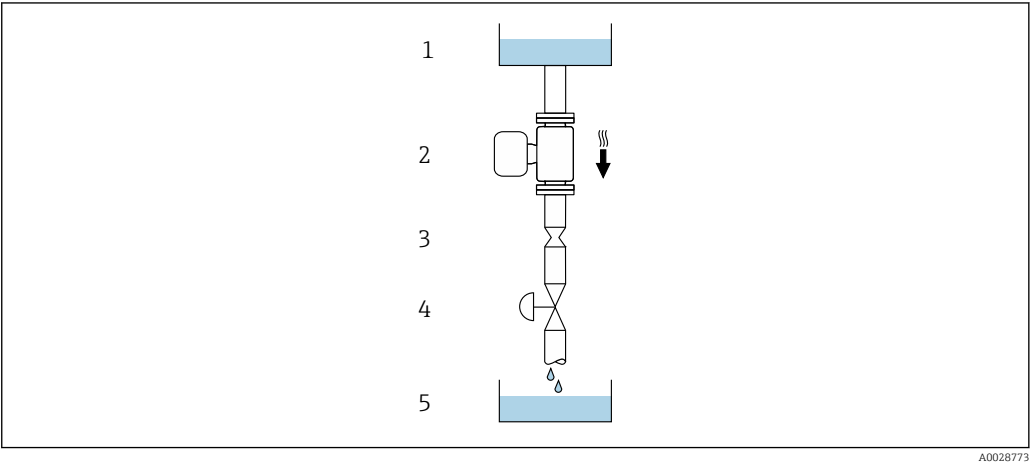


To prevent measuring errors arising from accumulation of gas bubbles in the measuring tube, avoid the following mounting locations in the pipe:

- Highest point of a pipeline.
- Directly upstream of a free pipe outlet in a down pipe.

Installation in down pipes

However, the following installation suggestion allows for installation in an open vertical pipeline. Pipe restrictions or the use of an orifice with a smaller cross-section than the nominal diameter prevent the sensor running empty while measurement is in progress.



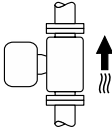
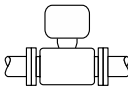
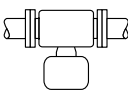
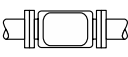
21 Installation in a down pipe (e.g. for batching applications)

- 1 Supply tank
- 2 Sensor
- 3 Orifice plate, pipe restriction
- 4 Valve
- 5 Batching tank

DN		Ø orifice plate, pipe restriction	
[mm]	[in]	[mm]	[in]
8	3/8	6	0.24
15	1/2	10	0.40
25	1	14	0.55
40	1 1/2	22	0.87
50	2	28	1.10
80	3	50	1.97

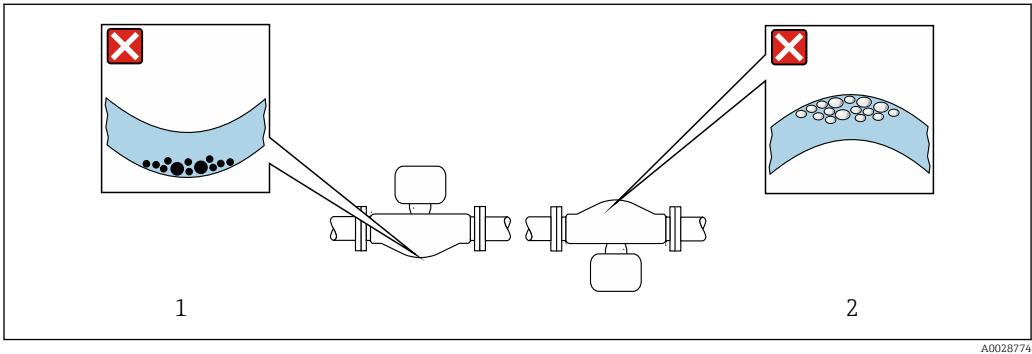
Orientation

The direction of the arrow on the sensor nameplate helps you to install the sensor according to the flow direction (direction of medium flow through the piping).

Orientation			Recommendation
A	Vertical orientation	 A0015591	✓✓✓ ¹⁾
B	Horizontal orientation, transmitter at top	 A0015589	✓✓✓ ²⁾ Exceptions: → ☒ 22, ☒ 48
C	Horizontal orientation, transmitter at bottom	 A0015590	✓✓✓ ³⁾ Exceptions: → ☒ 22, ☒ 48
D	Horizontal orientation, transmitter at side	 A0015592	✗

- 1) This orientation is recommended to ensure self-draining.
- 2) Applications with low process temperatures may decrease the ambient temperature. To maintain the minimum ambient temperature for the transmitter, this orientation is recommended.
- 3) Applications with high process temperatures may increase the ambient temperature. To maintain the maximum ambient temperature for the transmitter, this orientation is recommended.

If a sensor is installed horizontally with a curved measuring tube, match the position of the sensor to the fluid properties.



☒ 22 Orientation of sensor with curved measuring tube

- 1 Avoid this orientation for fluids with entrained solids: Risk of solids accumulating.
- 2 Avoid this orientation for outgassing fluids: Risk of gas accumulating.

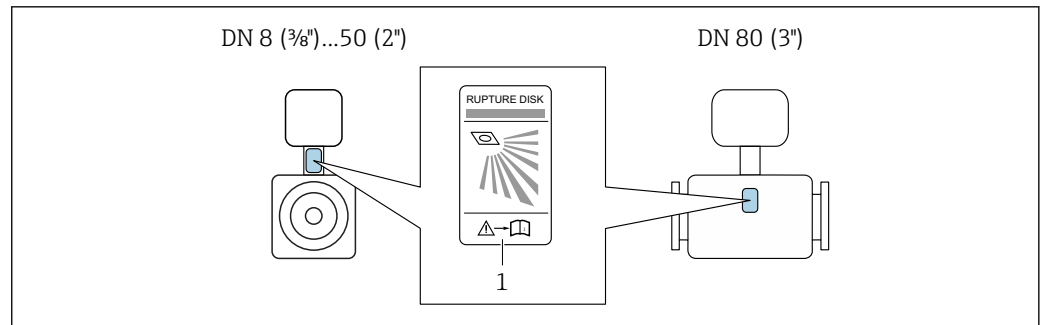
Inlet and outlet runs	No special precautions need to be taken for fittings which create turbulence, such as valves, elbows or T-pieces, as long as no cavitation occurs → ☒ 55.
Special mounting instructions	Drainability The measuring tubes can be completely drained and protected against solids build-up in vertical orientation. Rupture disk Information that is relevant to the process: → ☒ 55.

⚠ WARNING**Danger from medium escaping!**

Medium escaping under pressure can cause injury or material damage.

- ▶ Take precautions to prevent danger to persons and damage if the rupture disk is actuated.
- ▶ Observe information on the rupture disk sticker.
- ▶ Make sure that the function and operation of the rupture disk is not impeded through the installation of the device.
- ▶ Do not use a heating jacket.
- ▶ Do not remove or damage the rupture disk.

The position of the rupture disk is indicated on a sticker applied over it. If the rupture disk is triggered, the sticker is destroyed. The disk can therefore be visually monitored.



A0029956

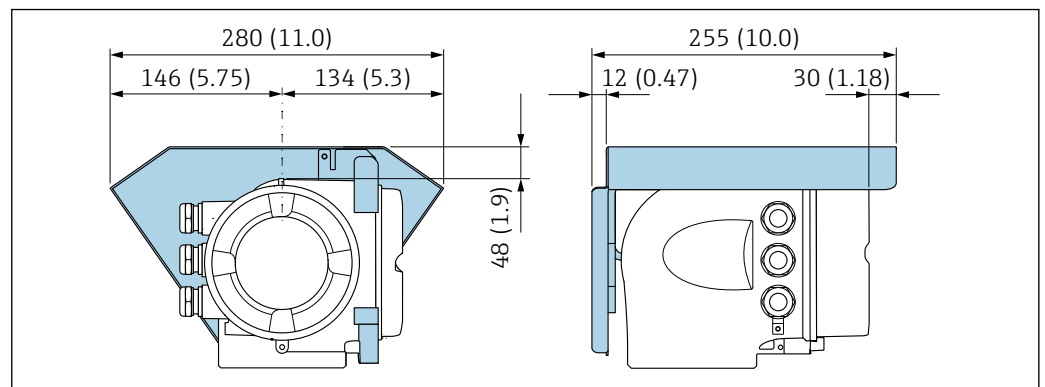
1 Rupture disk label

Zero point adjustment

All measuring devices are calibrated in accordance with state-of-the-art technology. Calibration takes place under reference conditions → 43. Therefore, a zero point adjustment in the field is generally not required.

Experience shows that zero point adjustment is advisable only in special cases:

- To achieve maximum measuring accuracy even with low flow rates.
- Under extreme process or operating conditions (e.g. very high process temperatures or very high-viscosity fluids).

Protective cover

A0029553

Environment

Ambient temperature range	Measuring device	■ -40 to +60 °C (-40 to +140 °F) ■ Order code for "Test, certificate", option JP: -50 to +60 °C (-58 to +140 °F)
	Readability of the local display	-20 to +60 °C (-4 to +140 °F) The readability of the display may be impaired at temperatures outside the temperature range.



Dependency of ambient temperature on medium temperature → 51

- If operating outdoors:
Avoid direct sunlight, particularly in warm climatic regions.



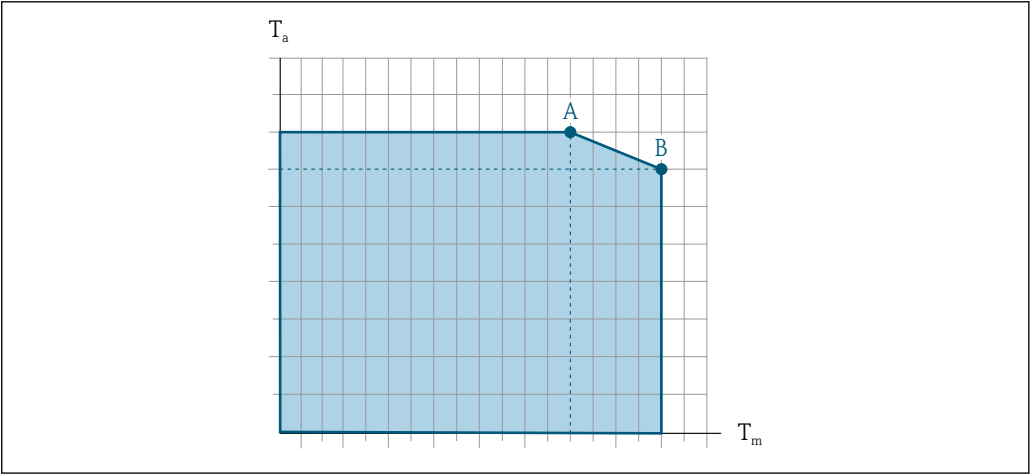
You can order a weather protection cover from Endress+Hauser. → 99.

Storage temperature	-50 to +80 °C (-58 to +176 °F)
Climate class	DIN EN 60068-2-38 (test Z/AD)
Degree of protection	Measuring device ■ As standard: IP66/67, type 4X enclosure ■ When housing is open: IP20, type 1 enclosure ■ Display module: IP20, type 1 enclosure ■ With the order code for "Sensor options", option CM: IP69 can also be ordered External WLAN antenna IP67
Vibration resistance	■ Oscillation, sinusoidal, following IEC 60068-2-6 - 2 to 8.4 Hz, 3.5 mm peak - 8.4 to 2 000 Hz, 1 g peak ■ Oscillation, broadband noise following IEC 60068-2-64 - 10 to 200 Hz, 0.003 g²/Hz - 200 to 2 000 Hz, 0.001 g²/Hz - Total: 1.54 g rms
Shock resistance	Shock, half-sine according to IEC 60068-2-27 6 ms 50 g
Shock resistance	Shock due to rough handling following IEC 60068-2-31
Interior cleaning	■ Cleaning in place (CIP) ■ Sterilization in place (SIP) Options Oil- and grease-free version for wetted parts, without declaration Order code for "Service", option HA
Electromagnetic compatibility (EMC)	■ As per IEC/EN 61326 and NAMUR Recommendation 21 (NE 21) ■ Device version with PROFIBUS DP: Complies with emission limits for industry as per EN 50170 Volume 2, IEC 61784 The following applies for PROFIBUS DP: If baud rates > 1.5 MBaud, an EMC cable entry must be used and the cable shield must continue as far as the terminal wherever possible. Details are provided in the Declaration of Conformity.

Process

Medium temperature range -40 to +150 °C (-40 to +302 °F)

Dependency of ambient temperature on medium temperature



A0031121

23 Exemplary representation, values in the table below.

- T_a Ambient temperature
 T_m Medium temperature
A Maximum permitted medium temperature T_m at $T_{a\max} = 60\text{ °C}$ (140 °F); higher medium temperatures T_m require a reduced ambient temperature T_a
B Maximum permitted ambient temperature T_a for the maximum specified medium temperature T_m of the sensor

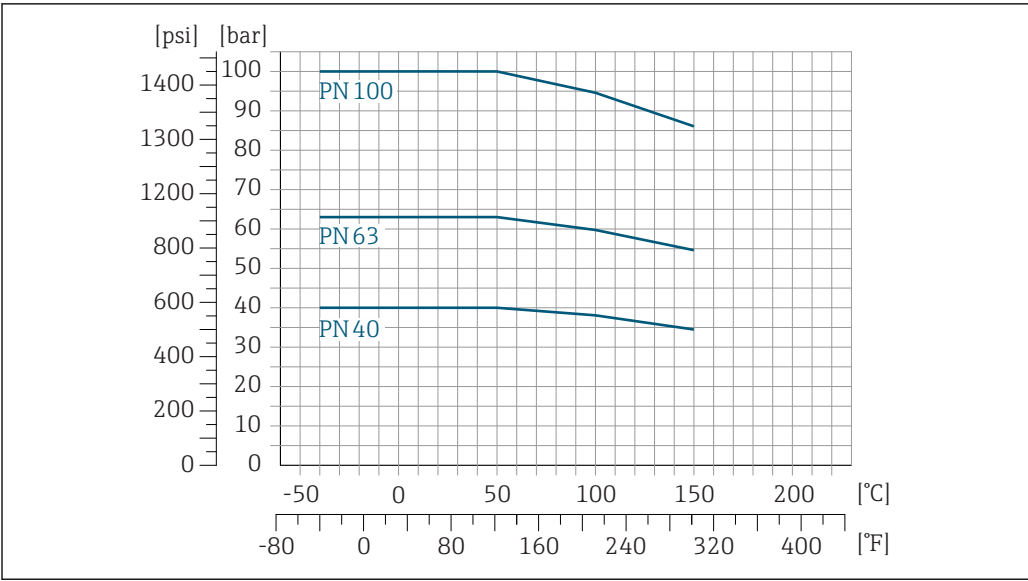
 Values for devices used in the hazardous area:
Separate Ex documentation (XA) for the device →  102.

Not insulated				Insulated			
A		B		A		B	
T_a	T_m	T_a	T_m	T_a	T_m	T_a	T_m
60 °C (140 °F)	150 °C (302 °F)	–	–	60 °C (140 °F)	110 °C (230 °F)	55 °C (131 °F)	150 °C (302 °F)

Density 0 to 5 000 kg/m³ (0 to 312 lb/cf)

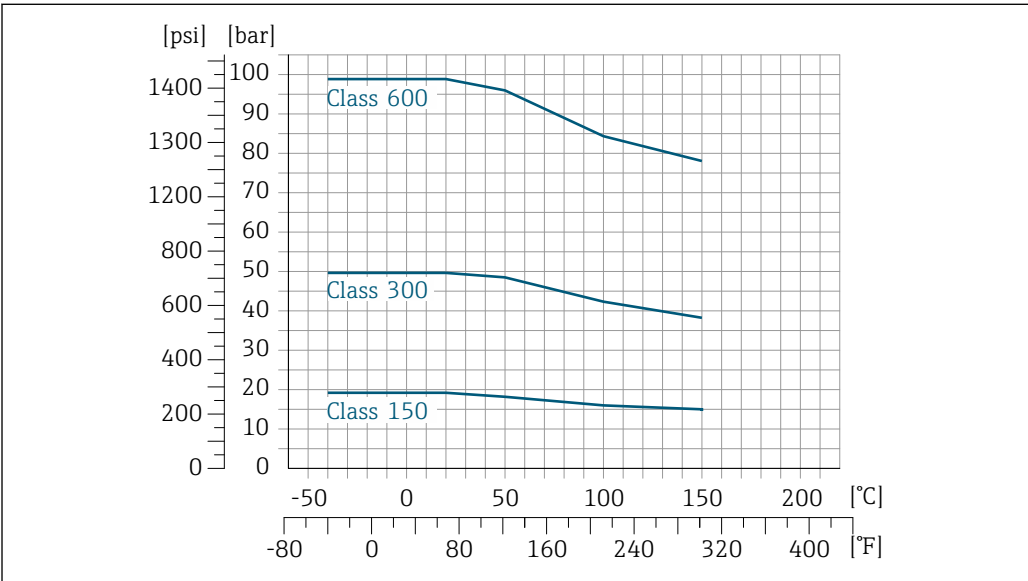
Pressure-temperature curves The following pressure/temperature diagrams apply to all pressure-bearing parts of the device and not just the process connection. The diagrams show the maximum permissible medium pressure depending on the specific medium temperature.

Flange according to EN 1092-1 (DIN 2501)



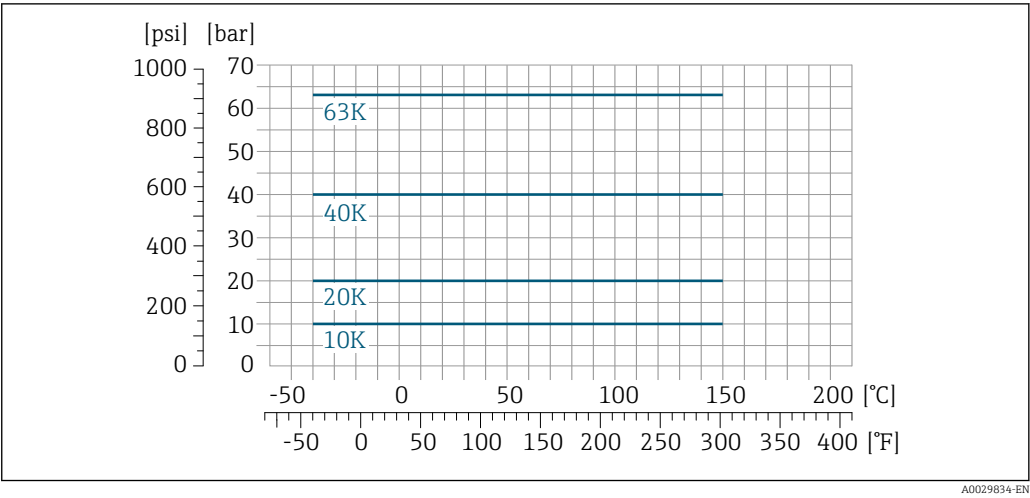
24 With flange material 1.4404 (F316/F316L)

Flange according to ASME B16.5



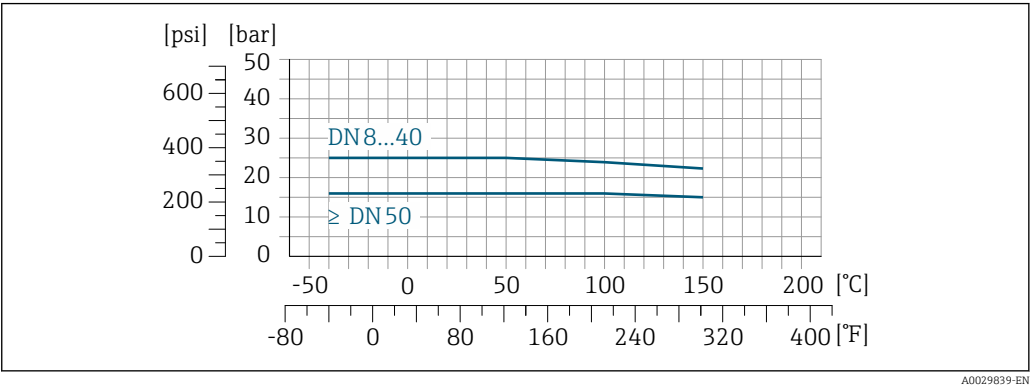
25 With flange material 1.4404 (F316/F316L)

Flange JIS B2220



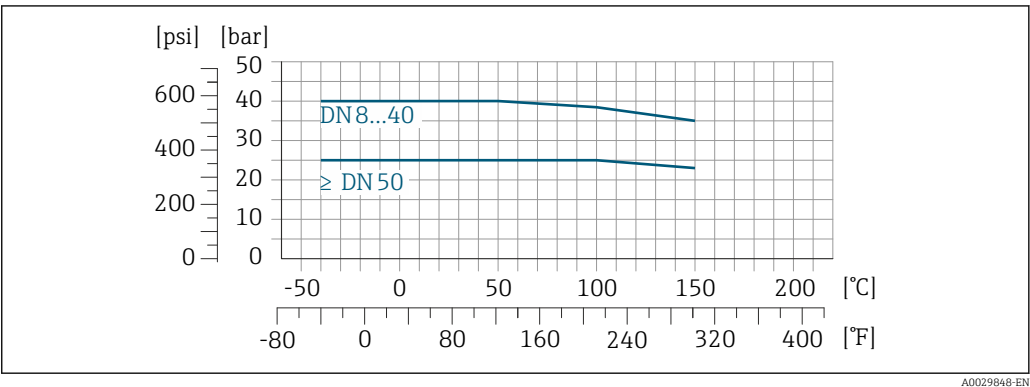
26 With flange material 1.4404 (F316/F316L)

Flange DIN 11864-2 Form A



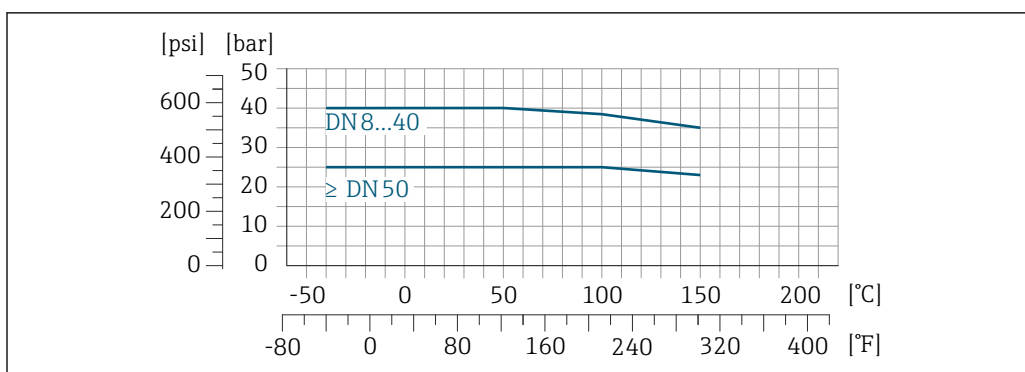
27 With flange material 1.4404 (316/316L)

Thread DIN 11851



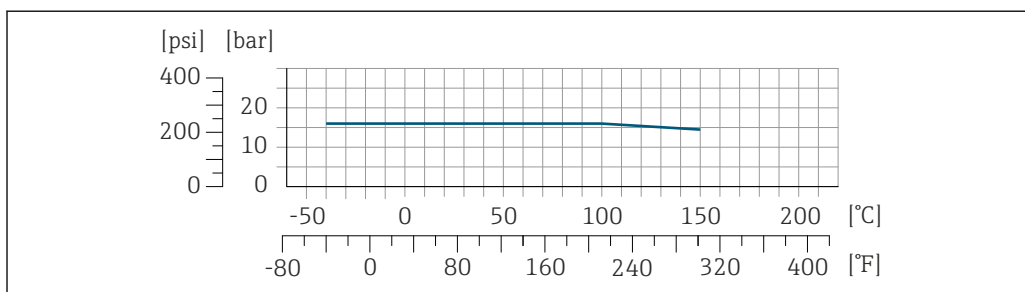
28 With connection material 1.4404 (316/316L)

DIN 11851 allows for applications up to +140 °C (+284 °F) if suitable sealing materials are used. Please take this into account when selecting seals and counterparts, as these components can limit the pressure and temperature range.

Thread DIN 11864-1 Form A

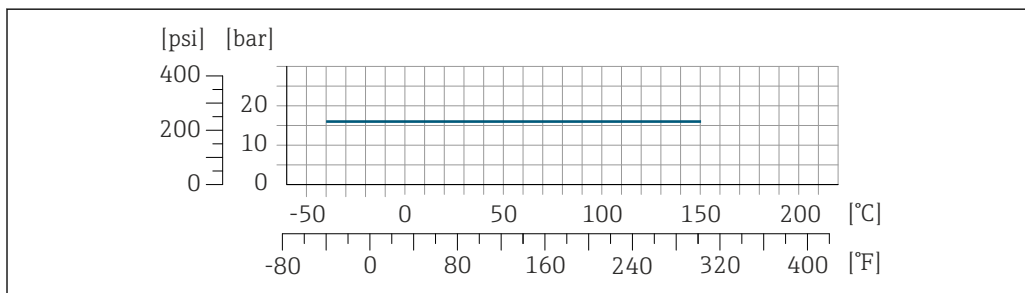
A0029848-EN

29 With connection material 1.4404 (316/316L)

Thread ISO 2853

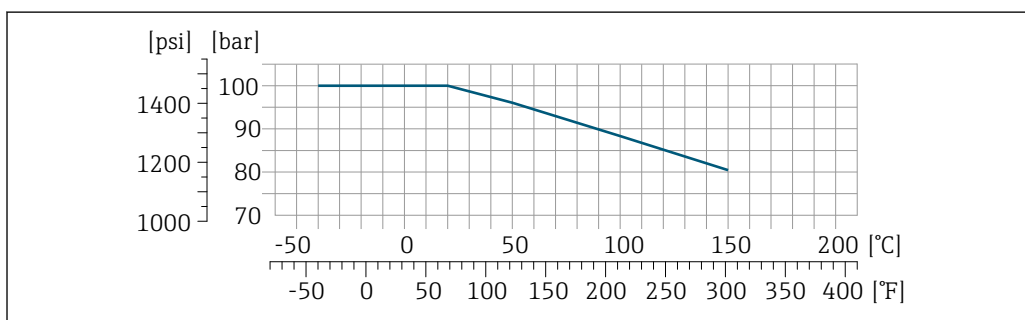
A0029853-EN

30 With connection material 1.4404 (316/316L)

Thread SMS 1145

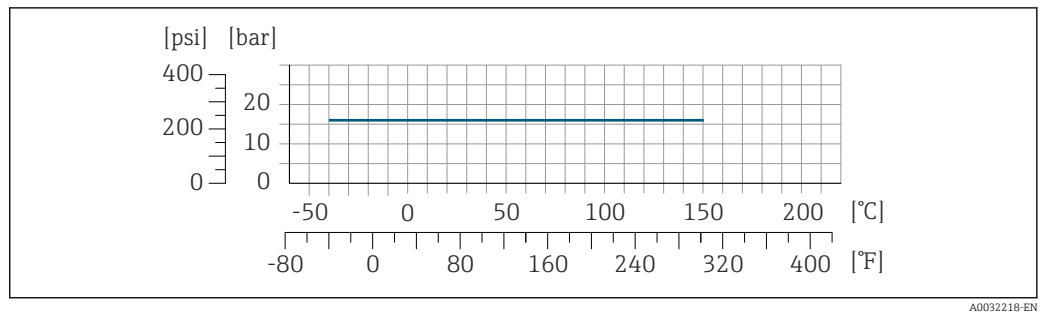
A0032218-EN

31 With connection material 1.4404 (316/316L)

VCO

A0029863-EN

32 With connection material 1.4404 (316/316L)

Tri-Clamp

A0032218-EN

The clamp connections are suitable up to a maximum pressure of 16 bar (232 psi). Please observe the operating limits of the clamp and seal used as they can be over 16 bar (232 psi). The clamp and seal are not included in the scope of supply.

Sensor housing

The sensor housing is filled with dry nitrogen gas and protects the electronics and mechanics inside.



If a measuring tube fails (e.g. due to process characteristics like corrosive or abrasive fluids), the fluid will initially be contained by the sensor housing.

Sensor housing nominal pressure rating and burst pressure

The secondary containment does not have a pressure rating classification.

Reference value for the pressure loading capacity of the sensor housing: 16 bar (232 psi)

If the device is fitted with a rupture disk (order code for "Sensor option", option CA "Rupture disk"), the rupture disk trigger pressure is decisive for the maximum nominal pressure.

Rupture disk

To increase the level of safety, a device version with a rupture disk with a trigger pressure of 10 to 15 bar (145 to 217.5 psi) can be used (order code for "Sensor option", option CA "rupture disk").

The use of rupture disks cannot be combined with the separately available heating jacket.

Flow limit

Select the nominal diameter by optimizing between the required flow range and permissible pressure loss.



For an overview of the full scale values for the measuring range, see the "Measuring range" section → 10

- The minimum recommended full scale value is approx. 1/20 of the maximum full scale value
- In most applications, 20 to 50 % of the maximum full scale value can be considered ideal
- A low full scale value must be selected for abrasive media (such as liquids with entrained solids): flow velocity < 1 m/s (< 3 ft/s).
- For gas measurement the following rules apply:
 - The flow velocity in the measuring tubes should not exceed half the sound velocity (0.5 Mach).
 - The maximum mass flow depends on the density of the gas: formula → 10



To calculate the flow limit, use the *Applicator* sizing tool → 101

Pressure loss

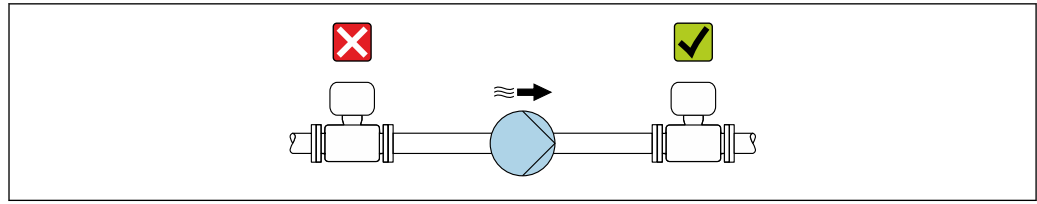
To calculate the pressure loss, use the *Applicator* sizing tool → 101

System pressure

It is important that cavitation does not occur, or that gases entrained in the liquids do not outgas. This is prevented by means of a sufficiently high system pressure.

For this reason, the following mounting locations are recommended:

- At the lowest point in a vertical pipe
- Downstream from pumps (no danger of vacuum)



A0028777

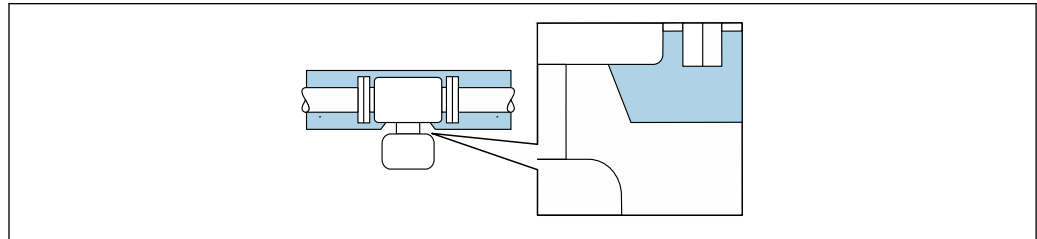
Thermal insulation

In the case of some fluids, it is important to keep the heat radiated from the sensor to the transmitter to a low level. A wide range of materials can be used for the required insulation.

NOTICE

Electronics overheating on account of thermal insulation!

- ▶ Recommended orientation: horizontal orientation, transmitter housing pointing downwards.
- ▶ Do not insulate the transmitter housing.
- ▶ Maximum permissible temperature at the lower end of the transmitter housing: 80 °C (176 °F)
- ▶ Thermal insulation with extended neck free: We recommend that you do not insulate the extended neck in order to ensure optimum dissipation of heat.



A0034391

33 Thermal insulation with extended neck free

Heating

Some fluids require suitable measures to avoid loss of heat at the sensor.

Heating options

- Electrical heating, e.g. with electric band heaters
- Via pipes carrying hot water or steam
- Via heating jackets



Heating jackets for the sensors can be ordered as accessories from Endress+Hauser. → 100

NOTICE

Danger of overheating when heating

- ▶ Ensure that the temperature at the lower end of the transmitter housing does not exceed 80 °C (176 °F).
- ▶ Ensure that sufficient convection takes place at the transmitter neck.
- ▶ Ensure that a sufficiently large area of the transmitter neck remains exposed. The uncovered part serves as a radiator and protects the electronics from overheating and excessive cooling.
- ▶ If using in potentially explosive atmospheres, observe the information in the device-specific Ex documentation. For detailed information on the temperature tables, see the separate document entitled "Safety Instructions" (XA) for the device.

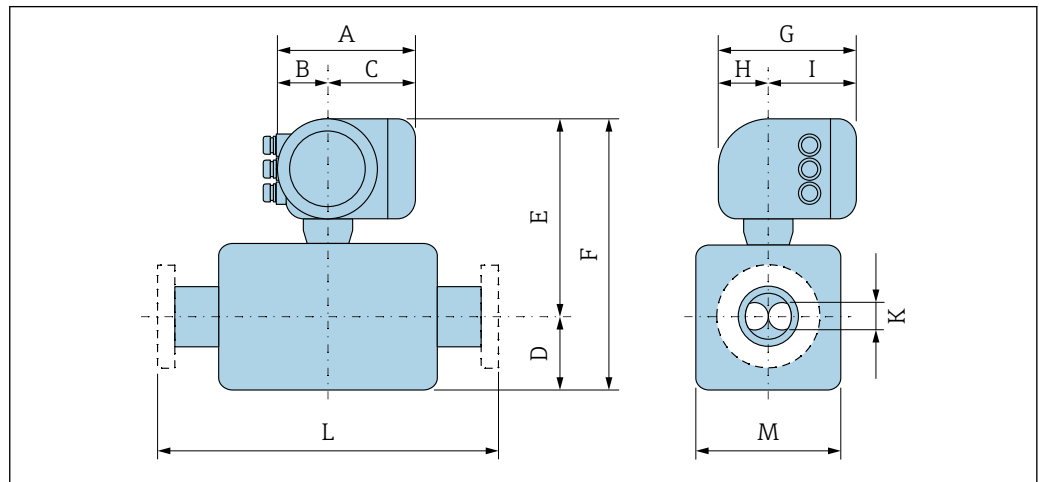
Vibrations

The high oscillation frequency of the measuring tubes ensures that the correct operation of the measuring system is not influenced by plant vibrations.

Mechanical construction

Dimensions in SI units

Compact version



A0033786

Order code for "Housing", option A "Aluminum, coated"

DN	A ¹⁾	B ¹⁾	C	D	E	F	G ²⁾	H	I ²⁾	K	L	M
[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]
8	169	68	101	89	252	341	200	59	141	5.35	³⁾	45
15	169	68	101	100	252	352	200	59	141	8.30	³⁾	45
25	169	68	101	102	249	351	200	59	141	12.0	³⁾	51
40	169	68	101	121	255	376	200	59	141	17.6	³⁾	65
50	169	68	101	175.5	269.5	445	200	59	141	26.0	³⁾	95
80	169	68	101	205	288.5	493.5	200	59	141	40.5	³⁾	127

1) Depending on the cable gland used: values up to + 30 mm

2) For version without local display: values - 30 mm

3) Dependent on respective process connection

Order code for "Housing", option A "Aluminum, coated"; Ex d

DN	A ¹⁾	B ¹⁾	C	D	E	F	G ²⁾	H	I ²⁾	K	L	M
[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]
8	188	85	103	89	282	371	206	58	148	5.35	³⁾	45
15	188	85	103	100	282	382	206	58	148	8.30	³⁾	45
25	188	85	103	102	279	381	206	58	148	12.0	³⁾	51
40	188	85	103	121	285	406	206	58	148	17.6	³⁾	65
50	188	85	103	175.5	299.5	475	206	58	148	26.0	³⁾	95
80	188	85	103	205	318.5	523.5	206	58	148	40.5	³⁾	127

1) Depending on the cable gland used: values up to + 30 mm

2) For version without local display: values - 38 mm

3) Dependent on respective process connection

Order code for "Housing", option B "Stainless, hygienic"

DN	A ¹⁾	B ¹⁾	C	D	E	F	G ²⁾	H	I ²⁾	K	L	M
[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]
8	183	73	110	89	252	341	207	65	142	5.35	³⁾	45
15	183	73	110	100	252	352	207	65	142	8.30	³⁾	45
25	183	73	110	102	249	351	207	65	142	12.0	³⁾	51
40	183	73	110	121	255	376	207	65	142	17.6	³⁾	65
50	183	73	110	175.5	269.5	445	207	65	142	26.0	³⁾	95
80	183	73	110	205	288.5	493.5	207	65	142	40.5	³⁾	127

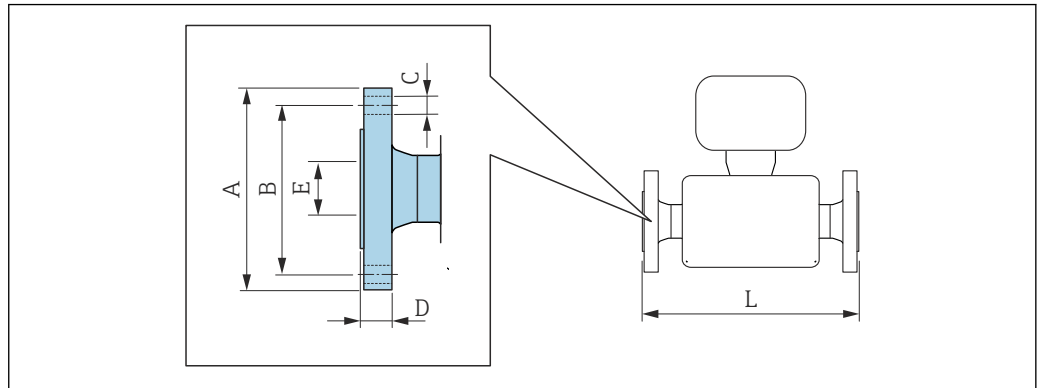
1) Depending on the cable gland used: values up to + 30 mm

2) For version without local display: values - 13 mm

3) Dependent on respective process connection

Flange connections

Fixed flange EN 1092-1, ASME B16.5, JIS B2220



A0015621



Length tolerance for dimension L in mm:
+1.5 / -2.0

Flange according to EN 1092-1 (DIN 2501 / DIN 2512N), PN 40

1.4404 (F316/F316L): order code for "Process connection", option D2S

Flange with groove according to EN 1092-1 Form D (DIN 2512N), PN 40

1.4404 (F316/F316L): order code for "Process connection", option D6S

DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
8 ¹⁾	95	65	4 × Ø14	16	17.3	232/510 ²⁾
15	95	65	4 × Ø14	16	17.3	279/510 ²⁾
25	115	85	4 × Ø14	18	28.5	329/600 ²⁾
40	150	110	4 × Ø18	18	43.1	445
50	165	125	4 × Ø18	20	54.5	556/715 ²⁾
80	200	160	8 × Ø18	24	82.5	611/915 ²⁾

Surface roughness (flange): EN 1092-1 Form B1 (DIN 2526 Form C), Ra 3.2 to 12.5 µm

- 1) DN 8 with DN 15 flanges as standard
- 2) Installation length in accordance with NAMUR recommendation NE 132 optionally available (order code for "Process connection", option D2N or D6N (with groove))

Flange according to EN 1092-1 (DIN 2501), PN 40 (with DN 25 flanges)

1.4404 (F316/F316L)

Order code for "Process connection", option R2S

DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
8	115	85	4 × Ø14	18	28.5	329
15	115	85	4 × Ø14	18	28.5	329

Surface roughness (flange): EN 1092-1 Form B1 (DIN 2526 Form C), Ra 3.2 to 12.5 µm

Flange according to EN 1092-1 (DIN 2501 / DIN 2512N), PN 63

1.4404 (F316/F316L): order code for "Process connection", option D3S

Flange with groove according to EN 1092-1 Form D (DIN 2512N), PN 63

1.4404 (F316/F316L): order code for "Process connection", option D7S

DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
50	180	135	4 × Ø22	26	54.5	565
80	215	170	8 × Ø22	28	81.7	646

Surface roughness (flange): EN 1092-1 Form B2 (DIN 2526 Form E), Ra 0.8 to 3.2 µm

Flange according to EN 1092-1 (DIN 2501 / DIN 2512N), PN 100

1.4404 (F316/F316L)

Order code for "Process connection", option D4S

Flange with groove according to EN 1092-1 Form D (DIN 2512N) available, PN 100

1.4404 (F316/F316L)

Order code for "Process connection", option D8S

DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
8 ¹⁾	105	75	4 × Ø14	20	17.3	261
15	105	75	4 × Ø14	20	17.3	295
25	140	100	4 × Ø18	24	28.5	360
40	170	125	4 × Ø22	26	42.5	486
50	195	145	4 × Ø26	28	53.9	581
80	230	180	8 × Ø26	32	80.9	656

Surface roughness (flange): EN 1092-1 Form B2 (DIN 2526 Form E), Ra 0.8 to 3.2 µm

1) DN 8 with DN 15 flanges as standard

Flange according to ASME B16.5, Class 150

1.4404 (F316/F316L)

Order code for "Process connection", option AAS

DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
8 ¹⁾	90	60.3	4 × Ø15.7	11.2	15.7	232
15	90	60.3	4 × Ø15.7	11.2	15.7	279
25	110	79.4	4 × Ø15.7	14.2	26.7	329
40	125	98.4	4 × Ø15.7	17.5	40.9	445
50	150	120.7	4 × Ø19.1	19.1	52.6	556
80	190	152.4	4 × Ø19.1	23.9	78.0	611

Surface roughness (flange): Ra 3.2 to 6.3 µm

1) DN 8 with DN 15 flanges as standard

Flange according to ASME B16.5, Class 300

1.4404 (F316/F316L)

Order code for "Process connection", option ABS

DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
8 ¹⁾	95	66.7	4 × Ø15.7	14.2	15.7	232
15	95	66.7	4 × Ø15.7	14.2	15.7	279

Flange according to ASME B16.5, Class 300
1.4404 (F316/F316L)
Order code for "Process connection", option ABS

DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
25	125	88.9	4 × Ø19.0	17.5	26.7	329
40	155	114.3	4 × Ø22.3	20.6	40.9	445
50	165	127	8 × Ø19.0	22.3	52.6	556
80	210	168.3	8 × Ø22.3	28.4	78.0	611
Surface roughness (flange): Ra 3.2 to 6.3 µm						

1) DN 8 with DN 15 flanges as standard

Flange according to ASME B16.5, Class 600
1.4404 (F316/F316L)
Order code for "Process connection", option ACS

DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
8 ¹⁾	95	66.7	4 × Ø15.7	20.6	13.9	261
15	95	66.7	4 × Ø15.7	20.6	13.9	295
25	125	88.9	4 × Ø19.1	23.9	24.3	380
40	155	114.3	4 × Ø22.4	28.7	38.1	496
50	165	127	8 × Ø19.1	31.8	49.2	583
80	210	168.3	8 × Ø22.4	38.2	73.7	671
Surface roughness (flange): Ra 3.2 to 6.3 µm						

1) DN 8 with DN 15 flanges as standard

Flange JIS B2220, 10K
1.4404 (F316/F316L)
Order code for "Process connection", option NDS

DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
50	155	120	4 × Ø19	16	50	556
80	185	150	8 × Ø19	18	80	603
Surface roughness (flange): Ra 3.2 to 6.3 µm						

Flange JIS B2220, 20K
1.4404 (F316/F316L)
Order code for "Process connection", option NES

DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
8 ¹⁾	95	70	4 × Ø15	14	15	232
15	95	70	4 × Ø15	14	15	279
25	125	90	4 × Ø19	16	25	329
40	140	105	4 × Ø19	18	40	445
50	155	120	8 × Ø19	18	50	556

Flange JIS B2220, 20K 1.4404 (F316/F316L) <i>Order code for "Process connection", option NES</i>						
DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
80	200	160	8 × Ø23	22	80	603
Surface roughness (flange): Ra 3.2 to 6.3 µm						

- 1) DN 8 with DN 15 flanges as standard

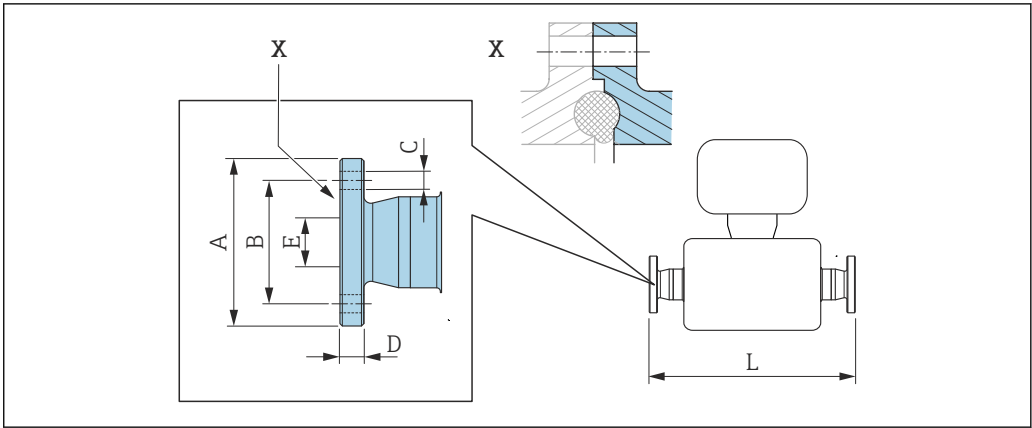
Flange JIS B2220, 40K 1.4404 (F316/F316L) <i>Order code for "Process connection", option NGS</i>						
DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
8 ¹⁾	115	80	4 × Ø19	20	15	261
15	115	80	4 × Ø19	20	15	300
25	130	95	4 × Ø19	22	25	375
40	160	120	4 × Ø23	24	38	496
50	165	130	8 × Ø19	26	50	601
80	210	170	8 × Ø23	32	75	661
Surface roughness (flange): Ra 3.2 to 6.3 µm						

- 1) DN 8 with DN 15 flanges as standard

Flange JIS B2220, 63K 1.4404 (F316/F316L) <i>Order code for "Process connection", option NHS</i>						
DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
8 ¹⁾	120	85	4 × Ø19	23	12	282
15	120	85	4 × Ø19	23	12	315
25	140	100	4 × Ø23	27	22	383
40	175	130	4 × Ø25	32	35	515
50	185	145	4 × Ø23	34	48	616
80	230	185	4 × Ø25	40	73	686
Surface roughness (flange): Ra 3.2 to 6.3 µm						

- 1) DN 8 with DN 15 flanges as standard

Fixed flange DIN 11864-2



A0015627

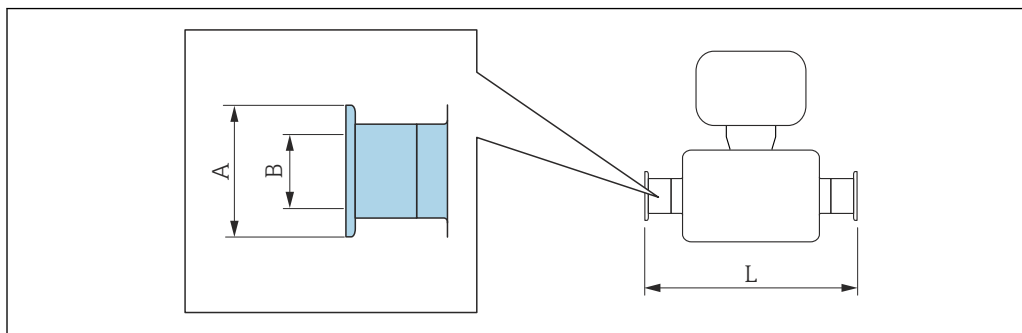
34 Detail X: Asymmetrical process connection; the part shown in blue is provided by the supplier.

i Length tolerance for dimension L in mm:
+1.5 / -2.0

Flange DIN11864-2 Form A, for pipe according to DIN11866 series A, flange with notch 1.4404 (316/316L) Order code for "Process connection", option KCS						
DN [mm]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	L [mm]
8	54	37	4 × Ø9	10	10	249
15	59	42	4 × Ø9	10	16	293
25	70	53	4 × Ø9	10	26	344
40	82	65	4 × Ø9	10	38	456
50	94	77	4 × Ø9	10	50	562
80	133	112	8 × Ø11	12	81	671
3-A version available: order code for "Additional approval", option LP in conjunction with Ra ≤ 0.8 µm: Order code for "Measuring tube material", option SB Ra ≤ 0.4 µm: Order code for "Measuring tube material", option SC						

Clamp connections

Tri-Clamp



A0015625

i Length tolerance for dimension L in mm:
+1.5 / -2.0

Tri-Clamp (½")

1.4404 (316/316L)

Order code for "Process connection", option **FDW**

DN [mm]	Clamp [in]	A [mm]	B [mm]	L [mm]
8	½	25.0	9.5	229
15	½	25.0	9.5	273

3-A version available: order code for "Additional approval", option LP in conjunction with

Ra ≤ 0.8 µm: Order code for "Measuring tube material", option **SB**

Ra ≤ 0.4 µm: Order code for "Measuring tube material", option **SC**

Tri-Clamp (≥ 1")

1.4404 (316/316L)

Order code for "Process connection", option **FTS**

DN [mm]	Clamp [in]	A [mm]	B [mm]	L [mm]
8	1	50.4	22.1	229
15	1	50.4	22.1	273
25	1	50.4	22.1	324
40	1½	50.4	34.8	456
50	2	63.9	47.5	562
80	3	90.9	72.9	671

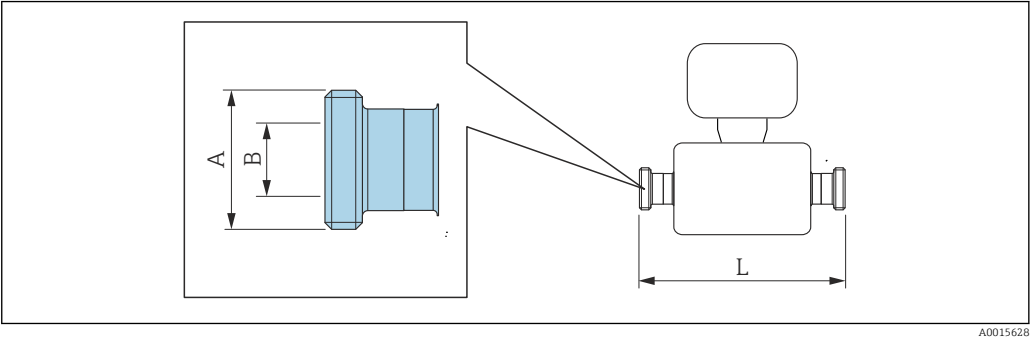
3-A version available: order code for "Additional approval", option LP in conjunction with

Ra ≤ 0.8 µm: Order code for "Measuring tube material", option **SB**

Ra ≤ 0.4 µm: Order code for "Measuring tube material", option **SC**

Threaded couplings

Thread DIN 11851, DIN11864-1, SMS 1145



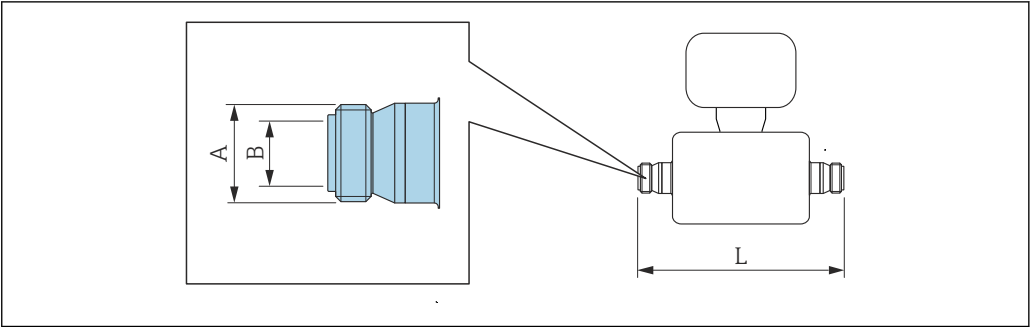
i Length tolerance for dimension L in mm:
+1.5 / -2.0

Thread DIN 11851, for pipe according to DIN11866, series A 1.4404 (316/316L) Order code for "Process connection", option FMW			
DN [mm]	A [in]	B [mm]	L [mm]
8	Rd 34 × 1/8	16	229
15	Rd 34 × 1/8	16	273
25	Rd 52 × 1/6	26	324
40	Rd 65 × 1/6	38	456
50	Rd 78 × 1/6	50	562
80	Rd 110 × 1/4	81	671
3-A version available: order code for "Additional approval", option LP in conjunction with Ra ≤ 0.8 µm: Order code for "Measuring tube material", option SB Ra ≤ 0.4 µm: Order code for "Measuring tube material", option SC			

Thread DIN11864-1 Form A, for pipe according to DIN11866, series A 1.4404 (316/316L) Order code for "Process connection", option FLW			
DN [mm]	A [in]	B [mm]	L [mm]
8	Rd 28 × 1/8	10	229
15	Rd 34 × 1/8	16	273
25	Rd 52 × 1/6	26	324
40	Rd 65 × 1/6	38	456
50	Rd 78 × 1/6	50	562
80	Rd 110 × 1/4	81	671
3-A version available: order code for "Additional approval", option LP in conjunction with Ra ≤ 0.8 µm: Order code for "Measuring tube material", option SB Ra ≤ 0.4 µm: Order code for "Measuring tube material", option SC			

Thread SMS 1145 1.4404 (316/316L) <i>Order code for "Process connection", option SCS</i>			
DN [mm]	A [in]	B [mm]	L [mm]
8	Rd 40 × 1/6	22.5	229
15	Rd 40 × 1/6	22.5	273
25	Rd 40 × 1/6	22.5	324
40	Rd 60 × 1/6	35.5	456
50	Rd 70 × 1/6	48.5	562
80	Rd 98 × 1/6	72.9	671
3-A version available: order code for "Additional approval", option LP in conjunction with Ra ≤ 0.8 µm: Order code for "Measuring tube material", option SB Ra ≤ 0.4 µm: Order code for "Measuring tube material", option SC			

Thread ISO 2853



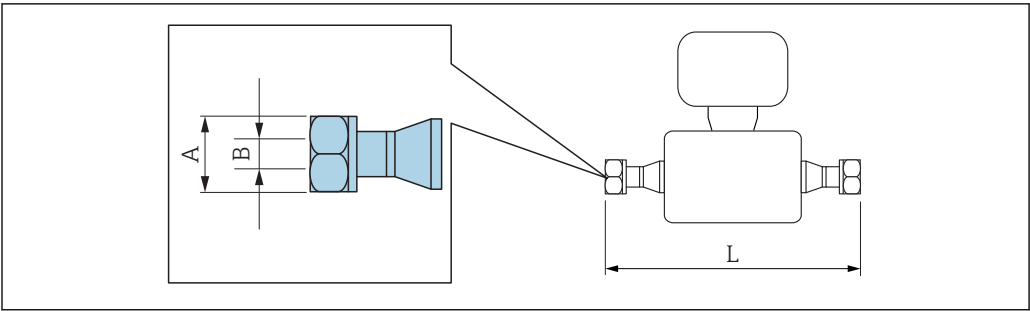
A0015623

 Length tolerance for dimension L in mm:
+1.5 / -2.0

Thread ISO 2853, for pipe according to ISO 2037 1.4404 (316/316L) Order code for "Process connection", option JSF			
DN [mm]	A ¹⁾ [mm]	B [mm]	L [mm]
8	37.13	22.6	229
15	37.13	22.6	273
25	37.13	22.6	324
40	50.68	35.6	456
50	64.16	48.6	562
80	91.19	72.9	671
3-A version available: order code for "Additional approval", option LP in conjunction with Ra ≤ 0.8 µm: Order code for "Measuring tube material", option SB Ra ≤ 0.4 µm: Order code for "Measuring tube material", option SC			

1) Max. thread diameter as per ISO 2853 annex A

VCO



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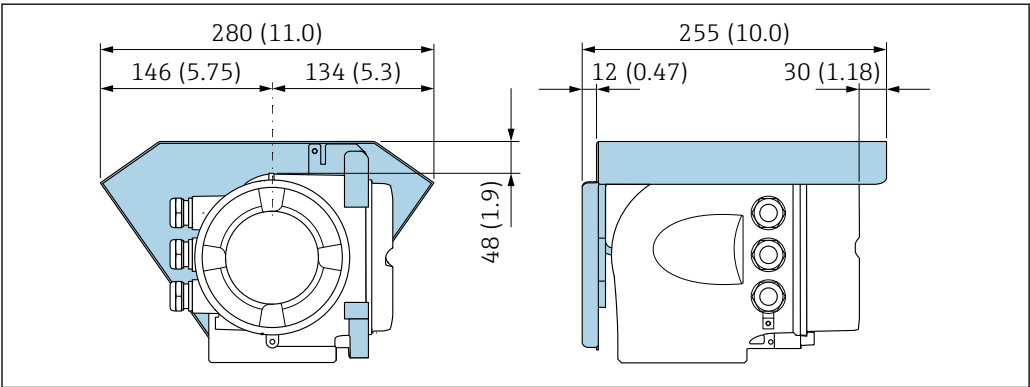
i Length tolerance for dimension L in mm:
+1.5 / -2.0

8-VCO-4 (½") 1.4404 (316/316L) Order code for "Process connection", option CVS			
DN [mm]	A [in]	B [mm]	L [mm]
8	AF 1	10.2	252

12-VCO-4 (¾") 1.4404 (316/316L) Order code for "Process connection", option CWS			
DN [mm]	A [in]	B [mm]	L [mm]
15	AF 1½	15.7	305

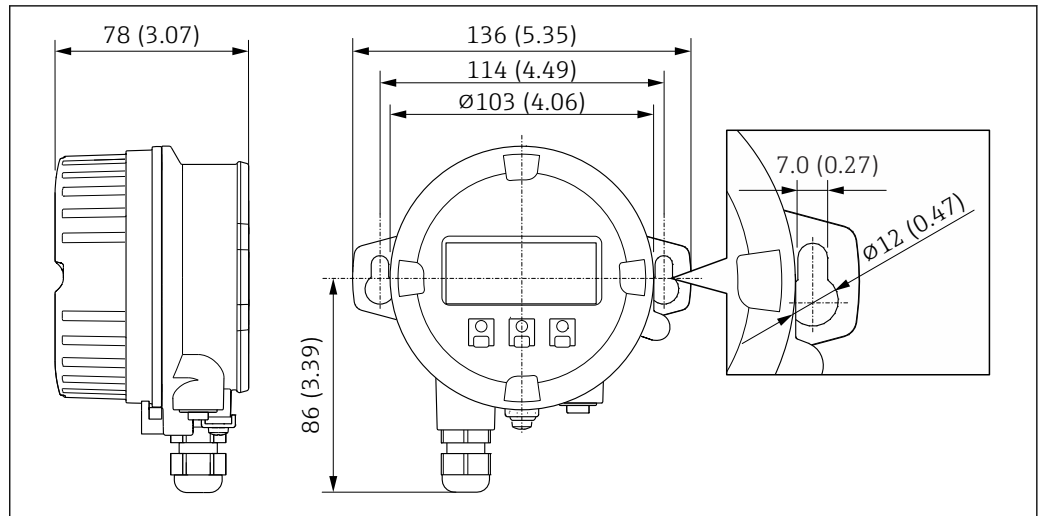
Accessories

Protective cover



A0029553

Remote display and operating module DKX001



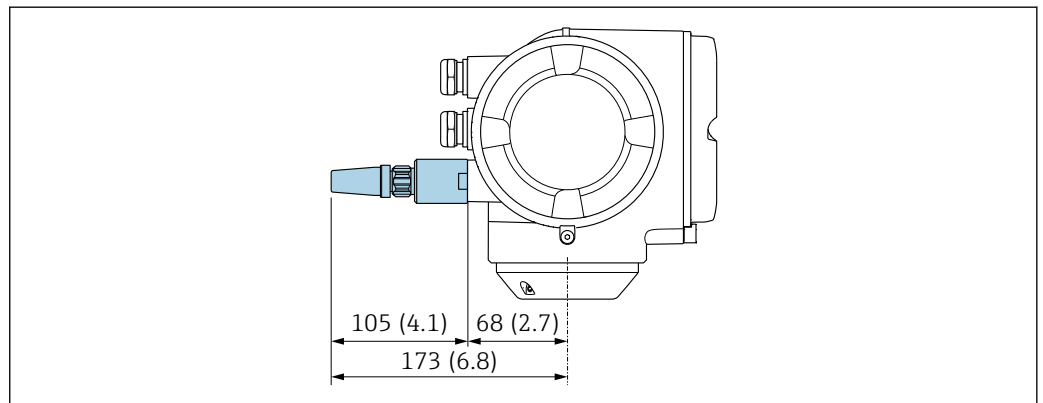
A0028921

35 Engineering unit mm (in)

External WLAN antenna

i The external WLAN antenna is not suitable for use in hygienic applications.

External WLAN antenna mounted on device

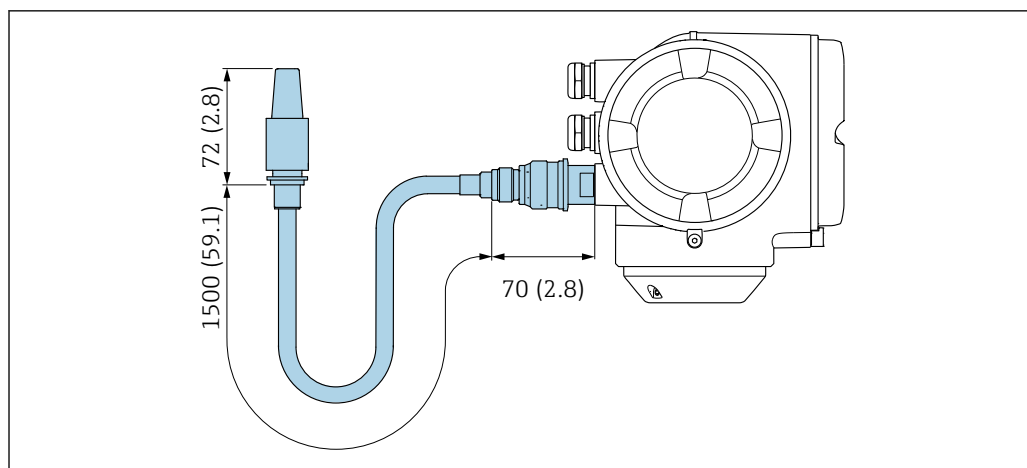


A0028923

36 Engineering unit mm (in)

External WLAN antenna mounted with cable

The external WLAN antenna can be mounted separately from the transmitter if the transmission/reception conditions at the transmitter mounting location are poor.

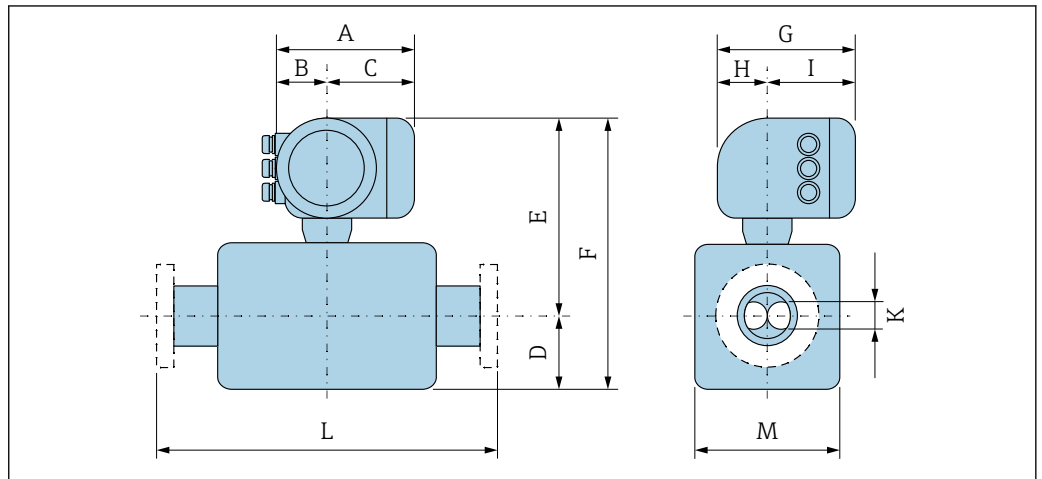


A0033597

37 Engineering unit mm (in)

Dimensions in US units

Compact version



A0033786

Order code for "Housing", option A "Aluminum, coated"

DN	A ¹⁾	B ¹⁾	C	D	E	F	G ²⁾	H	I ²⁾	K	L	M
[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]
3/8	6.65	2.68	3.98	3.5	9.92	13.43	7.87	2.32	5.55	0.211	³⁾	1.77
1/2	6.65	2.68	3.98	3.94	9.92	13.86	7.87	2.32	5.55	0.33	³⁾	1.77
1	6.65	2.68	3.98	4.02	9.8	13.82	7.87	2.32	5.55	0.47	³⁾	2.01
1 1/2	6.65	2.68	3.98	4.76	10.04	14.8	7.87	2.32	5.55	0.69	³⁾	2.56
2	6.65	2.68	3.98	6.91	10.61	17.52	7.87	2.32	5.55	1.02	³⁾	3.74
3	6.65	2.68	3.98	8.07	11.36	19.43	7.87	2.32	5.55	1.59	³⁾	5.00

- 1) Depending on the cable gland used: values up to + 1.18 in
 2) For version without local display: values - 1.18 in
 3) Dependent on respective process connection

Order code for "Housing", option A "Aluminum, coated"; Ex d

DN	A ¹⁾	B ¹⁾	C	D	E	F	G ²⁾	H	I ²⁾	K	L	M
[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]	[in]
3/8	7.40	3.35	4.06	3.5	11.1	14.61	8.11	2.28	5.83	0.211	³⁾	1.77
1/2	7.40	3.35	4.06	3.94	11.1	15.04	8.11	2.28	5.83	0.33	³⁾	1.77
1	7.40	3.35	4.06	4.02	10.98	15	8.11	2.28	5.83	0.47	³⁾	2.01
1 1/2	7.40	3.35	4.06	4.76	11.22	15.98	8.11	2.28	5.83	0.69	³⁾	2.56
2	7.40	3.35	4.06	6.91	11.79	18.7	8.11	2.28	5.83	1.02	³⁾	3.74
3	7.40	3.35	4.06	8.07	12.54	20.61	8.11	2.28	5.83	1.59	³⁾	5.00

- 1) Depending on the cable gland used: values up to + 1.18 in
 2) For version without local display: values - 1.49 in
 3) Dependent on respective process connection

Order code for "Housing", option B "Stainless, hygienic"

DN [in]	A ¹⁾ [in]	B ¹⁾ [in]	C [in]	D [in]	E [in]	F [in]	G ²⁾ [in]	H [in]	I ²⁾ [in]	K [in]	L [in]	M [in]
$\frac{3}{8}$	7.20	2.87	4.33	3.5	9.92	13.43	8.15	2.56	5.59	0.211	³⁾	1.77
$\frac{1}{2}$	7.20	2.87	4.33	3.94	9.92	13.86	8.15	2.56	5.59	0.33	³⁾	1.77
1	7.20	2.87	4.33	4.02	9.8	13.82	8.15	2.56	5.59	0.47	³⁾	2.01
1½	7.20	2.87	4.33	4.76	10.04	14.8	8.15	2.56	5.59	0.69	³⁾	2.56
2	7.20	2.87	4.33	6.91	10.61	17.52	8.15	2.56	5.59	1.02	³⁾	3.74
3	7.20	2.87	4.33	8.07	11.36	19.43	8.15	2.56	5.59	1.59	³⁾	5.00

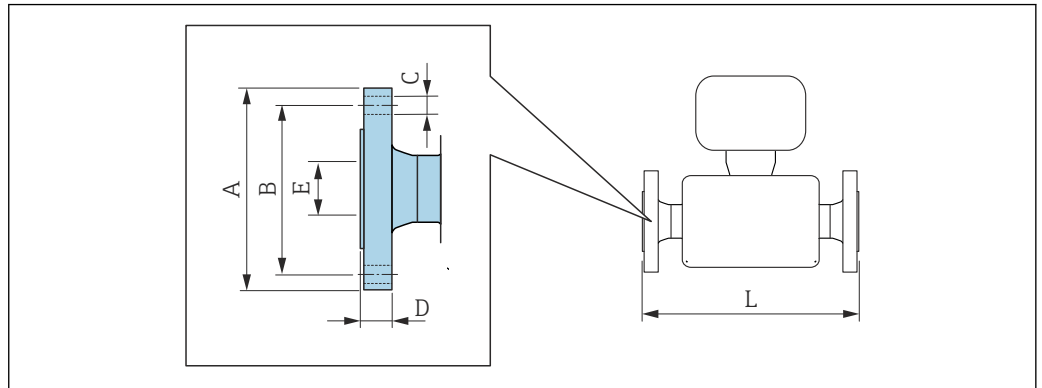
1) Depending on the cable gland used: values up to + 1.18 in

2) For version without local display: values - 0.51 in

3) Dependent on respective process connection

Flange connections

Fixed flange ASME B16.5



A0015621



Length tolerance for dimension L in inch:
+0.06 / -0.08

Flange according to ASME B16.5, Cl 150

1.4404 (F316/F316L)

Order code for "Process connection", option AAS

DN [in]	A [in]	B [in]	C [in]	D [in]	E [in]	L [in]
$\frac{3}{8}$ ¹⁾	3.54	2.37	4 × Ø0.62	0.44	0.62	9.13
$\frac{1}{2}$	3.54	2.37	4 × Ø0.62	0.44	0.62	10.98
1	4.33	3.13	4 × Ø0.62	0.56	1.05	12.95
1½	4.92	3.87	4 × Ø0.62	0.69	1.61	17.52
2	5.91	4.75	4 × Ø0.75	0.75	2.07	21.89
3	7.48	6.00	4 × Ø0.75	0.94	3.07	24.06

Surface roughness (flange): Ra 32 to 248 µin

- 1) DN $\frac{3}{8}$ " with DN $\frac{1}{2}$ " flanges as standard

Flange according to ASME B16.5, Class 300

1.4404 (F316/F316L)

Order code for "Process connection", option ABS

DN [in]	A [in]	B [in]	C [in]	D [in]	E [in]	L [in]
$\frac{3}{8}$ ¹⁾	3.74	2.63	4 × Ø0.62	0.56	0.62	9.13
$\frac{1}{2}$	3.74	2.63	4 × Ø0.62	0.56	0.62	10.98
1	4.92	3.50	4 × Ø0.75	0.69	1.05	12.95
1½	6.10	4.50	4 × Ø0.88	0.81	1.61	17.52
2	6.50	5.00	8 × Ø0.75	0.88	2.07	21.89
3	8.27	6.63	8 × Ø0.88	1.12	3.07	24.06

Surface roughness (flange): Ra 32 to 248 µin

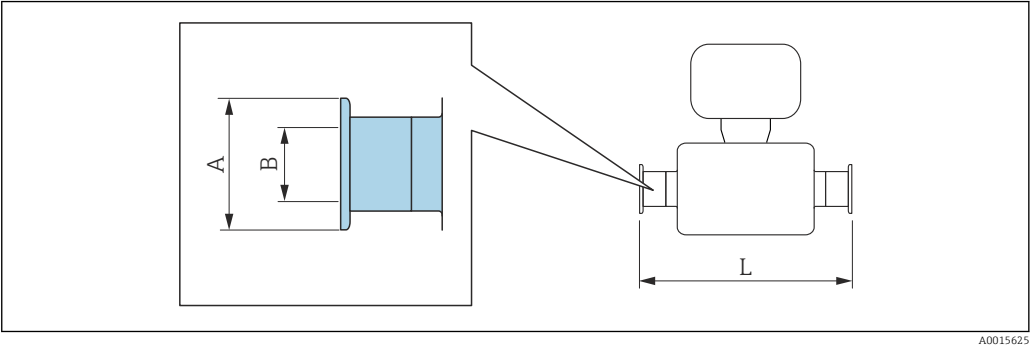
- 1) DN $\frac{3}{8}$ " with DN $\frac{1}{2}$ " flanges as standard

Flange according to ASME B16.5, Class 600 1.4404 (F316/F316L) <i>Order code for "Process connection", option ACS</i>						
DN [in]	A [in]	B [in]	C [in]	D [in]	E [in]	L [in]
$\frac{3}{8}$ ¹⁾	3.74	2.63	4 × Ø0.62	0.81	0.55	10.28
$\frac{1}{2}$	3.74	2.63	4 × Ø0.62	0.81	0.55	11.61
1	4.92	3.50	4 × Ø0.75	0.94	0.96	14.96
1½	6.10	4.50	4 × Ø0.88	1.13	1.50	19.53
2	6.50	5.00	8 × Ø0.75	1.25	1.94	22.95
3	8.27	6.63	8 × Ø0.88	1.50	2.9	26.42
Surface roughness (flange): Ra 32 to 248 µin						

1) DN $\frac{3}{8}$ " with DN $\frac{1}{2}$ " flanges as standard

Clamp connections

Tri-Clamp



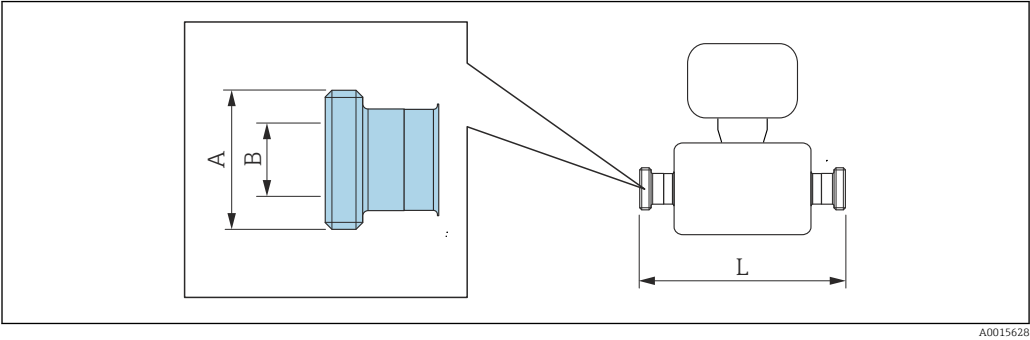
 Length tolerance for dimension L in inch:
+0.06 / -0.08

Tri-Clamp (½") 1.4404 (316/316L) Order code for "Process connection", option FDW				
DN [in]	Clamp [in]	A [in]	B [in]	L [in]
⅜	½	0.98	0.37	9.02
½	½	0.98	0.37	10.75
3-A version available: order code for "Additional approval", option LP in conjunction with Ra ≤ 32 µin: Order code for "Measuring tube material", option SB Ra ≤ 16 µin: Order code for "Measuring tube material", option SC				

Tri-Clamp (≥ 1") 1.4404 (316/316L) Order code for "Process connection", option FTS				
DN [in]	Clamp [in]	A [in]	B [in]	L [in]
⅜	1	1.98	0.87	9.02
½	1	1.98	0.87	10.75
1	1	1.98	0.87	12.76
1½	1½	1.98	1.37	17.95
2	2	2.52	1.87	22.13
3	3	3.58	2.87	26.42
3-A version available: order code for "Additional approval", option LP in conjunction with Ra ≤ 32 µin: Order code for "Measuring tube material", option SB Ra ≤ 16 µin: Order code for "Measuring tube material", option SC				

Threaded couplings

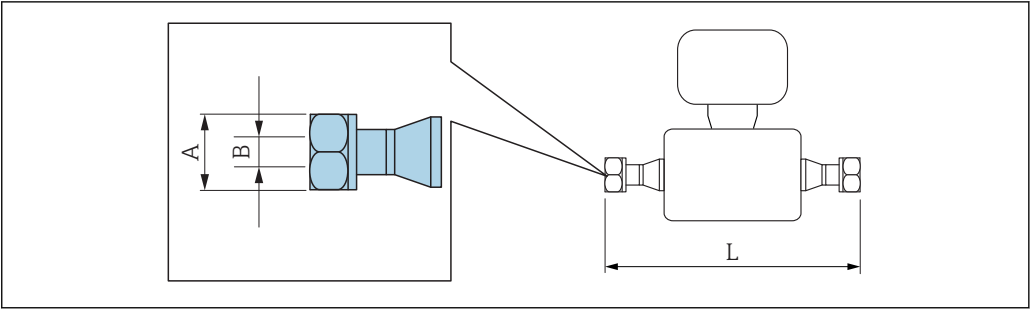
Thread SMS 1145



 Length tolerance for dimension L in inch:
+0.06 / -0.08

Thread SMS 1145 1.4404 (316/316L) Order code for "Process connection", option SCS			
DN [in]	A [in]	B [in]	L [in]
3/8	Rd 40 × 1/6	0.89	9.02
1/2	Rd 40 × 1/6	0.89	10.75
1	Rd 40 × 1/6	0.89	12.76
1 1/2	Rd 60 × 1/6	1.40	17.95
2	Rd 70 × 1/6	1.91	22.13
3	Rd 98 × 1/6	2.87	26.42
3-A version available: order code for "Additional approval", option LP in conjunction with Ra ≤ 32 µin: Order code for "Measuring tube material", option SB Ra ≤ 16 µin: Order code for "Measuring tube material", option SC			

VCO



A0015624

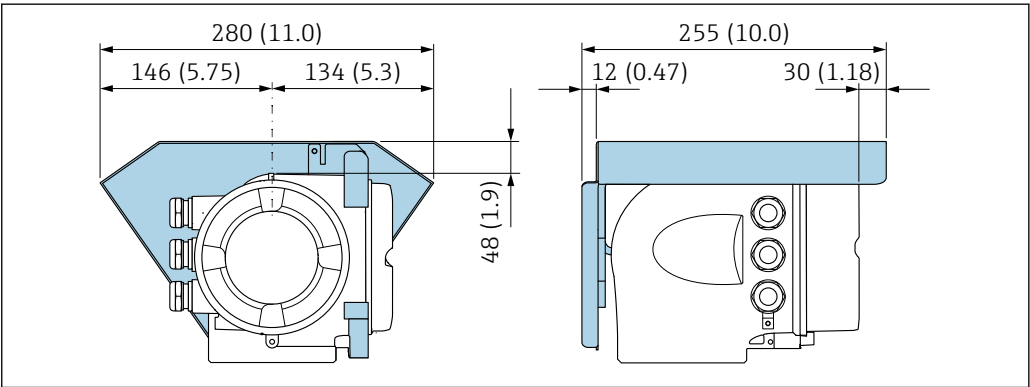
i Length tolerance for dimension L in inch:
+0.06 / -0.08

8-VCO-4 (½") 1.4404 (316/316L) <i>Order code for "Process connection", option CVS</i>			
DN [in]	A [in]	B [in]	L [in]
⅜	AF 1	0.40	9.92

12-VCO-4 (¾") 1.4404 (316/316L) <i>Order code for "Process connection", option CWS</i>			
DN [in]	A [in]	B [in]	L [in]
½	AF 1½	0.62	12.01

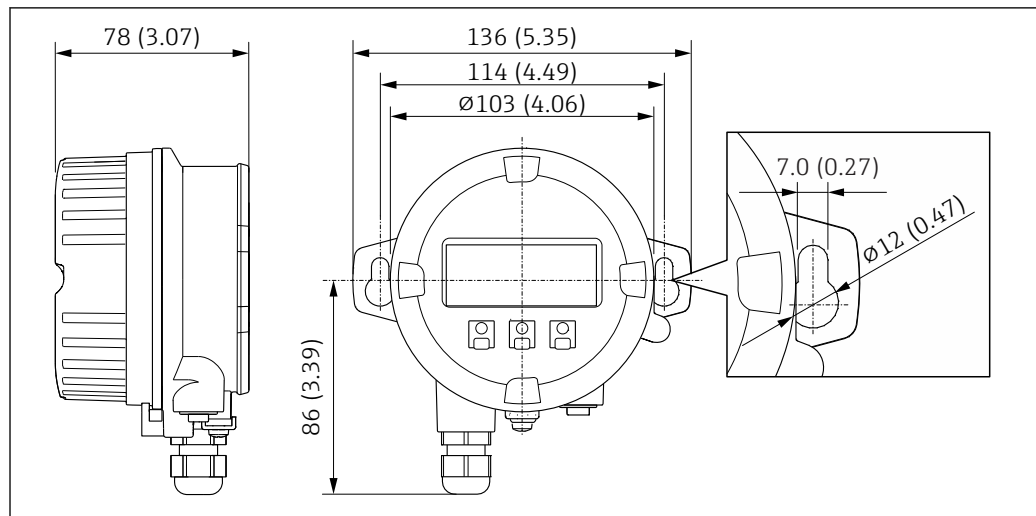
Accessories

Protective cover



A0029553

Remote display and operating module DKX001



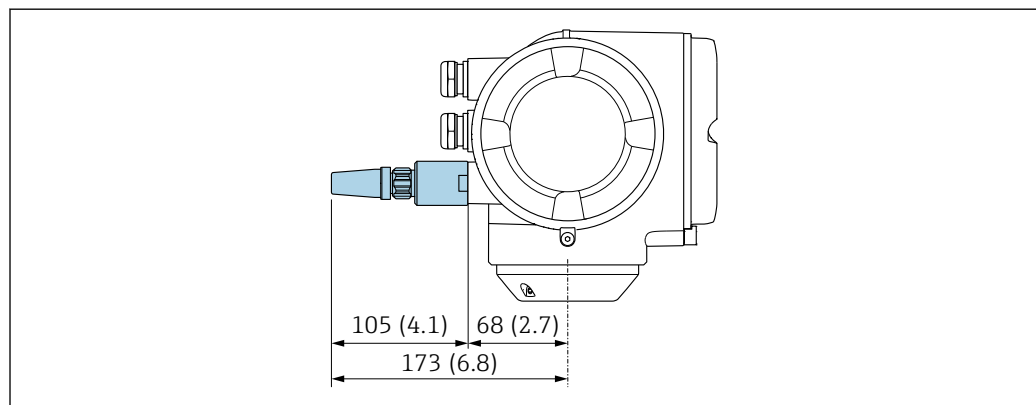
A0028921

38 Engineering unit mm (in)

External WLAN antenna

i The external WLAN antenna is not suitable for use in hygienic applications.

External WLAN antenna mounted on device

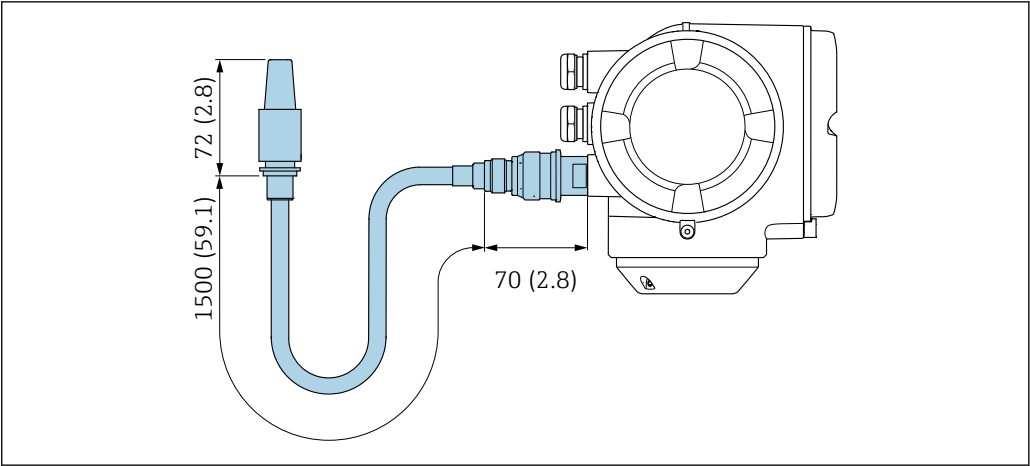


A0028923

39 Engineering unit mm (in)

External WLAN antenna mounted with cable

The external WLAN antenna can be mounted separately from the transmitter if the transmission/reception conditions at the transmitter mounting location are poor.



A0033597

40 Engineering unit mm (in)

Weight

All values (weight exclusive of packaging material) refer to devices with EN/DIN PN 40 flanges. Weight specifications including transmitter as per order code for "Housing", option A "Aluminum, coated".

Different values due to different transmitter versions:

- Transmitter version for the hazardous area
(Order code for "Housing", option A "Aluminum, coated"; Ex d): +2 kg (+4.4 lbs)
- Transmitter version for hygienic area
(Order code for "Housing", option B "Stainless, hygienic"): +0.2 kg (+0.44 lbs)

Weight in SI units

DN [mm]	Weight [kg]
8	5
15	5.5
25	7
40	11
50	16
80	32

Weight in US units

DN [in]	Weight [lbs]
3/8	11
1/2	12
1	15
1 1/2	24
2	35
3	71

Materials

Transmitter housing

Order code for "Housing":

- Option A "Aluminum, coated": aluminum, AlSi10Mg, coated
- Option B "Stainless, hygienic": stainless steel, 1.4404 (316L)

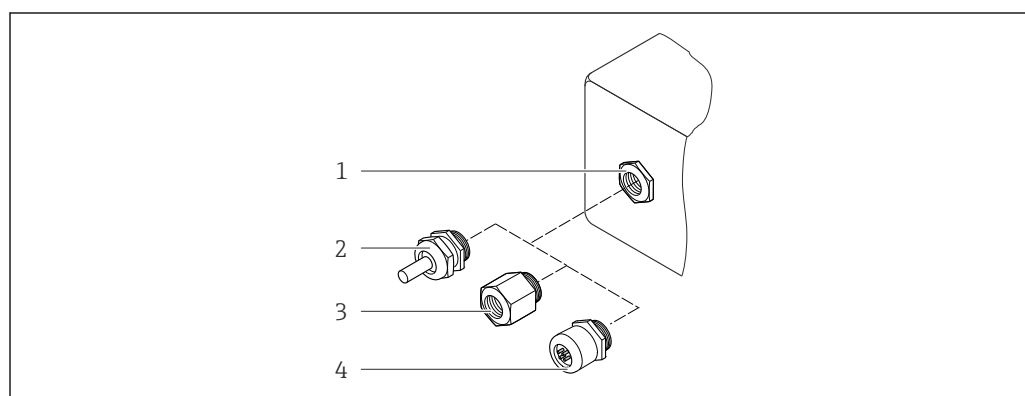
Window material

Order code for "Housing":

- Option **A** "Aluminum, coated": glass
- Option **B** "Stainless, hygienic": polycarbonate

Seals

Order code for "Housing":

Option **B** "Stainless, hygienic": EPDM and silicone**Cable entries/cable glands**

A0028352

 41 Possible cable entries/cable glands

- 1 Female thread M20 × 1.5
- 2 Cable gland M20 × 1.5
- 3 Adapter for cable entry with internal thread G ½" or NPT ½"
- 4 Device plugs

Order code for "Housing", option A "Aluminum, coated"

The various cable entries are suitable for hazardous and non-hazardous areas.

Cable entry/cable gland	Material
Cable gland M20 × 1.5	Plastic/nickel-plated brass
Adapter for cable entry with internal thread G ½"	Nickel-plated brass
Adapter for cable entry with internal thread NPT ½"	
Device plug	Plug M12 × 1 ■ Socket: Stainless steel, 1.4404 (316L) ■ Contact housing: Polyamide ■ Contacts: Gold-plated brass

Order code for "Housing", option B "Stainless, hygienic"

The various cable entries are suitable for hazardous and non-hazardous areas.

Cable entry/cable gland	Material
Cable gland M20 × 1.5	Plastic
Adapter for cable entry with internal thread G ½"	Nickel-plated brass
Adapter for cable entry with internal thread NPT ½"	
Device plug	Plug M12 × 1 ■ Socket: Stainless steel, 1.4404 (316L) ■ Contact housing: Polyamide ■ Contacts: Gold-plated brass

Device plug

Electrical connection	Material
Plug M12x1	■ Socket: Stainless steel, 1.4404 (316L) ■ Contact housing: Polyamide ■ Contacts: Gold-plated brass

Sensor housing

- Acid and alkali-resistant outer surface
- Stainless steel 1.4301 (304)

Measuring tubes

Stainless steel, 1.4539 (904L); manifold: stainless steel, 1.4404 (316L)

Process connections

- Flanges according to EN 1092-1 (DIN2501) / according to ASME B 16.5 / as per JIS B2220: Stainless steel, 1.4404 (F316/F316L)
- All other process connections: Stainless steel, 1.4404 (316/316L)



Available process connections → 81

Seals

Welded process connections without internal seals

Accessories*Protective cover*

Stainless steel, 1.4404 (316L)

External WLAN antenna

- Antenna: ASA plastic (acrylic ester-styrene-acrylonitrile) and nickel-plated brass
- Adapter: Stainless steel and nickel-plated brass
- Cable: Polyethylene
- Plug: Nickel-plated brass
- Angle bracket: Stainless steel

Process connections

- Fixed flange connections:
 - EN 1092-1 (DIN 2501) flange
 - EN 1092-1 (DIN 2512N) flange
 - Namur lengths in accordance with NE 132
 - ASME B16.5 flange
 - JIS B2220 flange
 - DIN 11864-2 Form A flange, DIN 11866 series A, flange with notch
- Clamp connections:
 - Tri-Clamp (OD tubes), DIN 11866 series C
- Thread:
 - DIN 11851 thread, DIN 11866 series A
 - SMS 1145 thread
 - ISO 2853 thread, ISO 2037
 - DIN 11864-1 Form A thread, DIN 11866 series A
- VCO connections:
 - 8-VCO-4
 - 12-VCO-4



Process connection materials → 81

Surface roughness

All data relate to parts in contact with fluid. The following surface roughness quality can be ordered.

- Not polished
- $Ra_{max} = 0.8 \mu m$ (32 μin)
- $Ra_{max} = 0.4 \mu m$ (16 μin)

Operability

Operating concept**Operator-oriented menu structure for user-specific tasks**

- Commissioning
- Operation
- Diagnostics
- Expert level

Fast and safe commissioning

- Guided menus ("Make-it-run" wizards) for applications
- Menu guidance with brief descriptions of the individual parameter functions
- Device access via Web server or SmartBlue app →  101
- WLAN access to the device via mobile handheld terminal, tablet or smart phone

Reliable operation

- Operation in local language →  82
- Uniform operating philosophy applied to device and operating tools
- If replacing electronic modules, transfer the device configuration via the integrated memory (HistoROM backup) which contains the process and measuring device data and the event logbook. No need to reconfigure.

Efficient diagnostics increase measurement availability

- Troubleshooting measures can be called up via the device and in the operating tools
- Diverse simulation options, logbook for events that occur and optional line recorder functions

Languages

Can be operated in the following languages:

- Via local operation
English, German, French, Spanish, Italian, Dutch, Portuguese, Polish, Russian, Turkish, Chinese, Japanese, Korean, Bahasa (Indonesian), Vietnamese, Czech, Swedish
- Via Web browser
English, German, French, Spanish, Italian, Dutch, Portuguese, Polish, Russian, Turkish, Chinese, Japanese, Korean, Bahasa (Indonesian), Vietnamese, Czech, Swedish
- Via "FieldCare", "DeviceCare" operating tool: English, German, French, Spanish, Italian, Chinese, Japanese

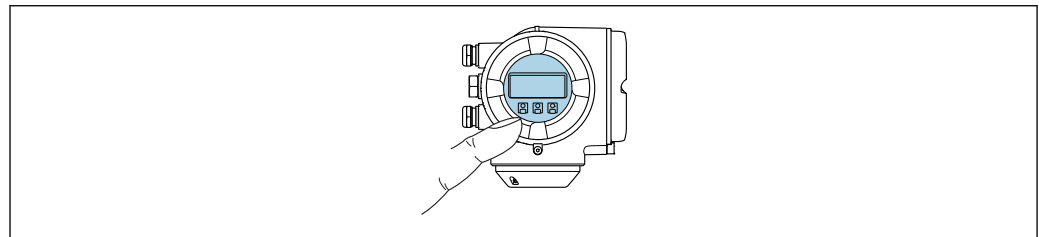
Local operation**Via display module**

Two display modules are available:

- Order code for "Display; operation", option F "4-line, illuminated, graphic display; touch control"
- Order code for "Display; operation", option G "4-line, illuminated, graphic display; touch control + WLAN"



Information about WLAN interface →  89



A0026785

 42 Operation with touch control

Display elements

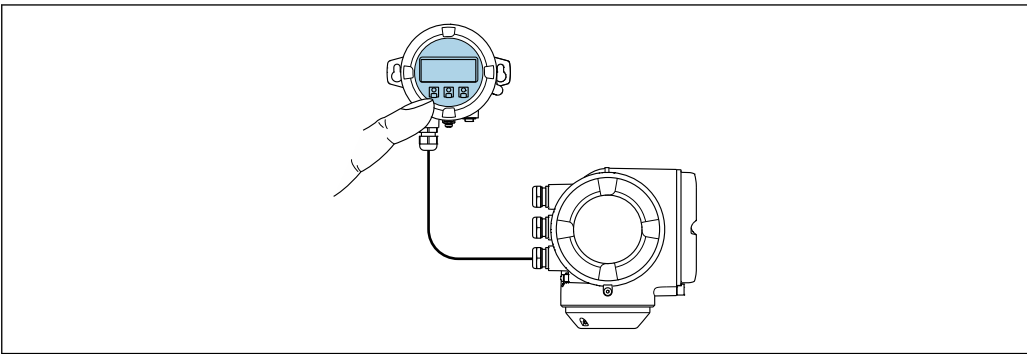
- 4-line, illuminated, graphic display
 - White background lighting; switches to red in event of device errors
 - Format for displaying measured variables and status variables can be individually configured
 - Permitted ambient temperature for the display: -20 to +60 °C (-4 to +140 °F)
- The readability of the display may be impaired at temperatures outside the temperature range.

Operating elements

- External operation via touch control (3 optical keys) without opening the housing: , , 
- Operating elements also accessible in the various zones of the hazardous area

Via remote display and operating module DKX001

-  The remote display and operating module DKX001 is available as an optional extra →  99.
- The remote display and operating module DKX001 is only available for the following housing version: order code for "Housing": option A "Aluminum, coated"
 - The measuring device is always supplied with a dummy cover when the remote display and operating module DKX001 is ordered directly with the measuring device. Display or operation at the transmitter is not possible in this case.
 - If ordered subsequently, the remote display and operating module DKX001 may not be connected at the same time as the existing measuring device display module. Only one display or operation unit may be connected to the transmitter at any one time.



A0026786

 43 Operation via remote display and operating module DKX001

Display and operating elements

The display and operating elements correspond to those of the display module →  82.

Material

The housing material of the display and operating module DKX001 depends on the choice of transmitter housing material.

Transmitter housing		Remote display and operating module
Order code for "Housing"	Material	Material
Option A "Aluminum, coated"	AlSi10Mg, coated	AlSi10Mg, coated

Cable entry

Corresponds to the choice of transmitter housing, order code for "Electrical connection".

Connecting cable

→  42

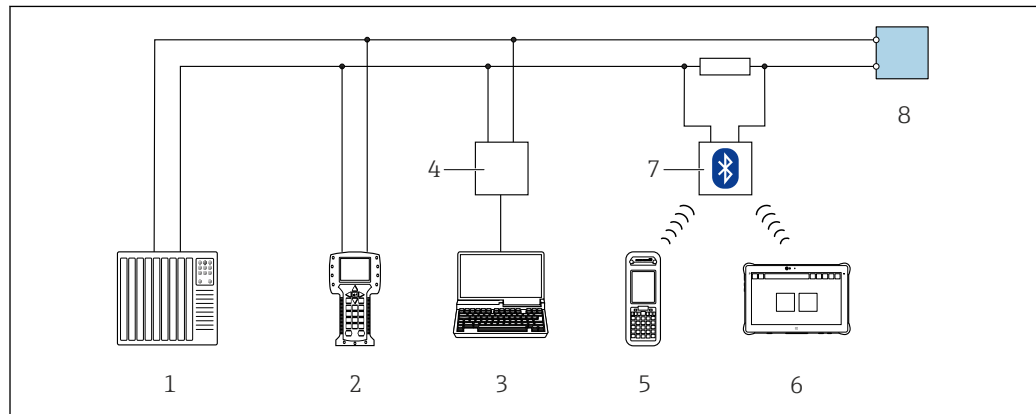
Dimensions

→  69

Remote operation

Via HART protocol

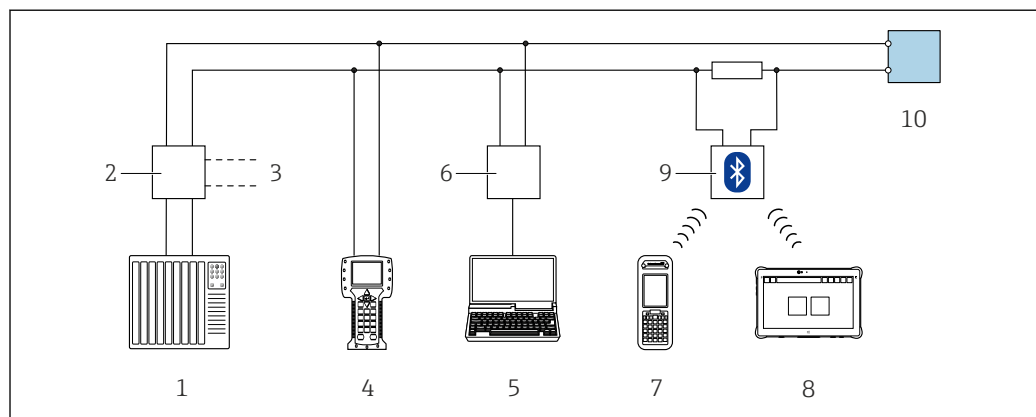
This communication interface is available in device versions with a HART output.



A0028747

44 Options for remote operation via HART protocol (active)

- 1 Control system (e.g. PLC)
- 2 Field Communicator 475
- 3 Computer with Web browser (e.g. Internet Explorer) for access to the integrated device Web server or computer with an operating tool (e.g. FieldCare, DeviceCare, AMS Device Manager, SIMATIC PDM) with COM DTM "CDI Communication TCP/IP"
- 4 Commubox FXA195 (USB)
- 5 Field Xpert SFX350 or SFX370
- 6 Field Xpert SMT70
- 7 VIATOR Bluetooth modem with connecting cable
- 8 Transmitter



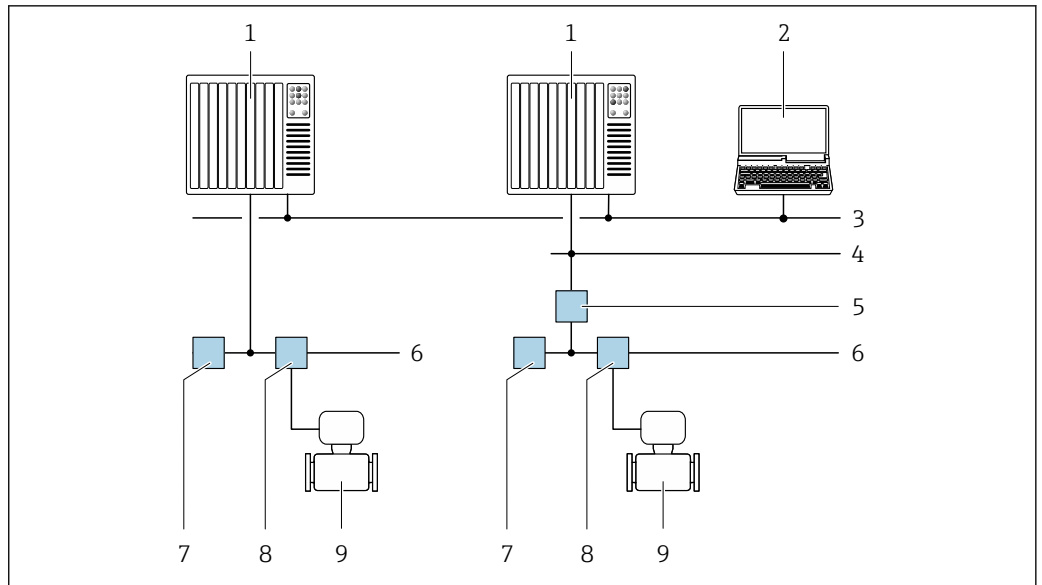
A0028746

45 Options for remote operation via HART protocol (passive)

- 1 Control system (e.g. PLC)
- 2 Transmitter power supply unit, e.g. RN221N (with communication resistor)
- 3 Connection for Commubox FXA195 and Field Communicator 475
- 4 Field Communicator 475
- 5 Computer with Web browser (e.g. Internet Explorer) for access to the integrated device Web server or computer with an operating tool (e.g. FieldCare, DeviceCare, AMS Device Manager, SIMATIC PDM) with COM DTM "CDI Communication TCP/IP"
- 6 Commubox FXA195 (USB)
- 7 Field Xpert SFX350 or SFX370
- 8 Field Xpert SMT70
- 9 VIATOR Bluetooth modem with connecting cable
- 10 Transmitter

Via FOUNDATION Fieldbus network

This communication interface is available in device versions with FOUNDATION Fieldbus.



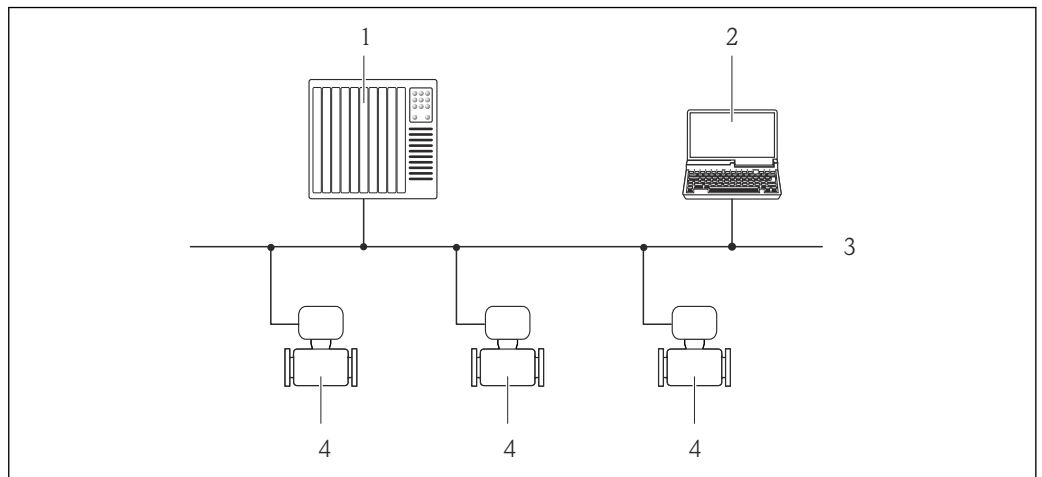
A0028837

46 Options for remote operation via FOUNDATION Fieldbus network

- 1 Automation system
- 2 Computer with FOUNDATION Fieldbus network card
- 3 Industry network
- 4 High Speed Ethernet FF-HSE network
- 5 Segment coupler FF-HSE/FF-H1
- 6 FOUNDATION Fieldbus FF-H1 network
- 7 Power supply FF-H1 network
- 8 T-box
- 9 Measuring device

Via PROFIBUS DP network

This communication interface is available in device versions with PROFIBUS DP.



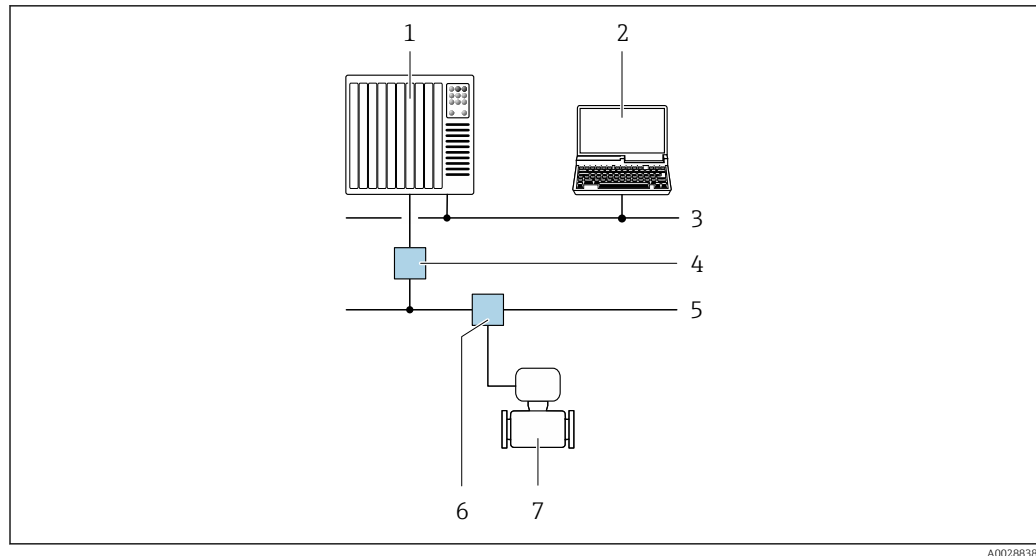
A0020903

47 Options for remote operation via PROFIBUS DP network

- 1 Automation system
- 2 Computer with PROFIBUS network card
- 3 PROFIBUS DP network
- 4 Measuring device

Via PROFIBUS PA network

This communication interface is available in device versions with PROFIBUS PA.



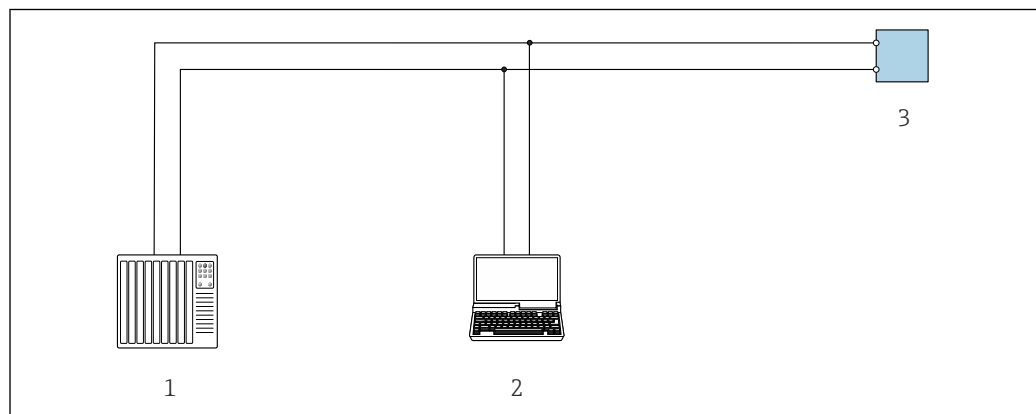
A0028838

48 Options for remote operation via PROFIBUS PA network

- 1 Automation system
- 2 Computer with PROFIBUS network card
- 3 PROFIBUS DP network
- 4 Segment coupler PROFIBUS DP/PA
- 5 PROFIBUS PA network
- 6 T-box
- 7 Measuring device

Via Modbus RS485 protocol

This communication interface is available in device versions with a Modbus-RS485 output.



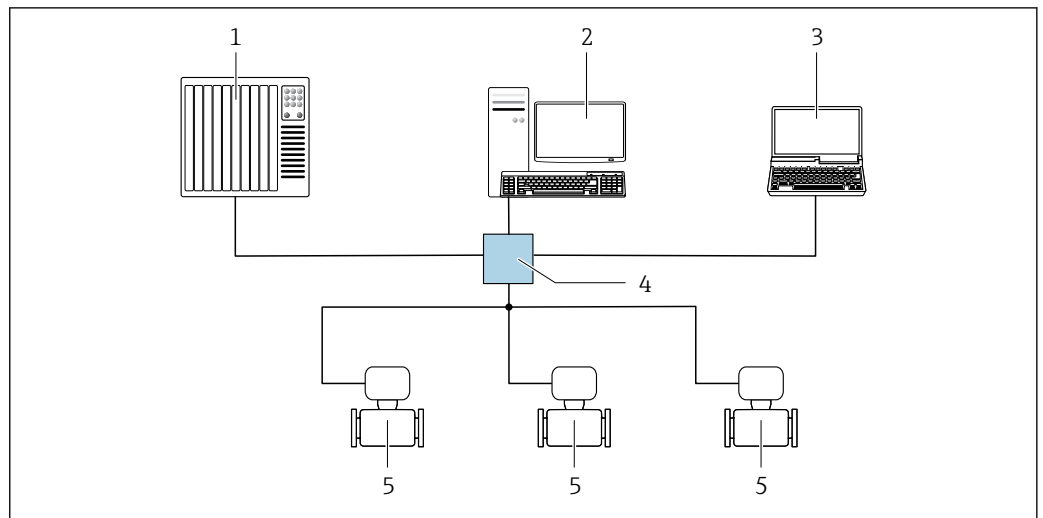
A0029437

49 Options for remote operation via Modbus-RS485 protocol (active)

- 1 Control system (e.g. PLC)
- 2 Computer with Web browser (e.g. Internet Explorer) for accessing the integrated device Web server or with operating tool (e.g. FieldCare, DeviceCare) with COM DTM "CDI Communication TCP/IP" or Modbus DTM
- 3 Transmitter

Via EtherNet/IP network

This communication interface is available in device versions with EtherNet/IP.

Star topology

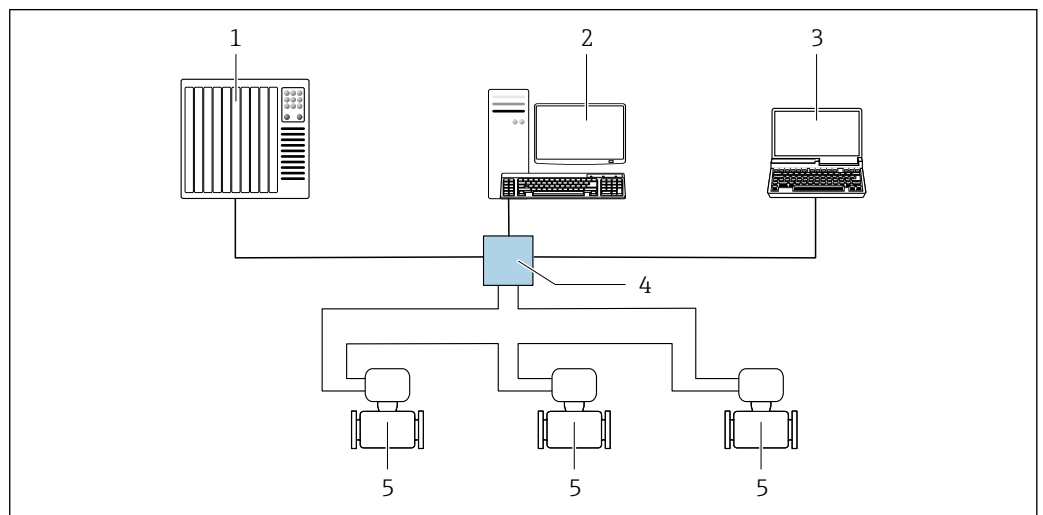
A0032078

50 Options for remote operation via EtherNet/IP network: star topology

- 1 Automation system, e.g. "RSLogix" (Rockwell Automation)
- 2 Workstation for measuring device operation: with Custom Add-On Profile for "RSLogix 5000" (Rockwell Automation) or with Electronic Data Sheet (EDS)
- 3 Computer with Web browser (e.g. Internet Explorer) for accessing the integrated device Web server or computer with operating tool (e.g. FieldCare, DeviceCare) with COM DTM "CDI Communication TCP/IP"
- 4 Ethernet switch
- 5 Measuring device

Ring topology

The device is integrated via the terminal connection for signal transmission (output 1) and the service interface (CDI-RJ45).



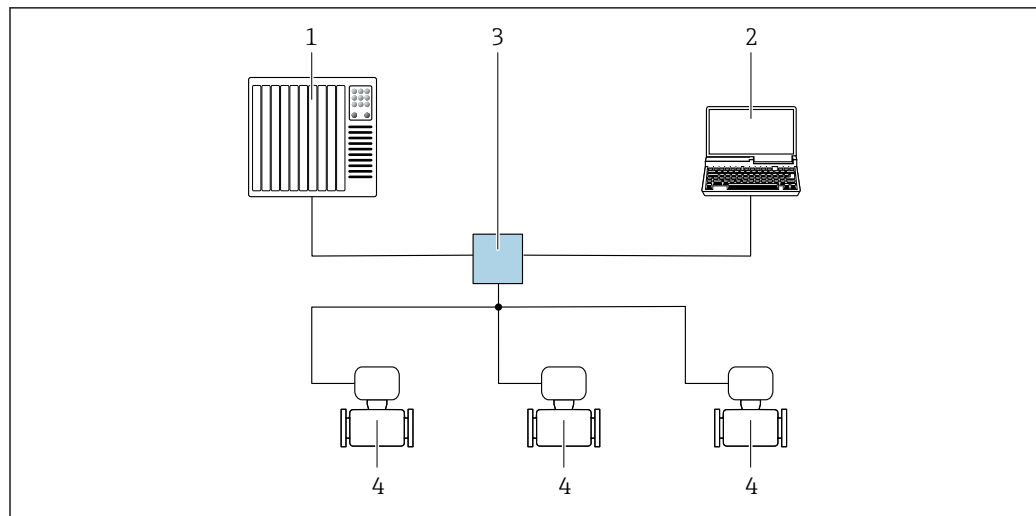
A0033725

51 Options for remote operation via EtherNet/IP network: ring topology

- 1 Automation system, e.g. "RSLogix" (Rockwell Automation)
- 2 Workstation for measuring device operation: with Custom Add-On Profile for "RSLogix 5000" (Rockwell Automation) or with Electronic Data Sheet (EDS)
- 3 Computer with Web browser (e.g. Internet Explorer) for accessing the integrated device Web server or computer with operating tool (e.g. FieldCare, DeviceCare) with COM DTM "CDI Communication TCP/IP"
- 4 Ethernet switch
- 5 Measuring device

Via PROFINET network

This communication interface is available in device versions with PROFINET.

Star topology

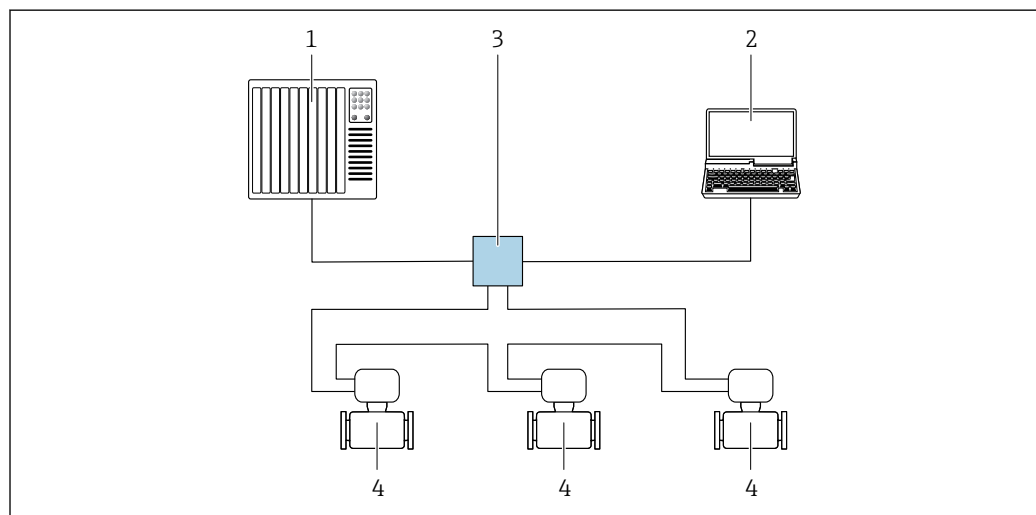
A0026545

52 Options for remote operation via PROFINET network: star topology

- 1 Automation system, e.g. Simatic S7 (Siemens)
- 2 Computer with Web browser (e.g. Internet Explorer) for accessing the integrated device Web server or computer with operating tool (e.g. FieldCare, DeviceCare, SIMATIC PDM) with COM DTM "CDI Communication TCP/IP"
- 3 Switch, e.g. Scalance X204 (Siemens)
- 4 Measuring device

Ring topology

This communication interface is available in device versions with PROFINET.



A0033719

53 Options for remote operation via PROFINET network: ring topology

- 1 Automation system, e.g. Simatic S7 (Siemens)
- 2 Computer with Web browser (e.g. Internet Explorer) for accessing the integrated device Web server or computer with operating tool (e.g. FieldCare, DeviceCare, SIMATIC PDM) with COM DTM "CDI Communication TCP/IP"
- 3 Switch, e.g. Scalance X204 (Siemens)
- 4 Measuring device

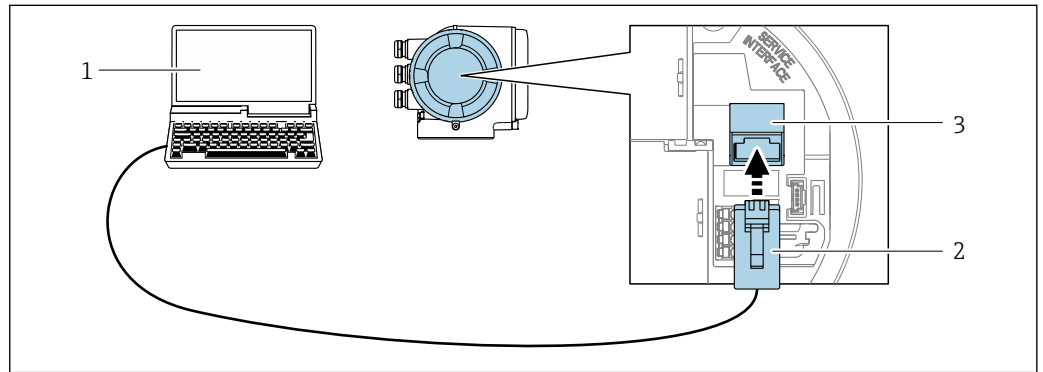
Service interface

Via service interface (CDI-RJ45)

A point-to-point connection can be established to configure the device onsite. With the housing open, the connection is established directly via the service interface (CDI-RJ45) of the device.

 An adapter for RJ45 and the M12 connector is optionally available:
Order code for "Accessories", option **NB**: "Adapter RJ45 M12 (service interface)"

The adapter connects the service interface (CDI-RJ45) to an M12 connector mounted in the cable entry. Therefore the connection to the service interface can be established via an M12 connector without opening the device.



A0027563

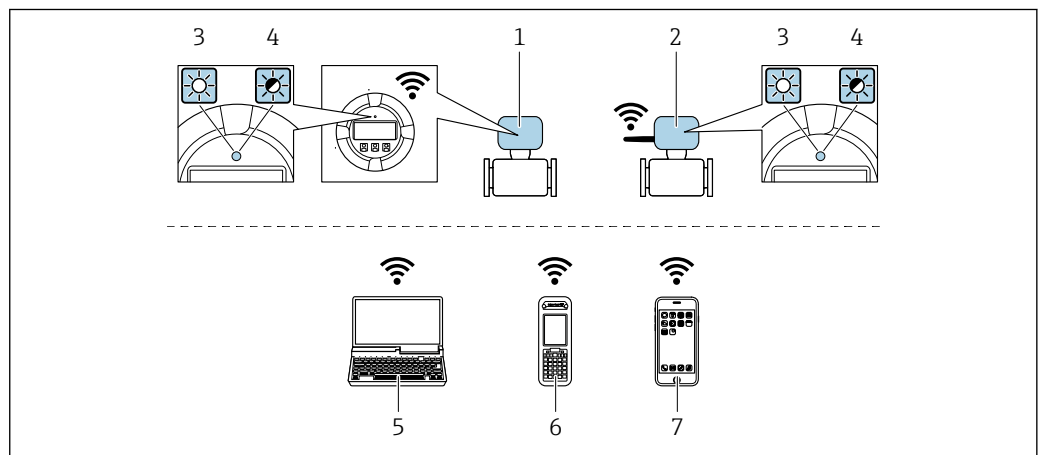
 54 Connection via service interface (CDI-RJ45)

- 1 Computer with Web browser (e.g. Microsoft Internet Explorer, Microsoft Edge) for accessing the integrated device Web server or with "FieldCare", "DeviceCare" operating tool with COM DTM "CDI Communication TCP/IP" or Modbus DTM
- 2 Standard Ethernet connecting cable with RJ45 connector
- 3 Service interface (CDI-RJ45) of the measuring device with access to the integrated Web server

Via WLAN interface

The optional WLAN interface is available on the following device version:

Order code for "Display; operation", option **G** "4-line, illuminated, graphic display; touch control + WLAN"



A0034570

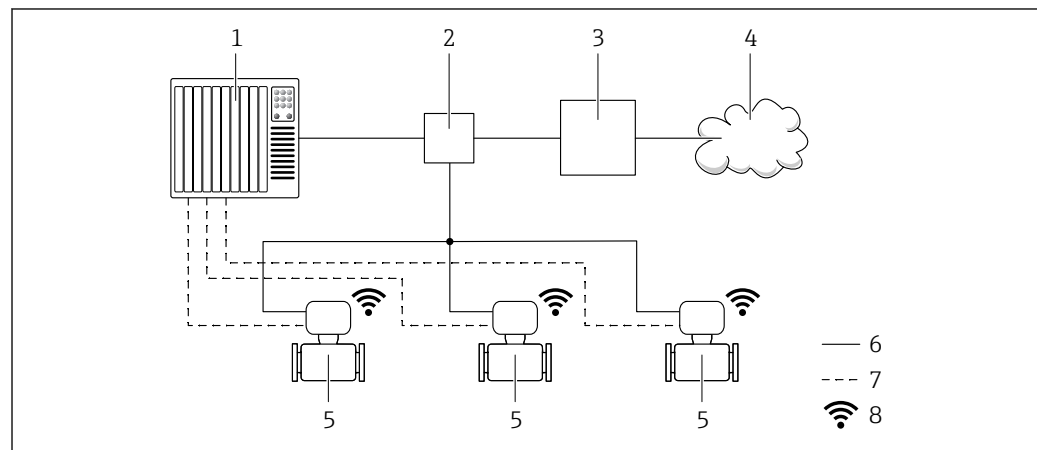
- 1 Transmitter with integrated WLAN antenna
- 2 Transmitter with external WLAN antenna
- 3 LED lit constantly: WLAN reception is enabled on measuring device
- 4 LED flashing: WLAN connection established between operating unit and measuring device
- 5 Computer with WLAN interface and Web browser (e.g. Microsoft Internet Explorer, Microsoft Edge) for accessing the integrated device Web server or with operating tool (e.g. FieldCare, DeviceCare)
- 6 Mobile handheld terminal with WLAN interface and Web browser (e.g. Microsoft Internet Explorer, Microsoft Edge) for accessing the integrated device Web server or operating tool (e.g. FieldCare, DeviceCare)
- 7 Smart phone or tablet (e.g. Field Xpert SMT70)

Function	WLAN: IEEE 802.11 b/g (2.4 GHz) ▪ Access point with DHCP server (default setting) ▪ Network
Encryption	WPA2-PSK AES-128 (in accordance with IEEE 802.11i)
Configurable WLAN channels	1 to 11
Degree of protection	IP67
Available antennas	▪ Internal antenna ▪ External antenna (optional) In the event of poor transmission/reception conditions at the place of installation. Available as an accessory → 99.  Only one antenna active in each case!
Max. range	50 m (164 ft)
Materials: External WLAN antenna	▪ Antenna: ASA plastic (acrylic ester-styrene-acrylonitrile) and nickel-plated brass ▪ Adapter: Stainless steel and nickel-plated brass ▪ Cable: Polyethylene ▪ Connector: Nickel-plated brass ▪ Angle bracket: Stainless steel

Network integration

With the optional OPC-UA-Server application package, the device can be integrated into an Ethernet network via the service interface (CDI-RJ45 and WLAN) and communicate with OPC-UA clients. If the device is used in this way, IT security must be considered.

For permanent access to device data and for device configuration via the Web server, the device is incorporated directly in a network via the service interface (CDI-RJ45). In this way, the device can be accessed any time from the control station. The measured values are processed separately via the inputs and outputs through the automation system.



A0033618

- 1 Automation system, e.g. Simatic S7 (Siemens)
- 2 Ethernet switch
- 3 Edge Gateway
- 4 Cloud
- 5 Measuring device
- 6 Ethernet network
- 7 Measured values via inputs and outputs
- 8 Optional WLAN interface

 The optional WLAN interface is available on the following device version:
Order code for "Display; operation", option **G** "4-line, illuminated, graphic display; touch control + WLAN"

 Special Documentation for the OPC-UA-Server application package → 103.

Supported operating tools

Different operating tools can be used for local or remote access to the measuring device. Depending on the operating tool used, access is possible with different operating units and via a variety of interfaces.

Supported operating tools	Operating unit	Interface	Additional information
Web browser	Notebook, PC or tablet with Web browser	■ CDI-RJ45 service interface ■ WLAN interface ■ Ethernet-based fieldbus (EtherNet/IP, PROFINET)	Special Documentation for device →  103
DeviceCare SFE100	Notebook, PC or tablet with Microsoft Windows system	■ CDI-RJ45 service interface ■ WLAN interface ■ Fieldbus protocol	→  101
FieldCare SFE500	Notebook, PC or tablet with Microsoft Windows system	■ CDI-RJ45 service interface ■ WLAN interface ■ Fieldbus protocol	→  101
Device Xpert	Field Xpert SFX 100/350/370	HART and FOUNDATION Fieldbus fieldbus protocol	Operating Instructions BA01202S Device description files: Use update function of handheld terminal



Other operating tools based on FDT technology with a device driver such as DTM/iDTM or DD/EDD can be used for device operation. These operating tools are available from the individual manufacturers. Integration into the following operating tools, among others, is supported:

- FactoryTalk AssetCentre (FTAC) by Rockwell Automation → www.rockwellautomation.com
- Process Device Manager (PDM) by Siemens → www.siemens.com
- Asset Management Solutions (AMS) by Emerson → www.emersonprocess.com
- FieldCommunicator 375/475 by Emerson → www.emersonprocess.com
- Field Device Manager (FDM) by Honeywell → www.honeywellprocess.com
- FieldMate by Yokogawa → www.yokogawa.com
- PACTWare → www.pactware.com

The associated device description files are available at: www.endress.com → Downloads

Web server

Thanks to the integrated Web server, the device can be operated and configured via a Web browser and via a service interface (CDI-RJ45) or via a WLAN interface. The structure of the operating menu is the same as for the local display. In addition to the measured values, status information on the device is also displayed and allows the user to monitor the status of the device. Furthermore the device data can be managed and the network parameters can be configured.

A device that has a WLAN interface (can be ordered as an option) is required for the WLAN connection: order code for "Display; operation", option G "4-line, illuminated; touch control + WLAN". The device acts as an Access Point and enables communication by computer or a mobile handheld terminal.

Supported functions

Data exchange between the operating unit (such as a notebook for example) and the measuring device:

- Upload the configuration from the measuring device (XML format, configuration backup)
- Save the configuration to the measuring device (XML format, restore configuration)
- Export event list (.csv file)
- Export parameter settings (.csv file or PDF file, document the measuring point configuration)
- Export the Heartbeat verification log (PDF file, only available with the "Heartbeat Verification" application package)

- Flash firmware version for device firmware upgrade, for instance
- Download driver for system integration
- Visualize up to 1000 saved measured values (only available with the **Extended HistoROM** application package →  97)

 Web server special documentation →  103

HistoROM data management

The measuring device features HistoROM data management. HistoROM data management comprises both the storage and import/export of key device and process data, making operation and servicing far more reliable, secure and efficient.

 When the device is delivered, the factory settings of the configuration data are stored as a backup in the device memory. This memory can be overwritten with an updated data record, for example after commissioning.

Additional information on the data storage concept

There are different types of data storage units in which device data are stored and used by the device:

	Device memory	T-DAT	S-DAT
Available data	▪ Event logbook such as diagnostic events for example ▪ Parameter data record backup ▪ Device firmware package ▪ Driver for system integration for exporting via Web server, e.g: – GSD for PROFIBUS DP – GSD for PROFIBUS PA – GSDML for PROFINET – EDS for EtherNet/IP – DD for FOUNDATION Fieldbus	▪ Measured value logging ("Extended HistoROM" order option) ▪ Current parameter data record (used by firmware at run time) ▪ Peakhold indicator (min/max values) ▪ Totalizer values	▪ Sensor data: nominal diameter etc. ▪ Serial number ▪ Calibration data ▪ Device configuration (e.g. SW options, fixed I/O or multi I/O)
Storage location	Fixed on the user interface board in the connection compartment	Attachable to the user interface board in the connection compartment	In the sensor plug in the transmitter neck part

Data backup

Automatic

- The most important device data (sensor and transmitter) are automatically saved in the DAT modules
- If the transmitter or measuring device is replaced: once the T-DAT containing the previous device data has been exchanged, the new measuring device is ready for operation again immediately without any errors
- If the sensor is replaced: once the sensor has been replaced, new sensor data are transferred from the S-DAT in the measuring device and the measuring device is ready for operation again immediately without any errors
- If exchanging the electronics module (e.g. I/O electronics module): Once the electronics module has been replaced, the software of the module is compared against the current device firmware. The module software is upgraded or downgraded where necessary. The electronics module is available for use immediately afterwards and no compatibility problems occur.

Manual

Additional parameter data record (complete parameter settings) in the integrated device memory HistoROM backup for:

- Data backup function
Backup and subsequent restoration of a device configuration in the device memory HistoROM backup
- Data comparison function
Comparison of the current device configuration with the device configuration saved in the device memory HistoROM backup

Data transfer

Manual

- Transfer of a device configuration to another device using the export function of the specific operating tool, e.g. with FieldCare, DeviceCare or Web server: to duplicate the configuration or to store in an archive (e.g. for backup purposes)
- Transmission of the drivers for system integration via Web server, e.g.:
 - GSD for PROFIBUS DP
 - GSD for PROFIBUS PA
 - GSDML for PROFINET
 - EDS for EtherNet/IP
 - DD for FOUNDATION Fieldbus

Event list

Automatic

- Chronological display of up to 20 event messages in the events list
- If the **Extended HistoROM** application package (order option) is enabled: up to 100 event messages are displayed in the events list along with a time stamp, plain text description and remedial measures
- The events list can be exported and displayed via a variety of interfaces and operating tools e.g. DeviceCare, FieldCare or Web server

Data logging

Manual

If the **Extended HistoROM** application package (order option) is enabled:

- Record up to 1 000 measured values via 1 to 4 channels
- User configurable recording interval
- Record up to 250 measured values via each of the 4 memory channels
- Export the measured value log via a variety of interfaces and operating tools e.g. FieldCare, DeviceCare or web server

Certificates and approvals



Currently available certificates and approvals can be called up via the product configurator.

CE mark

The device meets the legal requirements of the applicable EU Directives. These are listed in the corresponding EU Declaration of Conformity along with the standards applied.

Endress+Hauser confirms successful testing of the device by affixing to it the CE mark.

C-Tick symbol

The measuring system meets the EMC requirements of the "Australian Communications and Media Authority (ACMA)".

Ex approval

The measuring device is certified for use in hazardous areas and the relevant safety instructions are provided in the separate "Safety Instructions" (XA) document. Reference is made to this document on the nameplate.

Devices with the order code for "Approval; transmitter + sensor", option BA, BB, BC or BD have equipment protection level (EPL) Ga/Gb (Zone 0 in the measuring tube).



The separate Ex documentation (XA) containing all the relevant explosion protection data is available from your Endress+Hauser sales center.

ATEX/IECEx

Currently, the following versions for use in hazardous areas are available:

Ex db eb

Category	Type of protection
II1/2G	Ex db eb ia IIC T6...T1 Ga/Gb Ex db eb ia IIB T6...T1 Ga/Gb
II2G	Ex db eb ia IIC T6...T1 Gb Ex db eb ia IIB T6...T1 Gb

Ex db

Category	Type of protection
II1/2G	Ex db ia IIC T6...T1 Ga/Gb Ex db ia IIB T6...T1 Ga/Gb
II2G	Ex db ia IIC T6...T1 Gb Ex db ia IIB T6...T1 Gb

Ex ec

Category	Type of protection
II3G	Ex ec IIC T5...T1 Gc

Ex tb

Category	Type of protection
II2D	Ex tb IIIC T** °C Db

cCSA_{US}

Currently, the following versions for use in hazardous areas are available:

IS (Ex i) and XP (Ex d)

- Class I, III, III Division 1 Groups A-G
- Class I, III, III Division 1 Groups C-G

NI (Ex nA)

Class I Division 2 Groups A - D

Ex de

- Class I, Zone 1 AEx/ Ex de ia IIC T6...T1 Ga/Gb
Class I, Zone 1 AEx/ Ex de ia IIB T6...T1 Ga/Gb
- Class I, Zone 1 AEx/ Ex de ia IIC T6...T1 Gb
Class I, Zone 1 AEx/ Ex de ia IIB T6...T1 Gb

Ex d

- Class I, Zone 1 AEx/ Ex d ia IIC T6...T1 Ga/Gb
Class I, Zone 1 AEx/ Ex d ia IIB T6...T1 Ga/Gb
- Class I, Zone 1 AEx/ Ex d ia IIC T6...T1 Gb
Class I, Zone 1 AEx/ Ex d ia IIB T6...T1 Gb

Ex nA

Class I, Zone 2 AEx/ Ex nA IIC T5...T1 Gc

Ex tb

Zone 21 AEx/ Ex tb IIIC T** °C Db

Sanitary compatibility	■ 3-A approval Only devices with the order code for "Additional approval", option LP "3A" have 3-A approval. ■ EHEDG-tested Only devices with the order code for "Additional approval", option LT "EHEDG" have been tested and meet the requirements of the EHEDG. To meet the requirements for EHEDG certification, the device must be used with process connections in accordance with the EHEDG position paper entitled "Easy Cleanable Pipe Couplings and Process Connections" (www.ehedg.org). ■ FDA ■ Food Contact Materials Regulation (EC) 1935/2004
Pharmaceutical compatibility	■ FDA ■ USP Class VI ■ TSE/BSE Certificate of Suitability
Functional safety	The measuring device can be used for flow monitoring systems (min., max., range) up to SIL 2 (single-channel architecture; order code for "Additional approval", option LA) and SIL 3 (multichannel architecture with homogeneous redundancy) and is independently evaluated and certified by the TÜV in accordance with IEC 61508. The following types of monitoring in safety equipment are possible: ■ Mass flow ■ Volume flow ■ Density  Functional Safety Manual with information on the SIL device →  102
HART certification	HART interface The measuring device is certified and registered by the FieldComm Group. The measuring system meets all the requirements of the following specifications: ■ Certified according to HART 7 ■ The device can also be operated with certified devices of other manufacturers (interoperability)
FOUNDATION Fieldbus certification	FOUNDATION Fieldbus interface The measuring device is certified and registered by the FieldComm Group. The measuring system meets all the requirements of the following specifications: ■ Certified in accordance with FOUNDATION Fieldbus H1 ■ Interoperability Test Kit (ITK), revision version 6.2.0 (certificate available on request) ■ Physical Layer Conformance Test ■ The device can also be operated with certified devices of other manufacturers (interoperability)
Certification PROFIBUS	PROFIBUS interface The measuring device is certified and registered by the PNO (PROFIBUS User Organization). The measuring system meets all the requirements of the following specifications: ■ Certified in accordance with PROFIBUS PA Profile 3.02 ■ The device can also be operated with certified devices of other manufacturers (interoperability)
EtherNet/IP certification	The measuring device is certified and registered by the ODVA (Open Device Vendor Association). The measuring system meets all the requirements of the following specifications: ■ Certified in accordance with the ODVA Conformance Test ■ EtherNet/IP Performance Test ■ EtherNet/IP PlugFest compliance ■ The device can also be operated with certified devices of other manufacturers (interoperability)
Certification PROFINET	PROFINET interface The measuring device is certified and registered by the PNO (PROFIBUS User Organization). The measuring system meets all the requirements of the following specifications: ■ Certified according to: – Test specification for PROFINET devices – PROFINET Security Level 2 – Netload Class ■ The device can also be operated with certified devices of other manufacturers (interoperability)

Pressure Equipment Directive

The devices can be ordered with or without a PED approval. If a device with a PED approval is required, this must be explicitly stated in the order. For devices with nominal diameters less than or equal to DN 25 (1"), this is neither possible nor necessary.

- With the identification PED/G1/x (x = category) on the sensor nameplate, Endress+Hauser confirms conformity with the "Essential Safety Requirements" specified in Appendix I of the Pressure Equipment Directive 2014/68/EU.
- Devices bearing this marking (PED) are suitable for the following types of medium:
 - Media in Group 1 and 2 with a vapor pressure greater than, or smaller and equal to 0.5 bar (7.3 psi)
 - Unstable gases
- Devices not bearing this marking (PED) are designed and manufactured according to good engineering practice. They meet the requirements of Article 4 paragraph 3 of the Pressure Equipment Directive 2014/68/EU. The range of application is indicated in tables 6 to 9 in Annex II of the Pressure Equipment Directive 2014/68/EU.

Radio approval

The measuring device has radio approval.



For detailed information on the radio approval, see the Special Documentation → 103

Additional certification**CRN approval**

Some device versions have CRN approval. A CRN-approved process connection with a CSA approval must be ordered for a CRN-approved device.

Tests and certificates

- Pressure test, internal procedure, inspection certificate
- EN10204-3.1 material certificate, wetted parts and sensor housing
- PMI test (XRF), internal procedure, wetted parts, test report
- EN10204-2.1 confirmation of compliance with the order and EN10204-2.2 test report

Other standards and guidelines

- EN 60529
Degrees of protection provided by enclosures (IP code)
- IEC/EN 60068-2-6
Environmental influences: Test procedure - Test Fc: vibrate (sinusoidal).
- IEC/EN 60068-2-31
Environmental influences: Test procedure - Test Ec: shocks due to rough handling, primarily for devices.
- EN 61010-1
Safety requirements for electrical equipment for measurement, control and laboratory use - general requirements
- IEC/EN 61326
Emission in accordance with Class A requirements. Electromagnetic compatibility (EMC requirements).
- NAMUR NE 21
Electromagnetic compatibility (EMC) of industrial process and laboratory control equipment
- NAMUR NE 32
Data retention in the event of a power failure in field and control instruments with microprocessors
- NAMUR NE 43
Standardization of the signal level for the breakdown information of digital transmitters with analog output signal.
- NAMUR NE 53
Software of field devices and signal-processing devices with digital electronics
- NAMUR NE 80
The application of the pressure equipment directive to process control devices
- NAMUR NE 105
Specifications for integrating fieldbus devices in engineering tools for field devices
- NAMUR NE 107
Self-monitoring and diagnosis of field devices
- NAMUR NE 131
Requirements for field devices for standard applications

- NAMUR NE 132
Coriolis mass meter
- ETSI EN 300 328
Guidelines for 2.4 GHz radio components.
- EN 301489
Electromagnetic compatibility and radio spectrum matters (ERM).

Ordering information

Detailed ordering information is available from the following sources:

- In the Product Configurator on the Endress+Hauser website: www.endress.com -> Click "Corporate" -> Select your country -> Click "Products" -> Select the product using the filters and search field -> Open product page -> The "Configure" button to the right of the product image opens the Product Configurator.
- From your Endress+Hauser Sales Center: www.addresses.endress.com



Product Configurator - the tool for individual product configuration

- Up-to-the-minute configuration data
- Depending on the device: Direct input of measuring point-specific information such as measuring range or operating language
- Automatic verification of exclusion criteria
- Automatic creation of the order code and its breakdown in PDF or Excel output format
- Ability to order directly in the Endress+Hauser Online Shop

Application packages

Many different application packages are available to enhance the functionality of the device. Such packages might be needed to address safety aspects or specific application requirements.

The application packages can be ordered with the device or subsequently from Endress+Hauser. Detailed information on the order code in question is available from your local Endress+Hauser sales center or on the product page of the Endress+Hauser website: www.endress.com.



Detailed information on the application packages:
Special Documentation for the device → 102

Diagnostics functions

Package	Description
Extended HistoROM	Comprises extended functions concerning the event log and the activation of the measured value memory. Event log: Memory volume is extended from 20 message entries (standard version) to up to 100 entries. Data logging (line recorder): ■ Memory capacity for up to 1000 measured values is activated. ■ 250 measured values can be output via each of the 4 memory channels. The recording interval can be defined and configured by the user. ■ Measured value logs can be accessed via the local display or operating tool e.g. FieldCare, DeviceCare or Web server.

Heartbeat Technology	Package	Description
	Heartbeat Verification +Monitoring	Heartbeat Verification Meets the requirement for traceable verification to DIN ISO 9001:2008 Chapter 7.6 a) "Control of monitoring and measuring equipment". ■ Functional testing in the installed state without interrupting the process. ■ Traceable verification results on request, including a report. ■ Simple testing process via local operation or other operating interfaces. ■ Clear measuring point assessment (pass/fail) with high test coverage within the framework of manufacturer specifications. ■ Extension of calibration intervals according to operator's risk assessment. Heartbeat Monitoring Continuously supplies data, which are characteristic of the measuring principle, to an external condition monitoring system for the purpose of preventive maintenance or process analysis. These data enable the operator to: ■ Draw conclusions - using these data and other information - about the impact process influences (such as corrosion, abrasion, buildup etc.) have on the measuring performance over time. ■ Schedule servicing in time. ■ Monitor the process or product quality, e.g. gas pockets.
Concentration	Package	Description
	Concentration	Calculation and outputting of fluid concentrations The measured density is converted to the concentration of a substance of a binary mixture using the "Concentration" application package: ■ Choice of predefined fluids (e.g. various sugar solutions, acids, alkalis, salts, ethanol etc.) ■ Common or user-defined units ("Brix", "Plato", % mass, % volume, mol/l etc.) for standard applications. ■ Concentration calculation from user-defined tables.
Petroleum	Package	Description
	Petroleum	The most important parameters for the Oil & Gas Industry can be calculated and displayed with this application package. ■ Corrected volume flow and calculated reference density in accordance with the "API Manual of Petroleum Measurement Standards, Chapter 11.1" ■ Water content, based on density measurement ■ Weighted mean of the density and temperature
OPC-UA server	Package	Description
	OPC-UA-Server	The application package provides the user with an integrated OPC-UA server for comprehensive instrument services for IoT and SCADA applications.  Special Documentation for the "OPC-UA-Server" application package →  103.

Accessories

Various accessories, which can be ordered with the device or subsequently from Endress+Hauser, are available for the device. Detailed information on the order code in question is available from your local Endress+Hauser sales center or on the product page of the Endress+Hauser website: www.endress.com.

Device-specific accessories

For the transmitter

Accessories	Description
Proline 300 transmitter	Transmitter for replacement or storage. Use the order code to define the following specifications: ▪ Approvals ▪ Output ▪ Input ▪ Display/operation ▪ Housing ▪ Software  Order code: 8X3BXX  Installation Instructions EA01150
Remote display and operating module DKX001	▪ If ordered directly with the measuring device: Order code for "Display; operation", option O "Separate 4-line display, illum.; 10 m (30 ft) Cable; touch control". ▪ If ordered separately: – Measuring device: order code for "Display; operation", option M "None, prepared for separate display". – DKX001: Via the separate product structure DKX001. ▪ If ordered subsequently: DKX001: Via the separate product structure DKX001. Mounting bracket for DKX001 ▪ Ordered directly with the DKX001: Order code for "Enclosed accessories", option RA "Mounting bracket, 1 1/2" pipe". ▪ If ordered subsequently: order number: 71340960 Connecting cable (replacement cable) Via the separate product structure: DKX002  Further information on display and operating module DKX001 →  83.  Special Documentation SD01763D
External WLAN antenna	External WLAN antenna with 1.5 m (59.1 in) connecting cable and two angle brackets. Order code for "Enclosed accessories", option P8 "Wireless antenna wide area".  ▪ The external WLAN antenna is not suitable for use in hygienic applications. ▪ Further information on the WLAN interface →  89.  Order number: 71351317  Installation Instructions EA01238D
Protective cover	Is used to protect the measuring device from the effects of the weather: e.g. rainwater, excess heating from direct sunlight.  Order number: 71343505  Installation Instructions EA01160

For the sensor

Accessories	Description
Heating jacket	Is used to stabilize the temperature of the fluids in the sensor. Water, water vapor and other non-corrosive liquids are permitted for use as fluids.  If using oil as a heating medium, please consult with Endress+Hauser. Heating jackets cannot be used with sensors fitted with a rupture disk. ■ If ordered together with the measuring device: order code for "Enclosed accessories" – Option RB "heating jacket, G 1/2" internal thread" – Option RC "heating jacket, G 3/4" internal thread" – Option RD "Heating jacket, NPT 1/2" internal thread" – Option RE "Heating jacket, NPT 3/4" internal thread" ■ If ordered subsequently: Use the order code with the product root DK8003.  Special Documentation SD02151D

Communication-specific accessories

Accessories	Description
Commubox FXA195 HART	For intrinsically safe HART communication with FieldCare via the USB interface.  Technical Information TI00404F
HART Loop Converter HMX50	Is used to evaluate and convert dynamic HART process variables to analog current signals or limit values.  ■ Technical Information TI00429F ■ Operating Instructions BA00371F
Fieldgate FXA320	Gateway for the remote monitoring of connected 4-20 mA measuring devices via a Web browser.  - Technical Information TI00025S - Operating Instructions BA00053S
Fieldgate FXA520	Gateway for the remote diagnostics and remote configuration of connected HART measuring devices via a Web browser.  - Technical Information TI00025S - Operating Instructions BA00051S
Field Xpert SFX350	Field Xpert SFX350 is a mobile computer for commissioning and maintenance. It enables efficient device configuration and diagnostics for HART and FOUNDATION Fieldbus devices and can be used in non-hazardous areas.  Operating Instructions BA01202S
Field Xpert SFX370	Field Xpert SFX370 is a mobile computer for commissioning and maintenance. It enables efficient device configuration and diagnostics for HART and FOUNDATION Fieldbus devices and can be used in the non-hazardous area and in the hazardous area.  Operating Instructions BA01202S
Field Xpert SMT70	The Field Xpert SMT70 tablet PC for device configuration enables mobile plant asset management in hazardous and non-hazardous areas. It is suitable for commissioning and maintenance staff to manage field instruments with a digital communication interface and to record progress. This tablet PC is designed as an all-in-one solution with a preinstalled driver library and is an easy-to-use, touch-sensitive tool which can be used to manage field instruments throughout their entire life cycle.  ■ Technical Information TI01342S ■ Operating Instructions BA01709S ■ Product page: www.endress.com/smt70

Service-specific accessories

Accessories	Description
Applicator	Software for selecting and sizing Endress+Hauser measuring devices: - Choice of measuring devices for industrial requirements - Calculation of all the necessary data for identifying the optimum flowmeter: e.g. nominal diameter, pressure loss, flow velocity and accuracy. - Graphic illustration of the calculation results - Determination of the partial order code, administration, documentation and access to all project-related data and parameters over the entire life cycle of a project. Applicator is available: - Via the Internet: https://portal.endress.com/webapp/applicator - As a downloadable DVD for local PC installation.
W@M	W@M Life Cycle Management Improved productivity with information at your fingertips. Data relevant to a plant and its components is generated from the first stages of planning and during the asset's complete life cycle. W@M Life Cycle Management is an open and flexible information platform with online and on-site tools. Instant access for your staff to current, in-depth data shortens your plant's engineering time, speeds up procurement processes and increases plant uptime. Combined with the right services, W@M Life Cycle Management boosts productivity in every phase. For more information, visit www.endress.com/lifecyclemanagement
FieldCare	FDT-based plant asset management tool from Endress+Hauser. It can configure all smart field units in your system and helps you manage them. By using the status information, it is also a simple but effective way of checking their status and condition.  Operating Instructions BA00027S and BA00059S
DeviceCare	Tool to connect and configure Endress+Hauser field devices.  Innovation brochure IN01047S

System components

Accessories	Description
Memograph M graphic data manager	The Memograph M graphic data manager provides information on all the relevant measured variables. Measured values are recorded correctly, limit values are monitored and measuring points analyzed. The data are stored in the 256 MB internal memory and also on a SD card or USB stick.  - Technical Information TI00133R - Operating Instructions BA00247R
Cerabar M	The pressure transmitter for measuring the absolute and gauge pressure of gases, steam and liquids. It can be used to read in the operating pressure value.  - Technical Information TI00426P and TI00436P - Operating Instructions BA00200P and BA00382P
Cerabar S	The pressure transmitter for measuring the absolute and gauge pressure of gases, steam and liquids. It can be used to read in the operating pressure value.  - Technical Information TI00383P - Operating Instructions BA00271P
iTEMP	The temperature transmitters can be used in all applications and are suitable for the measurement of gases, steam and liquids. They can be used to read in the medium temperature.  "Fields of Activity" document FA00006T

Supplementary documentation



For an overview of the scope of the associated Technical Documentation, refer to the following:

- *W@M Device Viewer* (www.endress.com/deviceviewer): Enter the serial number from nameplate
- *Endress+Hauser Operations App*: Enter the serial number from the nameplate or scan the 2D matrix code (QR code) on the nameplate

Standard documentation

Brief Operating Instructions

Brief Operating Instructions for the sensor

Measuring device	Documentation code
Proline Promass E	KA01260D

Brief Operating Instructions for transmitter

Measuring device	Documentation code						
	HART	FOUNDATION Fieldbus	PROFIBUS PA	PROFIBUS DP	Modbus RS485	EtherNet/IP	PROFINET
Proline 300	KA01309D	KA01229D	KA01227D	KA01386D	KA01311D	KA01339D	KA01341D

Operating Instructions

Measuring device	Documentation code						
	HART	FOUNDATION Fieldbus	PROFIBUS PA	PROFIBUS DP	Modbus RS485	EtherNet/IP	PROFINET
Promass E 300	BA01484D	BA01517D	BA01506D	BA01855D	BA01495D	BA01727D	BA01738D

Description of Device Parameters

Measuring device	Documentation code						
	HART	FOUNDATION Fieldbus	PROFIBUS PA	PROFIBUS DP	Modbus RS485	EtherNet/IP	PROFINET
Promass 300	GP01057D	GP01094D	GP01058D	GP01134D	GP01059D	GP01114D	GP01115D

Device-dependent additional documentation

Safety instructions

Safety instructions for electrical equipment for hazardous areas.

Contents	Documentation code
ATEX/IECEX Ex d/Ex de	XA01405D
ATEX/IECEX Ex ec	XA01439D
cCSAus XP	XA01373D
cCSAus Ex d/ Ex de	XA01372D
cCSAus Ex nA	XA01507D
INMETRO Ex d/Ex de	XA01468D
INMETRO Ex ec	XA01470D
NEPSI Ex d/Ex de	XA01469D
NEPSI Ex nA	XA01471D

Remote display and operating module DKX001

Contents	Documentation code
ATEX/IECEX Ex i	XA01494D
ATEX/IECEX Ex ec	XA01498D
cCSAus IS	XA01499D
cCSAus Ex nA	XA01513D
INMETRO Ex i	XA01500D
INMETRO Ex ec	XA01501D
NEPSI Ex i	XA01502D
NEPSI Ex nA	XA01503D

Special Documentation

Contents	Documentation code
Information on the Pressure Equipment Directive	SD01614D
Functional Safety Manual	SD01727D
Remote display and operating module DKX001	SD01763D
Radio approvals for WLAN interface for A309/A310 display module	SD01793D
OPC-UA Server ¹⁾	SD02039D

1) This Special Documentation is only available for device versions with a HART output.

Contents	Documentation code						
	HART	FOUNDATION Fieldbus	PROFIBUS PA	PROFIBUS DP	Modbus RS485	PROFINET	EtherNet/IP
Web server	SD01662D	SD01665D	SD01664D	SD02226D	SD01663D	SD01969D	SD01968D
Heartbeat Technology	SD01642D	SD01696D	SD01698D	SD02202D	SD01697D	SD01988D	SD01982
Concentration measurement	SD01644D	SD01706D	SD01708D	SD02212D	SD01707D	SD02005D	SD02004D
Petroleum	SD02097D	–	SD02291D	SD02216D	SD02098D	SD02099D	SD02096D

Installation Instructions

Contents	Comment
Installation instructions for spare part sets and accessories	Documentation code: specified for each individual accessory .

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PROFIBUS®

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FOUNDATION™ Fieldbus

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Modbus®

Registered trademark of SCHNEIDER AUTOMATION, INC.

EtherNet/IP™

Trademark of ODVA, Inc.

PROFINET®

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TRI-CLAMP®

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